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Developed Markets
Global

Rates Vol Lab

In search of cheap vol

- **Weekly Focus**: January Formosa supply has somewhat exceeded our expectations, but the next wave of issuance in February should bring a smaller supply of vega. We continue to like our 10y20y/5y30y straddles switch. We also like buying 6m USDJPY put spreads vs 6m10y payer to monetize the relative richness of rates vol to FX vol. Finally, we initiate delta-hedged GBP 15y10y/25y10y flatteners as a long convexity buy with an attractive carry.
- **Vol Surface**: USD 10y vols still look rich on the surface. In EUR vol, the 30y tail now looks cheap, while 5-7y tails look rich.
- **Forward vol**: USD forward vols in the bottom-right part of the surface remain cheap, but longer-dated forward vols on shorter tails have richened. EUR forward vols have cheapened across the surface.
- **Board/swaption RV (US)**: Put skew looks rich to payer skew.
- **Midcurves**: Short-dated USD midcurves on 5y5y look cheap to plain vanilla swaptions. EUR midcurves on 1y2y look somewhat rich.
- **Curve vol**: The 10s30s spread vol looks very rich to recent realized vols in both USD and EUR.
- **Conditional flies**: We still like selling 6m10y payers on the 6m 2s10s30s payer flies in the US to monetize the relative richness of the 10y vol.
- *Note: Futures trading involves substantial risk of loss.*

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See Appendix A-1 for Analyst Certification, Important Disclosures and non-US research analyst disclosures.

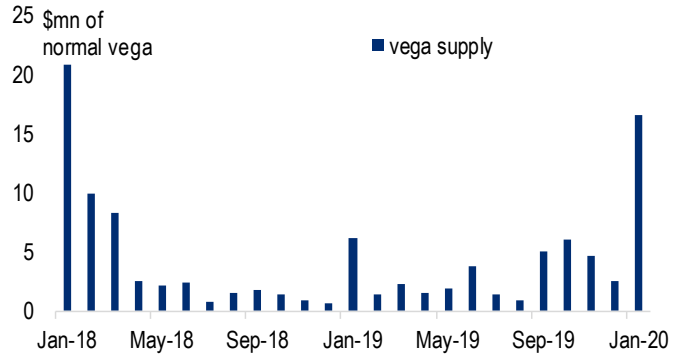
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Weekly Focus: In search of cheap vol

A strong start for Formosa supply

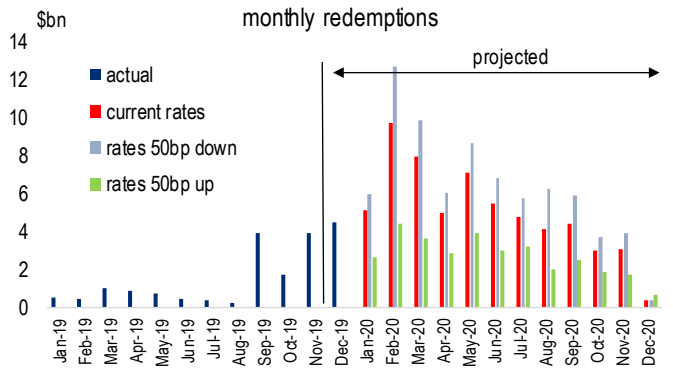
Formosa supply has had a strong start this year. We previously projected $5bn for January given an increased pace of redemptions and lower FX hedging costs, but $5.3bn already have been issued. Callable issuance should decline with the start of Taiwan holidays on January 23rd, but we think January issuance can reach up to $6bn by next week. With the issuance shifting toward 40NC5 structures, the supply of vega has surged notably. We estimate that about $16mn normal vega already has been swapped this month, the highest monthly supply since January 2018 (Figure 1).

Figure 1. Vega supply has surged to the highest levels since Jan 2018



Note: Normal parallel vega. Source: Citi Research

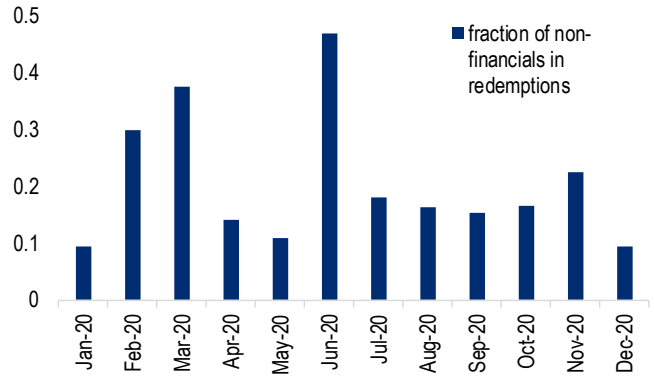
Figure 2. Expect more redemptions in February and March



Source: Citi Research

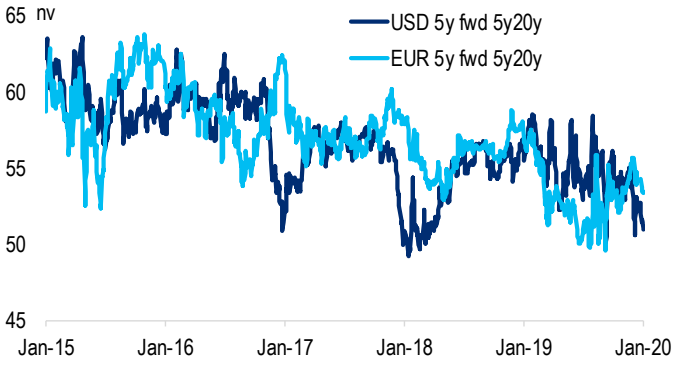
Supply should continue in February, after the lunar New Year holidays. Note that Formosa redemptions in February may be especially high (Figure 2). Still, Q1 Formosa supply tends to be front-loaded with at least 50% typically falling for January. In addition, a larger fraction of February issuance should be non-financials, who are less likely to swap vol (Figure 3). We therefore don't expect the same pace of vega supply as this month. We continue to like our 10y20y/5y30y straddle switch as way to buy the 5y fwd 5y20y volatility (Figure 4; 10y20y and 5y30y are at 54.6 and 56.6 nv, respectively, as of 1pm on 1/17/2020).

Figure 3. Expect more non-financial issuance in February and March



Source: Citi Research

Figure 4. USD forward vol continues to trade cheap



Source: Citi Research

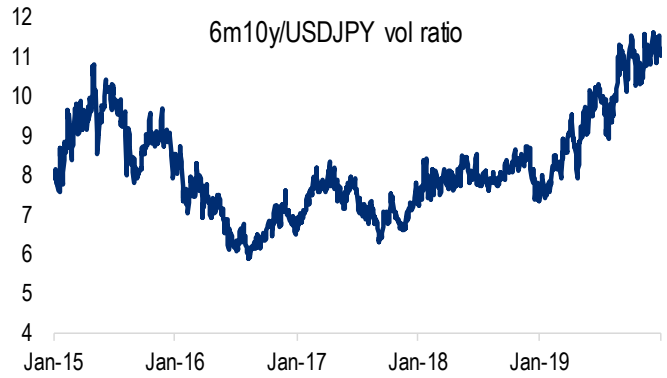
Buy 6m USDJPY put spreads vs 6m10y payers

Back in October we bought 3m USDJPY calls vs 3m10y payers as a way to monetize the relative dislocation of rates and FX vols (US Rates Weekly - Many ways to win). We then closed the payer leg of the trade keeping only $15mn of

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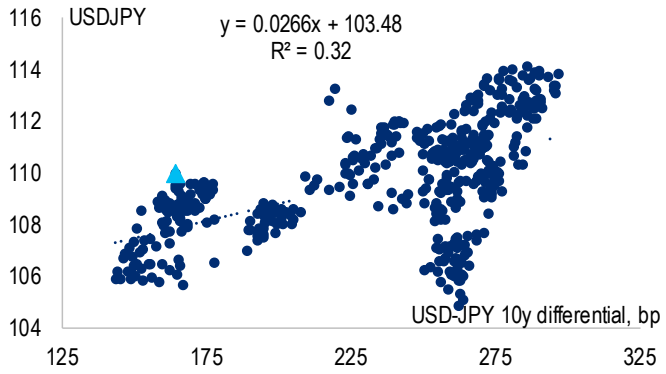
USDJPY 110 calls as a cheap option for a risk-on that could materialize if data improve or US-China trade negotiations show progress. As USDJPY rallied to about 110 we closed this trade earlier this week (Buy 6m USDJPY put spreads vs 6m10y payers).

Figure 5. USD rates vol continued to trade rich to USDJPY FX vol



Source: Citi Research

Figure 6. USDJPY looks too high to the rates differential



Sample: 2018-2019; Source: Citi Research

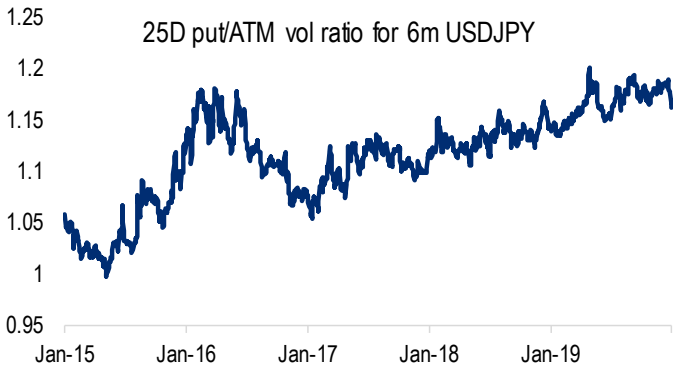
While rates and FX vol have normalized slightly, rates vol continues to trade very rich to FX vol by historical standards (Figure 5). At the same time, the yen is arguably too weak at these levels if evaluated by standard rates differentials (Figure 6). We believe this offers an attractive opportunity to position for a risk-off environment by buying USDJPY put spreads vs swaption payers - the yen tends to rally in risk off environments, as capital flows back to Japanese investors. Our base-case view assumes the return of growth concerns and further Fed cuts later in 2020 (US Rates Weekly: 2020 Outlook: Everything is awesome (when you're at the peak)). The dollar should sell off vs JPY in this scenario.

Figure 7. Macro-vol (rich/cheap to rates), normals (z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-7.6 (-0.3)	-3.5 (-0.2)	-3.7 (-0.2)	-3.1 (-0.2)	-1.9 (-0.1)	0.7 (0.0)	-2.6 (-0.2)
3m	-1.0 (-0.0)	1.4 (0.1)	0.8 (0.1)	1.3 (0.1)	2.7 (0.2)	4.4 (0.4)	0.5 (0.0)
6m	1.1 (0.1)	1.7 (0.1)	0.2 (0.0)	0.4 (0.0)	1.9 (0.2)	3.9 (0.4)	0.4 (0.0)
1y	3.4 (0.3)	2.0 (0.2)	1.1 (0.1)	0.3 (0.0)	1.2 (0.2)	2.0 (0.3)	-1.3 (-0.2)
2y	2.9 (0.3)	0.0 (0.0)	-0.2 (-0.0)	-0.7 (-0.1)	-0.4 (-0.1)	0.1 (0.0)	-2.7 (-0.4)
3y	0.4 (0.0)	-1.1 (-0.2)	-1.6 (-0.2)	-2.1 (-0.3)	-2.1 (-0.3)	-1.3 (-0.2)	-4.2 (-0.7)
5y	-1.3 (-0.2)	-2.6 (-0.4)	-3.0 (-0.5)	-3.0 (-0.5)	-3.1 (-0.5)	-2.8 (-0.5)	-5.7 (-1.0)
10y	-1.3 (-0.2)	-2.7 (-0.4)	-3.7 (-0.6)	-4.2 (-0.7)	-4.7 (-0.8)	-4.3 (-0.7)	-6.4 (-1.0)

Close of 1/14/20. We regress each ATM point on 3 principal components of the swap curve. We report rich/cheap for each vol as a residual from this regression together with z-scores in parentheses. We used 1996-2008 spliced with 2017-2019 data to exclude the near-zero rate policy regime. Source: Citi Research

Figure 8. USDJPY put skew looks optically rich



Source: Citi Research

We bought $30mn USDJPY 108/105 put spreads vs selling $50mn 6m10y ATMF+45bp (2.2%) payers for the net cost of $35K earlier this week (Buy 6m USDJPY put spreads vs 6m10y payers; USDJPY is at 110.17 as of 12pm on 1/17/20). The trade is structured similarly to a "quiet bull" familiar to rates investors, but we prefer buying USDJPY put spreads to rates receiver spreads given the relative cheapness of FX vol. We prefer selling 6m10y since 10y vol looks relatively

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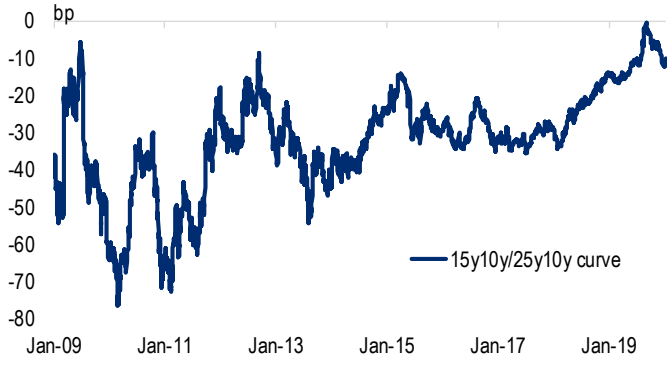
rich on the surface (Figure 7). The trade also benefits from the richness of the USDJPY put skew (Figure 8). The main risk to the trade is a return of inflation or a significantly improved global growth scenario which may cause the 10y to move above 2.2% by expiry.

Initiate 15y10y/25y10y delta-hedged GBP flatteners

In US Rates Vol Lab - Buying vol for less* we highlighted long-dated delta-hedged GBP flatteners as a long convexity trade, but did not initiate them at that time given the risks of a Labour win. With the UK election now behind us, the risk of a significant budget expansion has faded. While the GBP curve has flattened since then, long-dated forward curves, such as 15y10y/25y10y, remain quite steep from a historical standpoint given the richness of the 20y on the curve, likely because of receiving flows from ALM investors (Figure 9 and Figure 10).

In addition, the 15y10y/25y10y flatteners also carry positively by about 0.5bp over 1y. As a result, the flatteners offer an effectively free convexity buy. We initiate 50K DV01 of GBP 15y10y/25y10y flatteners at -11.2bp (as of 12pm on 1/17/2020). We will be delta-hedging the trade by adjusting the longer leg of the trade for DV01 neutral at each 25bp move in the 25y10y rate. We have previously found that hedging at 20-25bp offers an attractive balance between the accuracy of hedging and transaction costs (US Rates Vol Lab: Trading long-dated convexity). The risk to the trade is a continued richening of the 20y sector on ALM receiving.

Figure 9. The 15y10y/25y10y remains steep historically…



Source: Citi Research

Figure 10. … as 20s remain rich on the curve



Source: Citi Research

Figure 11 below reports a more systematic analysis of long-dated swap flatteners across USD, EUR, GBP, CAD and JPY curves. We evaluate steepeners by simple z-scores as well as the ratio of the daily breakeven (the move in rates that yields a convexity gain offsetting the negative daily roll) to realized daily vol of the underying rates (a smaller ratio indicates the cheapness of the embedded convexity and therefore a more attractive trade). GBP steepeners look especially attractive by these metrics. The 15y10y/25y10y flatteners, in particular, have both attractive valuation and carry.

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Figure 11. The 15y10y/25y10y GBP flatteners look especially attractive by z-scores and carry

curve	10y10y/ 15y15y	10y10y/ 20y5y	10y10y/ 20y10y	10y10y/ 20y15y	10y10y/ 25y5y	10y10y/ 25y10y	15y5y/ 20y5y	15y5y/ 20y10y	15y5y/ 20y15y	15y5y/ 25y5y	15y5y/ 25y10y	15y10y/ 25y5y	15y10y/ 25y10y	20y5y/ 25y5y	20y5y/ 25y10y
USD															
curve ,bp	-7.61	-8.77	-12.58	-16.57	-16.75	-21.02	-9.83	-13.65	-17.64	-17.81	-22.08	-13.13	-17.40	-7.98	-12.25
1y ZS	**0.04**	**0.24**	**-0.20**	-0.23	-0.73	-0.43	-0.73	-2.23	-1.66	-2.92	-1.76	-3.05	-1.93	-2.62	-2.25
3y ZS	**0.97**	**1.11**	**0.83**	0.72	0.58	0.56	0.53	0.18	0.02	-0.01	-0.11	-0.16	-0.35	-0.47	-0.96
ZS since 2000	**1.01**	**0.98**	**0.90**	0.86	0.71	0.69	0.68	0.58	0.53	0.41	0.39	0.19	0.14	-0.40	-0.38
1y carry, bp	-3.33	-4.30	-4.13	-4.13	-3.93	-4.02	-2.37	-2.20	-2.20	-2.00	-2.09	-0.87	-0.96	0.37	0.28
daily BE, bp	4.23	4.70	4.02	3.62	3.49	3.24	4.30	3.41	2.98	2.79	2.57	2.11	1.94	0.00	0.00
1y realized vol, bp	4.33	4.34	4.27	4.28	4.19	4.24	4.34	4.27	4.28	4.19	4.24	4.19	4.24	4.19	4.24
BE/realized vol	0.98	1.08	0.94	0.85	0.83	0.76	0.99	0.80	0.70	0.67	0.61	0.50	**0.46**	**0.00**	**0.00**
EUR															
curve ,bp	-17.05	-17.80	-27.18	-34.47	-36.83	-43.14	-20.04	-29.41	-36.70	-39.06	-45.37	-29.22	-35.53	-19.02	-25.34
1y ZS	**0.37**	**0.51**	**0.25**	0.10	0.02	-0.07	0.17	-0.13	-0.30	-0.34	-0.45	-0.59	-0.77	-1.53	-1.66
3y ZS	**1.30**	**1.51**	**1.14**	0.83	0.74	0.44	1.00	0.37	-0.18	-0.08	-0.56	-0.54	-1.16	-1.55	-1.89
ZS since 2000	**0.61**	**0.66**	**0.60**	0.51	0.57	0.48	0.49	0.42	0.31	0.40	0.29	0.38	0.23	0.17	-0.02
1y carry, bp	-7.60	-9.34	-9.24	-8.81	-9.13	-8.52	-4.88	-4.79	-4.36	-4.68	-4.07	-2.28	-1.67	0.20	0.82
daily BE, bp	6.29	6.89	5.96	5.22	5.28	4.67	6.10	4.96	4.11	4.22	3.53	3.39	2.52	0.00	0.00
1y realized vol, bp	4.66	4.67	4.51	4.44	4.34	4.30	4.18	4.24	4.08	4.44	4.34	4.30	4.18	4.24	4.08
BE/realized vol	1.35	1.48	1.32	1.18	1.22	1.09	1.46	1.17	1.01	0.95	0.81	0.79	**0.60**	**0.00**	**0.00**
GBP															
curve ,bp	-9.34	-11.01	-12.96	-14.96	-14.98	-17.04	-8.53	-10.48	-12.48	-12.50	-14.56	-8.31	-10.37	-3.97	-6.03
1y ZS	-0.42	-0.39	-0.30	-0.23	-0.23	-0.18	-0.10	-0.07	-0.06	-0.04	-0.04	-0.01	**-0.01**	**0.02**	**0.01**
3y ZS	0.94	0.91	0.98	1.02	1.03	1.06	0.99	1.06	1.09	1.11	1.12	**1.16**	1.15	**1.23**	**1.19**
ZS since 2000	1.17	1.12	1.26	1.40	1.39	1.54	1.23	1.46	1.67	1.61	**1.80**	**1.68**	**1.84**	1.36	1.36
1y carry, bp	-2.22	-2.77	-2.32	-2.19	-1.85	-1.88	-0.74	-0.29	-0.15	0.18	0.16	0.54	0.52	0.91	0.89
daily BE, bp	3.44	3.79	3.02	2.63	2.41	2.22	2.40	1.23	0.78	0.00	0.00	0.00	0.00	0.00	0.00
1y realized vol, bp	4.29	4.25	4.29	4.21	4.33	4.19	4.25	4.29	4.21	4.33	4.19	4.33	4.19	4.33	4.19
BE/realized vol	0.80	0.89	0.70	0.63	0.56	0.53	0.56	0.29	0.19	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
CAD															
curve ,bp	-25.20	-34.15	-39.56	-41.07	-45.44	-44.98	-33.97	-39.38	-40.89	-45.27	-44.81	-29.10	-28.64	-11.29	-10.83
1y ZS	-0.50	-0.48	-0.49	-0.44	-0.47	-0.41	-0.40	-0.42	-0.39	-0.41	-0.37	-0.40	**-0.34**	**-0.31**	**-0.23**
3y ZS	0.44	0.34	0.39	0.42	0.43	**0.45**	0.16	0.26	0.30	0.32	0.35	0.39	0.42	**0.52**	**0.56**
ZS since 2000	**0.55**	0.51	0.53	0.54	**0.55**	**0.56**	0.38	0.42	0.44	0.45	0.46	0.47	0.49	0.34	0.37
1y carry, bp	-6.68	-10.49	-8.34	-6.87	-5.95	-4.78	-6.73	-4.58	-3.11	-2.19	-1.02	1.01	2.18	4.54	5.71
daily BE, bp	5.99	7.28	5.68	4.67	4.28	3.53	7.15	4.89	3.53	2.92	1.79	0.00	0.00	0.00	0.00
1y realized vol, bp	4.11	4.11	3.95	3.92	3.95	3.94	3.93	3.97	3.94	3.92	3.95	3.94	3.93	3.97	3.94
BE/realized vol	1.46	1.77	1.44	1.19	1.09	0.90	1.82	1.23	0.90	0.74	0.45	**0.00**	**0.00**	**0.00**	**0.00**
JPY															
curve ,bp	10.43	11.64	12.40	8.70	13.19	7.16	5.00	5.76	2.06	6.55	0.52	4.07	-1.95	1.54	-4.48
1y ZS	0.12	0.13	0.46	0.37	0.72	0.46	**1.22**	**1.22**	1.05	**1.19**	0.99	1.14	0.90	1.02	0.72
3y ZS	-0.84	-0.79	-0.39	-0.65	-0.02	-0.58	0.69	0.92	0.55	**1.06**	0.49	**1.19**	0.36	**1.44**	0.00
ZS since 2000	0.55	0.53	0.79	0.60	0.98	0.63	1.08	1.27	1.05	**1.38**	1.02	**1.46**	0.93	**1.54**	0.55
1y carry, bp	-2.46	-2.79	-3.13	-4.13	-3.48	-4.83	-1.62	-1.97	-2.97	-2.32	-3.67	-1.51	-2.87	-0.69	-2.05
daily BE, bp	3.61	3.82	3.51	3.61	3.30	3.55	3.58	3.22	3.42	3.01	3.39	2.81	3.34	2.33	3.26
1y realized vol, bp	2.40	2.70	2.77	2.86	2.87	2.97	2.91	3.08	2.94	2.86	2.87	2.97	2.91	3.08	2.94
BE/realized vol	1.50	1.42	1.27	1.26	1.15	1.20	1.23	**1.05**	1.16	1.06	1.18	**0.95**	1.15	**0.76**	1.11

As of close of 1/16/2020. For each curve we calculate 1y, 3y and full sample (since 2000) z-scores. We also report 1y carry, daily breakeven (zero of carry is positive), and the ratio of the daily breakeven to realized vol. Three most attractive curves by each metric are marked in red. Source: Citi Research

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USD at-the-money swaption vols

USD relative value on the swaption surface

We report levels and z-scores of at-the-money vols, as well as the relative richness/cheapness of vols to rates (based on principal components of the curve).

- The vol surface has steepened across expiries since we last published our Vol Lab (Figure 13). Gamma has collapsed, while very long dated expiries have inched higher.
- The 10y tail looks relatively rich on the macro-vol model (Figure 14) and the PCA of the vol surface (Figure 15). For example, the 6m10y looks about 0.4 sigmas rich on the macro-vol model and 1.1 sigmas rich on PCA.
- Intermediate (1y) expiries also look rich on the surface (Figure 14 and Figure 15).
- 10y expiries still look cheap on both our macro-vol model (Figure 14) and the PCA of the vol surface (Figure 15). We have bought 10y20y/5y30y straddle switches to fade this (current levels are in Figure 12; the main risk is a big Tsy market selloff).

Figure 12. Implied vols, normals (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	26.0 (-1.2)	40.6 (-1.2)	46.4 (-1.4)	53.3 (-1.6)	54.0 (-1.7)	55.6 (-1.9)	54.1 (-1.8)
3m	35.1 (-0.9)	47.0 (-1.1)	52.3 (-1.1)	58.2 (-1.5)	59.8 (-1.6)	60.4 (-1.7)	58.0 (-1.7)
6m	40.8 (-1.2)	50.6 (-1.4)	53.9 (-1.6)	58.7 (-1.8)	60.3 (-1.8)	61.6 (-1.8)	58.9 (-1.8)
1y	51.1 (-1.4)	57.0 (-1.5)	59.1 (-1.6)	61.5 (-1.8)	62.4 (-1.9)	62.4 (-2.0)	58.7 (-2.1)
2y	60.9 (-1.4)	62.4 (-1.7)	63.3 (-1.5)	63.7 (-1.6)	63.5 (-1.6)	63.2 (-1.6)	58.8 (-1.3)
3y	64.8 (-0.9)	65.1 (-0.7)	64.9 (-0.9)	64.2 (-1.4)	63.5 (-1.4)	63.3 (-1.4)	57.9 (-1.1)
5y	66.5 (0.1)	65.5 (-0.1)	65.0 (-0.3)	64.6 (-0.6)	63.7 (-0.7)	63.1 (-0.8)	56.7 (-0.7)
10y	63.7 (0.3)	62.8 (-0.0)	62.1 (-0.3)	61.7 (-0.7)	60.6 (-0.8)	59.6 (-0.9)	53.7 (-0.4)

Close of 1/16/20. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 13. 2w change in implied vols, normals

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-5.6	-8.9	-10.4	-12.4	-12.5	-13.0	-11.4
3m	-3.9	-6.5	-6.7	-6.2	-6.5	-6.6	-6.0
6m	-5.0	-5.5	-5.6	-5.4	-4.7	-4.1	-2.9
1y	-3.8	-3.7	-3.6	-3.4	-3.4	-3.2	-3.2
2y	-2.6	-3.3	-3.0	-2.3	-2.0	-1.8	-1.1
3y	-1.9	-1.2	-1.3	-1.6	-1.4	-1.1	-0.5
5y	-0.6	-0.0	-0.1	-0.4	-0.3	-0.1	0.4
10y	0.1	0.6	0.7	0.6	0.5	0.4	1.3

Note: The table shows changes in ATM implied vols from 1/6/20 to 1/16/20. Source: Citi Research

Figure 14. Macro-vol (rich/cheap to rates), normals (z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-7.6 (-0.3)	-3.5 (-0.2)	-3.7 (-0.2)	-3.1 (-0.2)	-1.9 (-0.1)	0.7 (0.0)	-2.6 (-0.2)
3m	-1.0 (-0.0)	1.4 (0.1)	0.8 (0.1)	1.3 (0.1)	2.7 (0.2)	4.4 (0.4)	0.5 (0.0)
6m	1.1 (0.1)	1.7 (0.1)	0.2 (0.0)	0.4 (0.0)	1.9 (0.2)	3.9 (0.4)	0.4 (0.0)
1y	3.4 (0.3)	2.0 (0.2)	1.1 (0.1)	0.3 (0.0)	1.2 (0.2)	2.0 (0.3)	-1.3 (-0.2)
2y	2.9 (0.3)	0.0 (0.0)	-0.2 (-0.0)	-0.7 (-0.1)	-0.4 (-0.1)	0.1 (0.0)	-2.7 (-0.4)
3y	0.4 (0.0)	-1.1 (-0.2)	-1.6 (-0.2)	-2.1 (-0.3)	-2.1 (-0.3)	-1.3 (-0.2)	-4.2 (-0.7)
5y	-1.3 (-0.2)	-2.6 (-0.4)	-3.0 (-0.5)	-3.0 (-0.5)	-3.1 (-0.5)	-2.8 (-0.5)	-5.7 (-1.0)
10y	-1.3 (-0.2)	-2.7 (-0.4)	-3.7 (-0.6)	-4.2 (-0.7)	-4.7 (-0.8)	-4.3 (-0.7)	-6.4 (-1.0)

Close of 1/16/20. We regress each ATM point on 3 principal components of the swap curve. We report rich/cheap for each vol as a residual from this regression together with z-scores in parentheses. We used 1996-2008 spliced with 2017-2019 data to exclude the near-zero rate policy regime. Source: Citi Research

Figure 15. Rich/cheap to PCA of the vol surface, normals (z-score)

exp\tls	3m	1y	2y	3y	5y	7y	10y	30y
1m	-5.1 (-0.8)	-4.4 (-0.9)	-3.7 (-1.2)	-3.7 (-1.4)	-2.9 (-1.3)	-3.4 (-1.6)	-3.3 (-1.7)	-2.1 (-0.9)
3m	-4.0 (-0.8)	-0.4 (-0.1)	-0.5 (-0.3)	-0.0 (-0.0)	0.6 (0.6)	0.6 (0.7)	0.6 (0.8)	1.1 (1.1)
6m	-2.0 (-0.5)	-1.2 (-0.6)	-0.8 (-0.7)	-1.0 (-1.2)	-0.3 (-0.4)	0.2 (0.2)	0.8 (1.1)	1.4 (2.2)
1y	-1.0 (-0.5)	0.3 (0.2)	0.5 (0.5)	0.4 (0.5)	0.4 (0.5)	0.6 (0.7)	0.5 (0.6)	0.7 (1.1)
2y	3.6 (1.9)	1.7 (1.3)	0.9 (0.7)	1.0 (1.1)	0.1 (0.2)	0.0 (0.0)	0.0 (0.0)	0.7 (1.2)
3y	2.2 (1.7)	1.3 (1.1)	1.1 (1.1)	0.5 (0.5)	-0.4 (-0.8)	-0.6 (-1.4)	-0.3 (-0.7)	0.3 (0.6)
5y	0.8 (0.9)	0.2 (0.3)	-0.3 (-0.3)	-0.5 (-0.7)	-0.5 (-1.0)	-0.6 (-1.4)	-0.3 (-1.0)	-0.1 (-0.1)
10y	0.2 (0.3)	-0.4 (-0.6)	-0.9 (-1.1)	-1.2 (-1.6)	-1.0 (-1.5)	-1.1 (-1.7)	-0.6 (-0.9)	0.0 (0.0)

Close of 1/16/20. We calculate rich/cheap and z-scores in parentheses from PCA on the vol surface based on 3 principal components using 2y of data. By design, this framework only yields relative value. Source: Citi Research

USD implied vs realized volatilities

We calculate the relative richness of short-dated implied vols to 1m and 3m realized vols. We calculate realized vols using both intraday (hourly) and close-to-close sampling.

- Intraday realized volatilities have outperformed close-to-close volatilities over the last month, indicating a whipsawing intraday market (Figure 16).
- Consistent with the macro-vol model and the PCA of the vol surface, 1m expiries look cheap to recent realized vols (Figure 19).

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Figure 16. 1m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.22 (1.9)	1.14 (1.5)	1.11 (1.2)	1.16 (1.5)	1.14 (1.6)	1.13 (1.6)	1.11 (1.2)
3m	1.26 (2.3)	1.15 (1.6)	1.12 (1.2)	1.17 (1.6)	1.14 (1.5)	1.13 (1.6)	1.11 (1.2)
6m	1.17 (1.8)	1.13 (1.4)	1.11 (1.1)	1.16 (1.5)	1.14 (1.5)	1.12 (1.6)	1.11 (1.2)
1y	1.10 (1.2)	1.10 (1.0)	1.12 (1.1)	1.15 (1.5)	1.13 (1.5)	1.12 (1.5)	1.11 (1.2)

Note: Close of 1/16/20. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 17. 3m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.94 (-0.2)	0.91 (-0.5)	0.90 (-0.8)	0.90 (-0.9)	0.89 (-1.3)	0.89 (-1.8)	0.93 (-1.7)
3m	0.94 (-0.1)	0.91 (-0.5)	0.90 (-0.8)	0.90 (-0.9)	0.89 (-1.3)	0.89 (-1.8)	0.93 (-1.7)
6m	0.92 (-0.3)	0.90 (-0.6)	0.89 (-0.9)	0.90 (-1.0)	0.89 (-1.4)	0.89 (-1.9)	0.93 (-1.7)
1y	0.91 (-0.5)	0.89 (-0.8)	0.89 (-1.0)	0.90 (-1.2)	0.89 (-1.5)	0.89 (-2.0)	0.93 (-1.7)

Note: Close of 1/16/20. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 18. Intraday realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.01 (0.3)	1.12 (1.6)	1.08 (1.5)	1.03 (1.0)	0.99 (0.6)	0.98 (0.5)	0.91 (-0.1)
3m	1.15 (2.6)	1.17 (2.9)	1.14 (2.7)	1.09 (2.1)	1.07 (1.7)	1.05 (1.3)	0.97 (0.6)
6m	1.08 (3.0)	1.11 (2.7)	1.08 (2.3)	1.05 (1.7)	1.04 (1.5)	1.04 (1.2)	0.98 (0.6)
1y	1.06 (2.7)	1.08 (2.3)	1.05 (2.0)	1.03 (1.4)	1.03 (1.1)	1.02 (0.8)	0.96 (0.4)

Note: Close of 1/16/20. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 19. Intraday realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.77 (-0.6)	0.86 (-0.5)	0.86 (-0.4)	0.87 (-0.6)	0.85 (-0.8)	0.86 (-0.8)	0.82 (-0.8)
3m	0.84 (0.1)	0.89 (0.3)	0.90 (0.4)	0.92 (0.3)	0.92 (0.2)	0.92 (0.0)	0.88 (-0.3)
6m	0.81 (0.3)	0.87 (0.6)	0.87 (0.5)	0.89 (0.3)	0.90 (0.2)	0.92 (0.2)	0.88 (-0.2)
1y	0.83 (0.9)	0.88 (0.9)	0.88 (0.7)	0.89 (0.3)	0.90 (0.2)	0.91 (0.0)	0.87 (-0.3)

Note: Close of 1/16/20. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 20. Close-to-close realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.23 (1.9)	1.28 (2.8)	1.20 (2.5)	1.20 (2.5)	1.13 (2.0)	1.11 (1.9)	1.02 (1.0)
3m	1.45 (4.5)	1.34 (4.1)	1.27 (3.6)	1.27 (3.5)	1.22 (3.1)	1.18 (2.7)	1.08 (1.5)
6m	1.27 (4.3)	1.25 (4.1)	1.20 (3.3)	1.21 (3.1)	1.19 (2.8)	1.17 (2.5)	1.09 (1.3)
1y	1.17 (3.9)	1.18 (3.3)	1.18 (3.0)	1.18 (2.7)	1.17 (2.4)	1.15 (1.9)	1.07 (1.0)

Note: Close of 1/16/20. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 21. Close-to-close realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.72 (-0.6)	0.78 (-0.6)	0.77 (-0.7)	0.78 (-0.9)	0.76 (-1.2)	0.77 (-1.4)	0.77 (-1.4)
3m	0.79 (0.1)	0.81 (-0.0)	0.81 (-0.2)	0.83 (-0.4)	0.82 (-0.7)	0.82 (-0.9)	0.82 (-1.1)
6m	0.74 (0.0)	0.78 (-0.0)	0.78 (-0.3)	0.80 (-0.6)	0.80 (-0.7)	0.82 (-0.8)	0.82 (-0.9)
1y	0.75 (0.2)	0.78 (-0.0)	0.78 (-0.3)	0.80 (-0.6)	0.81 (-0.7)	0.81 (-0.8)	0.81 (-0.9)

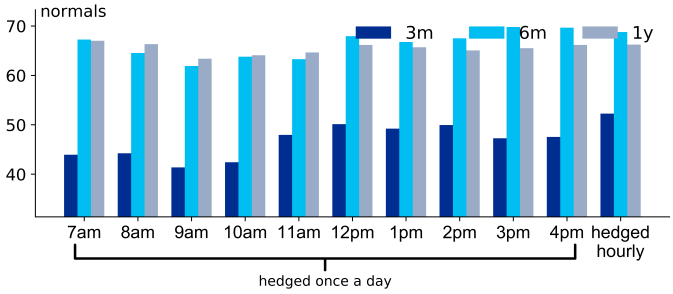
Note: Close of 1/16/20. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

USD day-to-day realized volatilities for a given time snapshot

We calculate day-to-day realized volatilities measured at a given hour within a day for the last 3m, 6m and 1y (EST time)

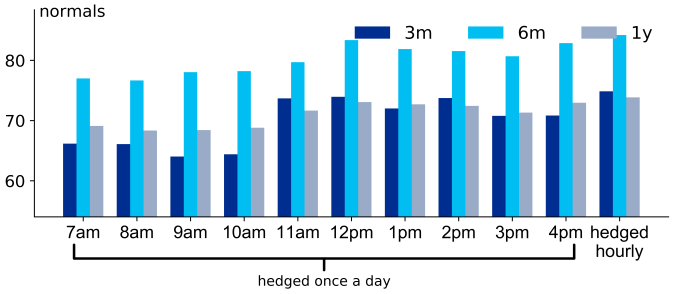
- Hedging around 9am has delivered relatively low realized volatility of the 10y rate over the past 3 and 12 months. In contrast, hedging around 12pm has delivered the highest realized vol over the recent past.

Figure 22. Realized vol of the 2y rate



Note: Close of 1/16/20, time is NY. Source: Citi Research

Figure 23. Realized vol of the 10y rate



Note: Close of 1/16/20, time is NY. Source: Citi Research

USD expiry and tail switches

We analyze ATM vol switches across expiries and tails.

- The 5y expiries look rich against neighboring expiries. For example, the 3y10y/5y10y ratio is 2.6 sigmas low while the 5y10y/10y10y ratio is 0.4 sigmas high (Figure 24).

- The 5y to 10y vols remain rich to shorter tails, especially in 1y and 2y expiries. For example, the 2y2y/2y5y vol ratio is 1.4 sigmas low (Figure 25).

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Figure 24. Short/long expiry ratio (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m/3m	0.74 (-1.8)	0.86 (-1.8)	0.89 (-2.0)	0.92 (-1.8)	0.90 (-2.0)	0.92 (-2.2)	0.93 (-1.9)
3m/6m	0.86 (-0.2)	0.93 (-0.3)	0.97 (0.1)	0.99 (-0.3)	0.99 (-0.8)	0.98 (-1.2)	0.99 (-1.0)
6m/1y	0.80 (-0.9)	0.89 (-1.1)	0.91 (-1.3)	0.95 (-1.7)	0.97 (-1.3)	0.99 (-0.9)	1.00 (-0.7)
1y/3y	0.79 (-1.3)	0.88 (-1.6)	0.91 (-1.7)	0.96 (-1.7)	0.98 (-2.1)	0.99 (-2.7)	1.01 (-2.8)
3y/5y	0.97 (-2.0)	0.99 (-1.7)	1.00 (-2.0)	0.99 (-2.4)	1.00 (-2.5)	1.00 (-2.6)	1.02 (-1.9)
5y/10y	1.04 (-0.2)	1.04 (-0.2)	1.05 (-0.1)	1.05 (0.2)	1.05 (0.4)	1.06 (0.4)	1.06 (-0.8)

Note: Close of 1/16/20. The table reports shorter expiry/longer expiry ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

Figure 25. Short/long tail ratio (3m z-score)

exp\tails	1y/2y	1y/3y	2y/5y	2y/10y	5y/10y	5y/30y	10y/30y
1m	0.64 (-1.1)	0.56 (-1.1)	0.76 (-0.8)	0.73 (-0.8)	0.96 (-0.5)	0.99 (-0.8)	1.03 (-1.4)
3m	0.75 (-0.6)	0.67 (-0.7)	0.81 (-0.7)	0.78 (-0.7)	0.96 (-0.8)	1.00 (-0.7)	1.04 (-0.3)
1y	0.90 (-1.2)	0.87 (-1.2)	0.93 (-1.0)	0.91 (-1.0)	0.99 (-1.0)	1.05 (-0.5)	1.06 (0.5)
2y	0.98 (-0.4)	0.96 (-1.1)	0.98 (-1.4)	0.99 (-1.3)	1.01 (-0.6)	1.08 (-0.1)	1.08 (0.1)
5y	1.01 (0.7)	1.02 (1.1)	1.01 (1.6)	1.04 (1.3)	1.02 (0.5)	1.14 (0.4)	1.11 (0.3)
10y	1.01 (0.9)	1.03 (1.2)	1.02 (1.9)	1.05 (2.4)	1.03 (1.4)	1.15 (-0.3)	1.11 (-0.9)

Note: Close of 1/16/20. The table reports shorter tail/longer tail ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

USD forward volatilities

We calculate forward 1y2y, 1y5y, 1y10y and 1y30y volatilities implied from the spot swaption surface. Forward volatilities reflect market expectations about future short-dated vols and may highlight dislocations on the spot vol surface.

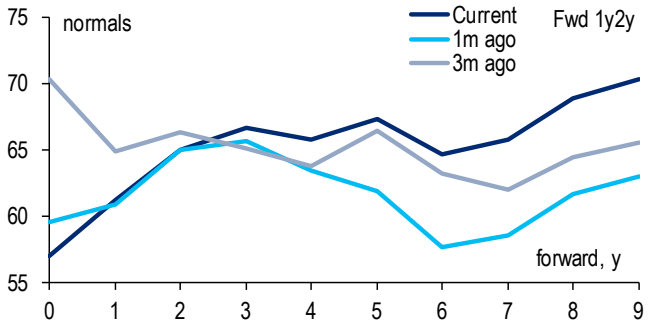
- Longer-dated forward vols on 2-5y tails have richened over the past week and month, consistent with the steepening of the vol surface (Figure 26).
- The term structure of 1y2y vols in long expiries remains very steep (Figure 27). This suggests 7y2y spot vol may be cheap to 10y2y.

Figure 26. Forward volatilities

	1y2y							1y5y							1y10y							1y30y						
Yrs		1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y				1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y	
Fwd	Vol	chg	chg	avg	z-scr	spt	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt	z-scr
spt	57.0	(1.4)	(2.5)	(9.0)	(1.5)	(10.2)	(0.5)	61.5	(2.2)	(2.5)	(4.1)	(0.9)	(19.0)	(1.1)	62.4	(2.3)	(2.1)	(1.0)	(0.2)	(22.1)	(1.4)	58.7	(2.3)	(2.1)	(0.1)	(0.0)	(21.7)	(1.4)
1	61.2	(1.8)	0.3	(6.5)	(1.4)	(6.0)	(0.3)	62.7	(0.4)	0.3	(3.6)	(1.1)	(17.9)	(1.1)	62.5	(0.2)	0.1	(1.5)	(0.6)	(21.9)	(1.4)	58.3	0.3	0.3	0.2	0.1	(22.1)	(1.4)
2	64.9	3.3	0.0	(4.0)	(0.7)	(2.3)	(0.1)	63.2	0.2	0.3	(4.6)	(1.0)	(17.4)	(1.0)	63.1	0.1	0.4	(1.8)	(0.6)	(21.4)	(1.3)	55.7	0.4	0.7	(1.4)	(0.5)	(24.6)	(1.5)
3	66.6	2.6	1.0	(3.2)	(0.4)	(0.6)	(0.0)	65.6	1.5	1.2	(1.5)	(0.3)	(15.0)	(0.9)	64.8	0.9	2.2	(0.8)	(0.2)	(19.7)	(1.2)	55.4	1.1	1.2	(1.7)	(0.6)	(25.0)	(1.6)
4	65.8	1.7	2.3	(1.4)	(0.2)	(1.4)	(0.1)	65.5	2.0	1.5	(1.2)	(0.2)	(15.1)	(0.9)	63.4	1.4	1.0	(2.4)	(0.6)	(21.1)	(1.3)	55.5	1.8	1.8	(1.7)	(0.6)	(24.8)	(1.5)
5	67.3	(0.1)	5.4	(1.2)	(0.2)	0.1	0.0	66.0	1.3	2.3	(1.8)	(0.4)	(14.5)	(0.9)	62.5	1.0	1.3	(2.6)	(0.7)	(21.9)	(1.4)	55.1	3.1	2.6	(1.1)	(0.4)	(25.2)	(1.6)
6	64.7	(0.5)	7.0	(0.8)	(0.2)	(2.5)	(0.1)	64.7	1.1	2.5	(3.7)	(0.7)	(15.8)	(1.0)	62.6	0.7	0.8	(3.2)	(1.0)	(21.9)	(1.4)	55.2	3.3	2.3	(0.9)	(0.4)	(25.2)	(1.6)
7	65.7	2.0	7.2	1.0	0.2	(1.5)	(0.1)	64.9	2.0	2.7	(2.6)	(0.6)	(15.6)	(0.9)	64.4	0.9	0.9	(3.0)	(0.9)	(20.1)	(1.3)	54.7	2.3	1.1	(2.0)	(0.9)	(25.7)	(1.6)
8	68.9	3.0	7.3	2.0	0.4	1.7	0.1	64.3	2.7	2.9	(1.9)	(0.4)	(16.2)	(1.0)	63.1	1.1	0.8	(3.2)	(0.9)	(21.4)	(1.3)	50.4	1.9	0.4	(1.7)	(0.8)	(30.0)	(1.9)
9	70.3	2.4	7.3	1.7	0.3	3.1	0.2	64.0	2.4	2.7	(1.2)	(0.4)	(16.5)	(1.0)	63.9	0.9	0.4	(3.6)	(1.2)	(20.6)	(1.3)	51.5	1.0	(1.0)	(4.7)	(1.7)	(28.8)	(1.8)

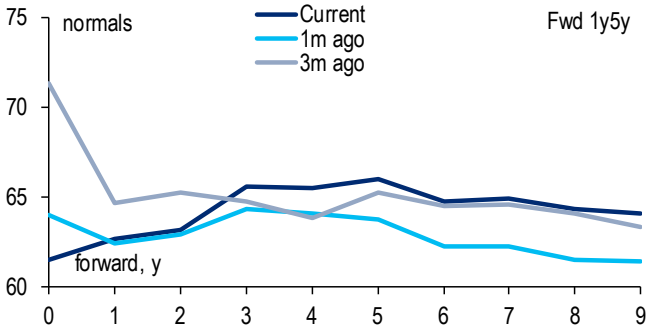
Note: Close of 1/16/20. Forward vols are computed using the standard triangular approximation from spot vols and using 6m realized correlations as a proxy for implied correlations. In addition to levels of forward vols, we calculate 1w and 1m changes, the difference from 1y average (with corresponding z-scores), and the difference from the 10y average of the corresponding 1y expiry spot vol (together with z-scores). Source: Citi Research

Figure 27. Forward vol of 1y2y



Note: Close of 1/16/20. Source: Citi Research

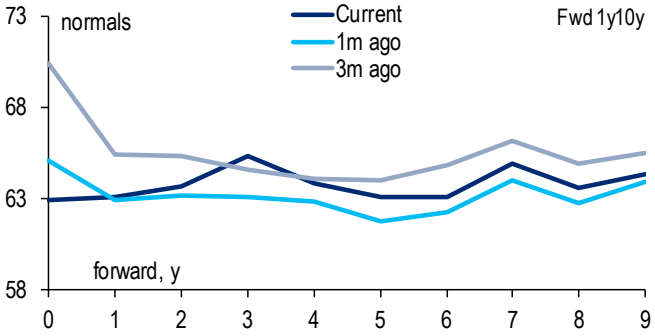
Figure 28. Forward vol of 1y5y



Note: Close of 1/16/20. Source: Citi Research

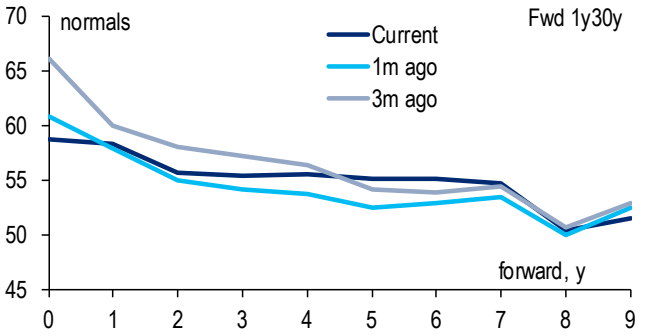
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Figure 29. Forward vol of 1y10y



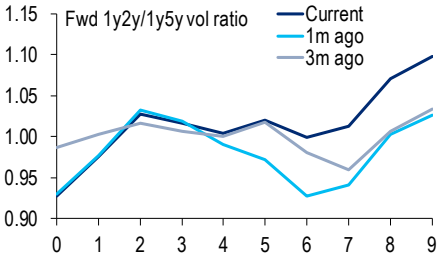
Note: Close of 1/16/20. Source: Citi Research

Figure 30. Forward vol of 1y30y



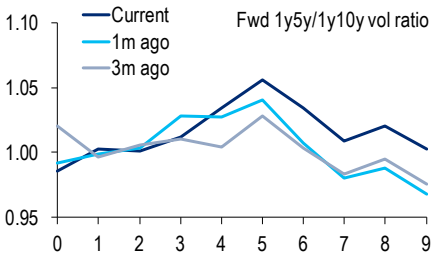
Note: Close of 1/16/20. Source: Citi Research

Figure 31. Fwd 1y2y/1y5y vol ratio



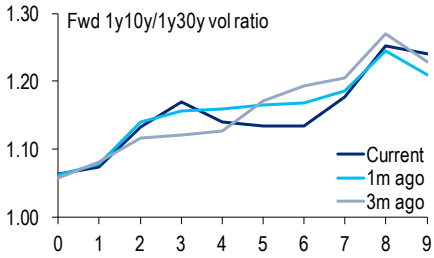
Note: Close of 1/16/20. Source: Citi Research

Figure 32. Fwd 1y5y/1y10y vol ratio



Note: Close of 1/16/20. Source: Citi Research

Figure 33. Fwd 1y10y/1y30y vol ratio



Note: Close of 1/16/20. Source: Citi Research

The efficient frontier of USD forward vol trades

We analyze all exactable forward vol trades, expressed using gamma-neutral calendar spreads and triangles, and report 20 trades on the efficient frontier based on ex-ante Sharpe (measured by the vol-adjusted roll) and value (measured by 1y-zcore).

- We still like 10y20y/5y30y straddle switches as long 5y fwd 5y20y position. See the Weekly Focus for more detail. Calendar switches in the upper left, such as 1y2y2y/18m5y, also started to look attractive due to the recent cheapening of upper-left vols (Figure 34).

Figure 34. Forward vol trades on the efficient frontier (Sharpe vs z-score)

short expry	long expry	fwd vol	gamma-ntrl wghts	short vol, nv	long vol, nv	fwd vol, nv	3m roll, nv	Shrp	1y ZS	3m ZS	short expry	long expry	fwd vol	gamma-ntrl wghts	short vol, nv	long vol, nv	fwd vol, nv	3m roll, nv	Shrp	1y ZS	3m ZS
1y3y2y	18m7y	1y6m7y	316.5/ 100	69.5	62.9	51.1	2.28	0.26	-1.8	-0.9	5y25y	15y5y	5y10y5y	12.6/ 100	57.1	57.1	57.3	0.22	0.09	-1.9	-1.6
1y2y2y	18m5y	1y6m5y	224.4/ 100	68.6	62.6	49.8	2.23	0.24	-1.6	-0.5	7y25y	10y20y	7y3y20y	70.1/ 100	56.0	54.5	52.0	0.25	0.09	-1.7	-1.1
1y3y1y	2y5y	1y1y5y	386.0/ 100	69.8	63.7	58.9	0.65	0.12	-1.5	-0.7	5y30y	15y10y	5y10y10y	21.6/ 100	56.7	55.6	55.4	0.21	0.09	-1.8	-1.9
2y8y20y	15y10y	2y13y10y	22.3/ 100	58.5	55.6	55.1	0.25	0.12	-1.6	-1.4	7y25y	15y10y	7y8y10y	29.6/ 100	56.0	55.6	55.6	0.20	0.09	-1.9	-2.5
10y30y	15y20y	10y5y20y	59.8/ 100	53.7	51.1	46.6	0.28	0.11	-2.2	-3.3	5y10y/ 5y30y	15y20y	5y10y20y	68.4/ 100	52.4	51.1	50.4	0.19	0.09	-2.3	-2.0
7y30y	10y25y	7y3y25y	73.9/ 100	55.7	53.8	50.3	0.29	0.11	-1.8	-1.1	5y15y/ 5y20y	20y5y	5y15y5y	58.0/ 100	50.8	53.5	54.4	0.18	0.08	-1.7	-1.5
7y30y	15y15y	7y8y15y	38.7/ 100	55.7	52.8	51.3	0.24	0.11	-2.2	-2.6	7y20y	15y5y	7y8y5y	18.0/ 100	56.6	57.1	57.8	0.18	0.08	-2.0	-2.0
3y30y	15y5y	3y12y5y	8.4/ 100	57.9	57.1	57.0	0.24	0.10	-1.8	-1.4	10y25y	15y15y	10y5y15y	52.4/ 100	53.8	52.8	52.1	0.19	0.08	-1.7	-3.2
10y5y/ 10y10y	15y5y	10y5y5y	85.8/ 100	57.3	57.1	56.7	0.31	0.10	-1.6	-1.5	10y5y/ 10y30y	15y25y	10y5y25y	89.8/ 100	51.6	50.5	48.1	0.17	0.07	-3.1	-3.1
7y30y	20y5y	7y13y5y	11.7/ 100	55.7	53.5	53.0	0.21	0.10	-1.5	-1.8	10y5y/ 10y25y	15y20y	10y5y20y	88.0/ 100	51.4	51.1	50.5	0.04	0.01	-2.6	-3.2

Note: Close of 1/16/20. The table reports 20 best forward vol trades on the efficient frontier based on ex-ante Sharpe ratio and 1y z-score. We analyze both calendar spreads (underlying forwards with matching mid-points) and triangles. The ex-ante Sharpe is calculated as vol-adjusted roll on the surface. Source: Citi Research

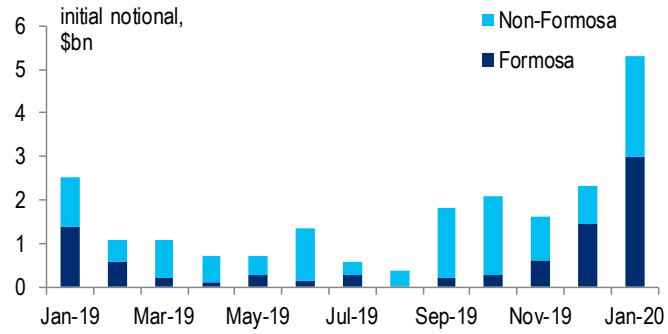
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Long-dated callable USD supply

We report supply and redemptions data for long-dated callable bonds.

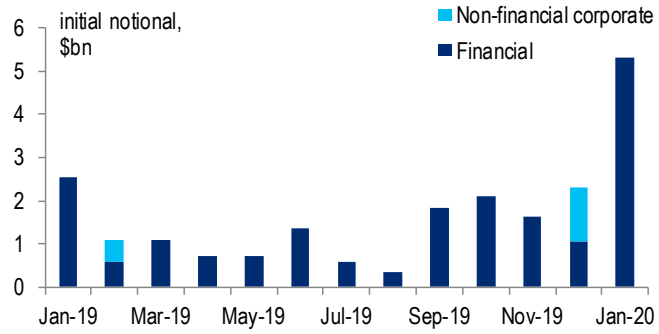
- Almost $5.3bn of long-dated callables has been issued this month so far, which already has exceeded our expectations of $5bn. The shift in the issuance to 40NC5 has caused a significant surge of vega supply (Figure 38). See the Weekly Focus for more detail.

Figure 35. Gross callable supply: Formosa vs Non-Formosa



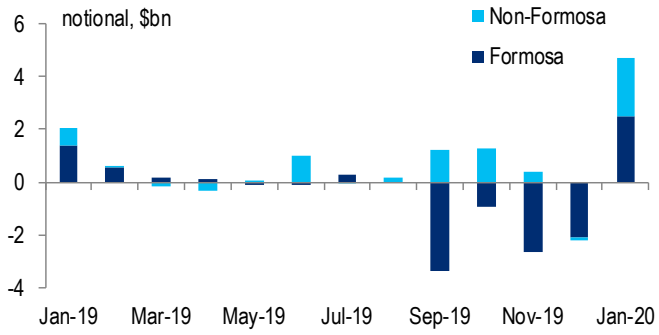
Note: We include all callable Formosa (zero-coupon and fixed-rate) and zero-coupon non-Formosa bonds. Source: Citi Research

Figure 36. Gross callable supply: Financial vs Non-Financial



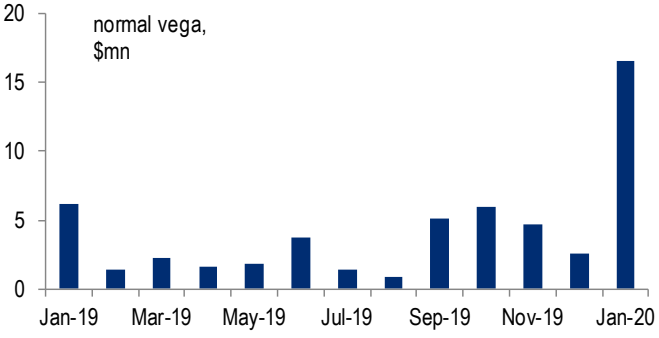
Note: We include all callable Formosa (zero-coupon and fixed-rate) and zero-coupon non-Formosa bonds. Source: Citi Research

Figure 37. Net callable notional supply



Source: Citi Research

Figure 38. Vega supply (normal parallel vega)



Note: We assume (simplistically) that all financial and government bonds are swapped, while all non-financial bonds are not swapped. Source: Citi Research

Figure 39. Recent callable supply estimates

	gross issued, initial notional, $bn					redeemed, $bn					net issued, $bn					Vega, $mn
	Formosa	Non-Formosa	Financial	Non-Financial	All	Formosa	Non-Formosa	Financial	Non-Financial	All	Formosa	Non-Formosa	Financial	Non-Financial	All	
Jan-20 MTD	2.99	2.32	5.31	0.00	5.31	0.49	0.13	0.62	0.00	0.62	2.51	2.19	4.69	0.00	4.69	16.6
Dec-19 (prev month)	1.46	0.86	1.05	1.27	2.31	3.57	0.93	3.58	0.92	4.49	-2.11	-0.07	-2.53	0.35	-2.18	2.6
Jan-19 (prev year)	1.40	1.13	2.53	0.00	2.53	0.00	0.49	0.49	0.00	0.49	1.40	0.64	2.04	0.00	2.04	6.1
2020 YTD	2.99	2.32	5.31	0.00	5.31	0.49	0.13	0.62	0.00	0.62	2.51	2.19	4.69	0.00	4.69	16.6
2019 YTD	1.40	1.13	2.53	0.00	2.53	0.00	0.49	0.49	0.00	0.49	1.40	0.64	2.04	0.00	2.04	6.1
2019 (all year)	5.56	10.78	14.57	1.78	16.34	12.22	6.55	14.50	4.27	18.77	-6.66	4.23	0.07	-2.49	-2.43	37.8

Note: To compute vega we assume (simplistically) that all financial and government bonds are swapped, while all non-financial bonds are not swapped. Source: Citi Research

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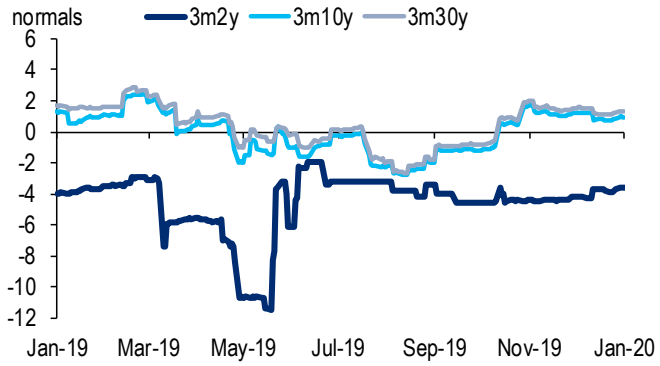
USD swaption skew

USD skew history and z-scores

We calculate z-scores of payer and receiver swaption skews across the surface.

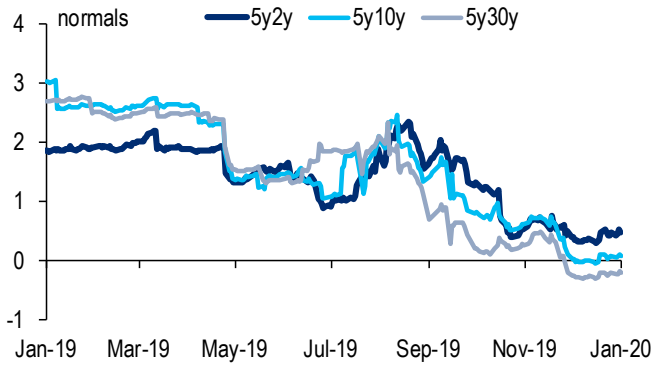
- Short-dated risk reversals have been little changed and remain slightly inverted (Figure 44).
- Longer-dated payer skews remain very cheap historically (Figure 41 and Figure 43).

Figure 40. 3m expiry payer skew



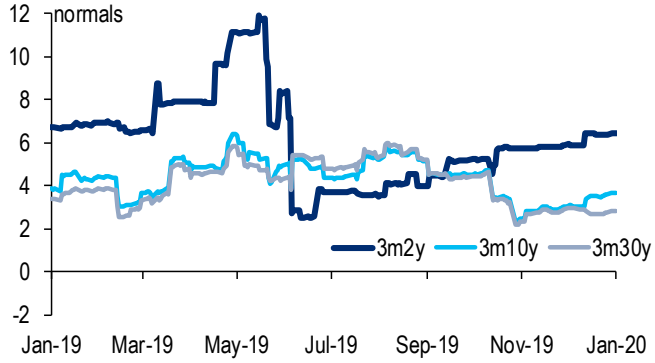
Note: 25bp-out payer skews for 3m expiry payers. Source: Citi Research

Figure 41. 5y expiry payer skew



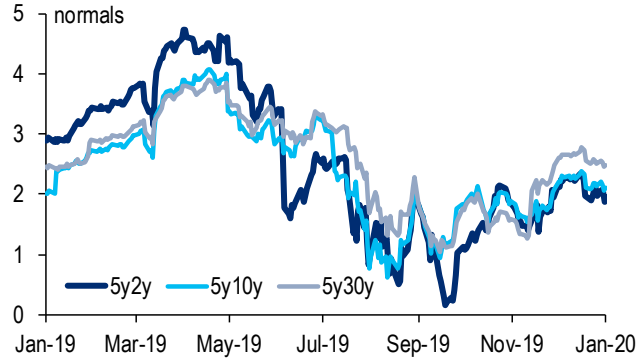
Note: 100bp-out payer skews for 5y expiry payers. Source: Citi Research

Figure 42. 3m expiry receiver skew



Note: 25bp-out receiver skews for 3m expiry receivers. Source: Citi Research

Figure 43. 5y expiry receiver skew



Note: 100bp-out receiver skews for 5y expiry receivers. Source: Citi Research

Figure 44. Skews (1y z-scores) of risk reversals, OTM payer and OTM receivers

	Risk Reversals						Payer skew						Receiver skew					
expr\tail	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y
3m	-10.9 (0.1)	-10.1 (0.2)	-5.5 (0.8)	-4.2 (0.9)	-2.7 (0.8)	-1.6 (1.0)	-3.8 (0.1)	-3.6 (0.5)	-0.8 (1.2)	0.0 (1.1)	1.0 (0.8)	1.3 (0.7)	7.2 (0.0)	6.5 (0.2)	4.7 (-0.2)	4.2 (-0.5)	3.7 (-0.8)	2.8 (-1.3)
6m	-17.0 (0.0)	-15.1 (0.1)	-8.1 (0.6)	-6.4 (0.8)	-4.6 (0.7)	-3.2 (0.8)	-5.6 (-0.1)	-4.9 (0.4)	-0.2 (1.3)	1.0 (1.1)	2.2 (0.7)	2.9 (0.4)	11.3 (0.0)	10.2 (0.2)	7.9 (0.2)	7.4 (0.0)	6.8 (-0.5)	6.1 (-1.2)
1y	-12.3 (0.3)	-10.2 (0.4)	-6.6 (0.4)	-5.6 (0.3)	-4.6 (0.0)	-2.8 (0.8)	-4.6 (0.4)	-3.5 (0.7)	-1.2 (0.8)	-0.7 (0.5)	-0.1 (0.1)	0.2 (-0.4)	7.7 (-0.1)	6.7 (0.0)	5.4 (0.0)	4.9 (0.1)	4.5 (0.2)	3.0 (-2.0)
2y	-4.6 (0.3)	-3.9 (0.3)	-3.4 (-0.4)	-3.6 (-1.0)	-3.0 (-1.1)	-2.5 (-0.6)	-1.3 (0.6)	-0.9 (0.6)	-0.9 (-0.4)	-1.0 (-1.3)	-0.7 (-1.3)	-0.4 (-1.2)	3.3 (0.0)	2.9 (0.0)	2.5 (0.4)	2.5 (0.5)	2.4 (0.5)	2.1 (-0.3)
3y	-3.4 (0.5)	-3.3 (0.2)	-2.8 (0.1)	-3.1 (-0.7)	-3.1 (-1.2)	-3.5 (-1.2)	0.3 (0.8)	0.4 (0.2)	0.5 (-0.8)	0.5 (-1.2)	0.5 (-1.6)	0.8 (-1.6)	3.8 (-0.4)	3.7 (-0.2)	3.3 (-0.5)	3.5 (-0.1)	3.7 (0.0)	4.2 (0.1)
5y	-0.7 (0.5)	-1.5 (-0.3)	-1.1 (-0.2)	-1.6 (-1.1)	-2.0 (-1.2)	-2.7 (-1.8)	1.0 (-1.0)	0.5 (-1.7)	0.5 (-1.7)	0.3 (-1.8)	0.1 (-1.7)	-0.2 (-1.7)	1.7 (-0.7)	1.9 (-0.5)	1.6 (-0.9)	1.9 (-0.1)	2.1 (-0.4)	2.5 (-0.1)
10y	1.2 (0.1)	0.9 (0.2)	0.5 (0.0)	-0.3 (-0.8)	-1.4 (-1.3)	-1.1 (-1.4)	1.6 (-1.7)	1.4 (-1.6)	1.1 (-1.7)	0.8 (-1.7)	0.3 (-1.6)	0.6 (-1.4)	0.4 (-1.8)	0.5 (-1.2)	0.6 (-1.0)	1.1 (-0.6)	1.7 (-0.1)	1.6 (-0.6)

Note: Close of 1/16/20. The table reports skews and 1y z-scores in parentheses for risk reversals, payer and receiver skews, defined in normal vol terms. The following OTM conventions are used: 25bp out for 3m expiries, 50bp out for 6m-2y expiries, 100bp out for 3y+ expiries. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

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USD implied vs realized skew

We compare implied swaption skews with realized skews. We calculate realized skews as the model value of OTM options from the empirical SABR model calibrated to the recent (3m) realized dynamics of ATM volatilities and rates.

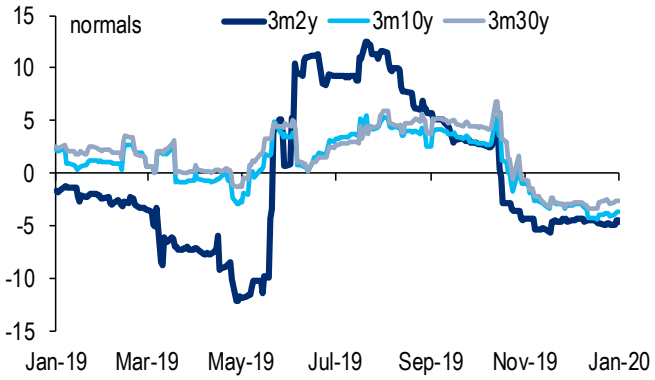
- Risk reversals in shorter expiries continue to trade cheap to realized skew. For example, the 6m10y risk reversals trade about two sigmas cheap to realized skew (Figure 45).

Figure 45. Implied vs realized skews (spread to the model for risk reversals, price/model ratio for payers and receivers)

	Risk Reversals (1Y z-score)					OTM payers (1Y z-score)					OTM receivers (1Y z-score)				
expr\tail	1y	2y	5y	10y	30y	1y	2y	5y	10y	30y	1y	2y	5y	10y	30y
3m	-4.3 (-0.8)	-4.6 (-0.6)	-5.5 (-1.3)	-3.7 (-1.8)	-2.6 (-1.7)	0.96 (-0.9)	0.97 (-0.6)	0.98 (-1.2)	1.00 (-1.8)	1.01 (-1.9)	1.08 (0.7)	1.07 (0.9)	1.08 (1.5)	1.07 (1.7)	1.05 (1.3)
6m	-8.2 (-0.9)	-7.5 (-0.8)	-8.2 (-1.5)	-6.4 (-2.0)	-4.7 (-1.8)	0.93 (-1.2)	0.96 (-0.8)	0.99 (-1.3)	1.02 (-2.0)	1.03 (-2.1)	1.12 (0.8)	1.11 (0.9)	1.13 (1.8)	1.12 (1.9)	1.11 (1.6)
1y	-5.3 (-0.9)	-4.8 (-0.8)	-5.8 (-1.5)	-4.1 (-1.9)	-2.3 (-1.6)	0.97 (-1.1)	0.98 (-0.7)	0.98 (-1.4)	1.00 (-2.0)	1.00 (-2.1)	1.07 (0.9)	1.06 (1.1)	1.08 (1.6)	1.06 (1.8)	1.04 (1.0)
2y	-0.5 (-0.7)	0.0 (-0.7)	-1.1 (-1.1)	-1.9 (-1.6)	-1.2 (-1.4)	1.01 (-0.6)	1.02 (-0.5)	1.00 (-1.1)	1.00 (-1.8)	1.00 (-1.6)	1.02 (0.8)	1.02 (0.8)	1.02 (1.2)	1.03 (1.4)	1.02 (1.2)
3y	1.7 (-0.3)	2.2 (-0.5)	0.9 (-0.9)	-0.1 (-1.2)	-0.7 (-1.2)	1.06 (-0.4)	1.07 (-0.5)	1.05 (-0.9)	1.04 (-1.3)	1.04 (-1.4)	1.03 (0.3)	1.03 (0.5)	1.03 (0.6)	1.04 (1.0)	1.05 (1.1)
5y	3.8 (-0.1)	3.4 (-0.4)	2.8 (-0.7)	2.3 (-1.0)	2.1 (-0.9)	1.08 (-0.2)	1.08 (-0.4)	1.06 (-0.8)	1.05 (-1.1)	1.05 (-1.1)	1.02 (0.0)	1.02 (0.2)	1.01 (0.4)	1.01 (0.7)	1.01 (0.6)
10y	5.6 (-0.3)	5.4 (-0.3)	5.0 (-0.4)	3.5 (-0.6)	4.6 (-0.5)	1.08 (-0.3)	1.08 (-0.2)	1.07 (-0.4)	1.06 (-0.7)	1.08 (-0.7)	0.99 (0.2)	0.99 (0.2)	0.99 (0.1)	1.00 (0.5)	0.99 (0.2)

Note: Close of 1/16/20. We calibrate an empirical SABR model to the recent observed dynamics of ATM vols and rates. We use a 2y rolling window to estimate a backbone parameter, and a 3m window to estimate correlation and vol-of-vol parameters. The empirical model matches ATM implied vols exactly, so it's different from the market only in skew parameters. We calculate "realized skew" as the theoretical values of OTM payers, receivers and risk reversals from this empirical model. The table shows market-"realized" spreads of risk reversals, and market/"realized" ratios of OTM payers and receivers together with 1y z-scores of these spreads and ratios in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 46. 3m expiry risk reversals/model spread



Note: The table shows model-"realized" spreads for 25bp-out risk reversals in 3m expiries using the same methodology as in Figure 45. Source: Citi Research.

Figure 47. 5y expiry risk reversals/model spread



Note: The table shows model-"realized" spreads for 100bp-out risk reversals in 5y expiries using the same methodology as in Figure 45. Source: Citi Research.

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USD Board/swaption relative value

We evaluate the relative pricing of Board and swaption volatilities.

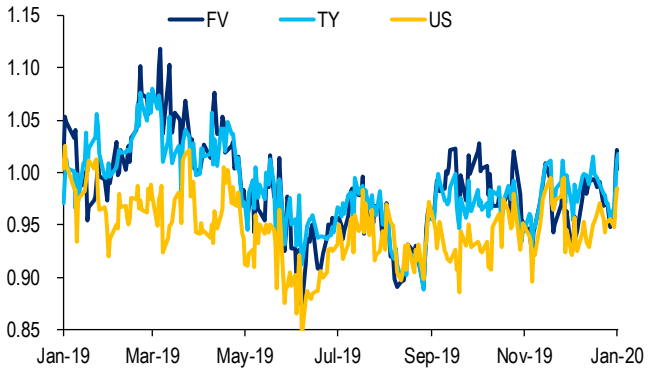
- Board vol trade largely on par with swaptions, which is consistent with recent realized vols. FVH0 may be slightly rich to swaptions (Figure 48).
- Put skew looks rich to payer skew (Figure 50, Figure 51).
- *Note: Futures trading involves substantial risk of loss.*

Figure 48. Board vols vs swaptions

option	FVH0	FVJ0	TYH0	TYJ0	USH0	USJ0
Board vol	54.3	56.6	57.4	59.3	57.2	59.9
1w chg	-6.2	-3.5	-5.1	-3.8	-4.8	-2.1
Board/ 1m RV	1.18	1.13	1.14	1.15	1.05	1.06
Board/ 3m RV	0.74	0.77	0.75	0.77	0.75	0.78
CTD-matched swpn vol	54.1	59.0	56.4	60.9	58.1	60.3
1w chg	-8.4	-7.0	-7.9	-5.7	-7.0	-4.6
swpn/ 1m RV	1.22	1.25	1.13	1.17	1.08	1.11
swpn/ 3m RV	0.74	0.77	0.74	0.77	0.75	0.77
Board/swpn	1.00	0.96	1.02	0.97	0.99	0.99
1w ago	0.97	0.91	0.97	0.95	0.95	0.96
ftrs 1m RV/swap 1m RV	1.04	1.06	1.00	0.99	1.02	1.03
ftrs 3m RV/swap 3m RV	1.00	0.96	1.00	0.98	1.00	0.99

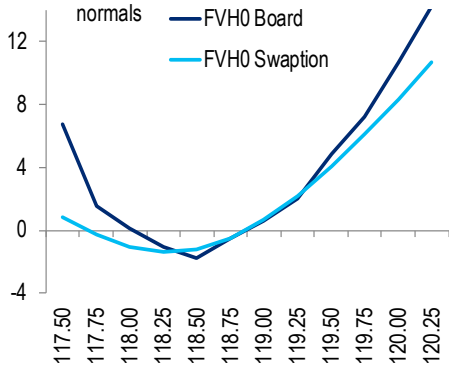
Note: Close of 1/16/20. The table shows implied ATM (interpolated) Board vols vs CTD-matched swaption vols with corresponding realized volatilities. Source: Citi Research

Figure 49. Board/swaption vol ratio



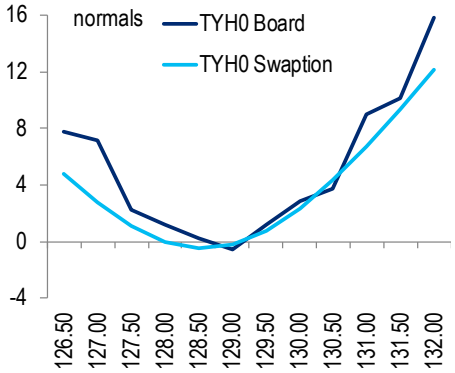
Note: The chart shows the ratio of ATM Board vol to CTD-matched swaption vols. The second available expiry (rolled on the expiration of front options) is used for the chart. Source: Citi Research

Figure 50. FV Board vs swaption skew



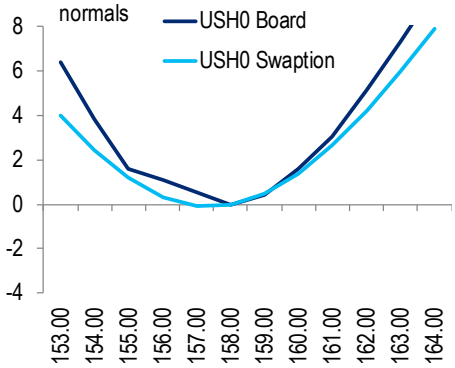
Note: Close of 1/16/20. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

Figure 51. TY Board vs swaption skew



Note: Close of 1/16/20. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

Figure 52. US Board vs swaption skew



Note: Close of 1/16/20. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

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Figure 53. Conditional spread trades

							puts/payers								calls/receivers							
optn	expiry	futures price	fut DV01	swap DV01	swpn ntl, $mn	invc sprd, bp	put strike	strike yield, %	payer strike, %	strike spread, bp	bp pick-up	put vol	payer vol	vol ratio	call strike	strike yield, %	costless receiver strike, %	strike spread, bp	bp pick-up	call vol	rcvr vol	vol ratio
FVH0	02/21/20	118-27.75	4.73	4.01	117.9	-0.6	118.75	1.64	1.63	-0.8	0.2	53.7	53.6	1.00	119.00	1.58	1.58	-0.4	-0.2	54.9	54.7	1.00
FVH0	02/21/20	118-27.75	4.73	4.01	117.9	-0.6	118.50	1.69	1.68	-0.7	0.0	52.5	52.9	0.99	119.25	1.53	1.53	-0.4	-0.3	56.3	56.2	1.00
FVH0	02/21/20	118-27.75	4.73	4.01	117.9	-0.6	118.25	1.74	1.73	-1.1	0.4	53.2	52.7	1.01	119.50	1.48	1.48	0.2	-0.8	59.1	57.8	1.02
FVH0	02/21/20	118-27.75	4.73	4.01	117.9	-0.6	118.00	1.79	1.78	-1.6	1.0	54.4	52.9	1.03	119.75	1.42	1.43	0.5	-1.1	61.5	59.7	1.03
FVH0	02/21/20	118-27.75	4.73	4.01	117.9	-0.6	117.75	1.85	1.83	-2.2	1.6	55.8	53.5	1.04	120.00	1.37	1.39	1.5	-2.1	64.9	61.5	1.06
FVH0	02/21/20	118-27.75	4.73	4.01	117.9	-0.6	117.50	1.90	1.85	-5.2	4.6	61.0	53.9	1.13	120.25	1.32	1.35	2.7	-3.3	68.5	63.3	1.08
FVJ0	03/27/20	119-5.25	4.80	4.00	120.0	-1.4	119.00	1.64	1.64	-0.7	-0.6	56.4	58.3	0.97	119.25	1.59	1.57	-2.3	0.9	56.8	59.5	0.95
FVJ0	03/27/20	119-5.25	4.80	4.00	120.0	-1.4	118.75	1.70	1.69	-0.9	-0.5	56.4	57.7	0.98	119.50	1.54	1.52	-2.1	0.8	58.4	60.6	0.96
FVJ0	03/27/20	119-5.25	4.80	4.00	120.0	-1.4	118.50	1.75	1.74	-0.7	-0.7	55.9	57.4	0.97	119.75	1.49	1.46	-2.8	1.5	58.7	62.1	0.95
FVJ0	03/27/20	119-5.25	4.80	4.00	120.0	-1.4	118.25	1.80	1.79	-0.7	-0.6	56.0	57.3	0.98	120.00	1.44	1.41	-2.4	1.1	61.1	63.4	0.96
FVJ0	03/27/20	119-5.25	4.80	4.00	120.0	-1.4	118.00	1.85	1.85	-0.6	-0.7	56.2	57.5	0.98	120.25	1.38	1.36	-2.5	1.1	62.7	65.0	0.97
FVJ0	03/27/20	119-5.25	4.80	4.00	120.0	-1.4	117.75	1.91	1.88	-2.3	1.0	58.9	57.8	1.02	120.50	1.33	1.30	-3.0	1.6	63.9	66.8	0.96
TYH0	02/21/20	129-5	7.99	6.28	127.2	-6.3	129.00	1.75	1.68	-6.6	0.3	56.9	56.2	1.01	129.50	1.69	1.63	-5.6	-0.7	58.6	57.0	1.03
TYH0	02/21/20	129-5	7.99	6.28	127.2	-6.3	128.50	1.81	1.74	-7.0	0.7	57.6	55.9	1.03	130.00	1.62	1.57	-5.4	-0.9	60.2	58.5	1.03
TYH0	02/21/20	129-5	7.99	6.28	127.2	-6.3	128.00	1.87	1.80	-7.4	1.2	58.6	56.3	1.04	130.50	1.56	1.51	-5.7	-0.6	61.2	60.6	1.01
TYH0	02/21/20	129-5	7.99	6.28	127.2	-6.3	127.50	1.94	1.86	-7.7	1.4	59.6	57.3	1.04	131.00	1.50	1.46	-3.8	-2.5	66.4	62.2	1.07
TYH0	02/21/20	129-5	7.99	6.28	127.2	-6.3	127.00	2.00	1.90	-10.1	3.9	64.6	58.2	1.11	131.50	1.44	1.40	-4.3	-1.9	67.6	65.0	1.04
TYH0	02/21/20	129-5	7.99	6.28	127.2	-6.3	126.50	2.07	1.97	-9.9	3.6	65.2	60.1	1.08	132.00	1.38	1.36	-1.7	-4.6	73.2	66.5	1.10
TYJ0	03/27/20	128-31	7.93	6.26	126.7	-5.7	128.50	1.79	1.74	-5.2	-0.5	59.1	60.4	0.98	129.00	1.72	1.66	-6.2	0.5	59.3	61.0	0.97
TYJ0	03/27/20	128-31	7.93	6.26	126.7	-5.7	128.00	1.85	1.80	-5.0	-0.7	58.7	60.3	0.97	129.50	1.66	1.60	-6.0	0.3	60.7	61.9	0.98
TYJ0	03/27/20	128-31	7.93	6.26	126.7	-5.7	127.50	1.91	1.86	-5.5	-0.2	60.1	60.6	0.99	130.00	1.60	1.54	-5.8	0.1	62.4	63.1	0.99
TYJ0	03/27/20	128-31	7.93	6.26	126.7	-5.7	127.00	1.98	1.92	-5.8	0.1	61.2	61.3	1.00	130.50	1.54	1.48	-6.0	0.4	63.6	64.7	0.98
TYJ0	03/27/20	128-31	7.93	6.26	126.7	-5.7	126.50	2.04	1.98	-6.4	0.7	63.0	62.2	1.01	131.00	1.47	1.41	-6.5	0.8	64.9	66.6	0.97
TYJ0	03/27/20	128-31	7.93	6.26	126.7	-5.7	126.00	2.11	2.04	-6.5	0.8	64.3	63.5	1.01	131.50	1.41	1.35	-6.4	0.7	67.0	68.5	0.98
USH0	02/21/20	158-3	19.08	13.95	136.8	-9.5	158.00	1.99	1.89	-9.4	-0.1	57.2	58.0	0.99	159.00	1.94	1.84	-9.6	0.1	57.7	58.6	0.98
USH0	02/21/20	158-3	19.08	13.95	136.8	-9.5	157.00	2.04	1.95	-9.6	0.1	57.7	58.0	1.00	160.00	1.88	1.79	-9.5	-0.1	58.8	59.5	0.99
USH0	02/21/20	158-3	19.08	13.95	136.8	-9.5	156.00	2.09	2.00	-9.7	0.1	58.3	58.4	1.00	161.00	1.83	1.74	-9.4	-0.1	60.3	60.7	0.99
USH0	02/21/20	158-3	19.08	13.95	136.8	-9.5	155.00	2.15	2.05	-9.5	0.0	58.9	59.3	0.99	162.00	1.78	1.69	-9.0	-0.5	62.4	62.2	1.00
USH0	02/21/20	158-3	19.08	13.95	136.8	-9.5	154.00	2.20	2.10	-10.1	0.6	61.0	60.4	1.01	163.00	1.73	1.64	-8.6	-0.9	64.6	63.8	1.01
USH0	02/21/20	158-3	19.08	13.95	136.8	-9.5	153.00	2.26	2.15	-11.0	1.4	63.6	61.6	1.03	164.00	1.68	1.60	-8.3	-1.2	66.9	65.5	1.02
USJ0	03/27/20	157-5	18.73	13.70	136.7	-9.8	157.00	2.00	1.90	-9.7	-0.1	59.7	60.3	0.99	158.00	1.94	1.85	-9.6	-0.2	60.6	60.6	1.00
USJ0	03/27/20	157-5	18.73	13.70	136.7	-9.8	156.00	2.05	1.96	-9.5	-0.3	59.3	60.2	0.98	159.00	1.89	1.79	-9.7	-0.1	61.1	61.3	1.00
USJ0	03/27/20	157-5	18.73	13.70	136.7	-9.8	155.00	2.10	2.01	-9.5	-0.3	59.7	60.4	0.99	160.00	1.84	1.74	-9.7	-0.1	61.8	62.2	0.99
USJ0	03/27/20	157-5	18.73	13.70	136.7	-9.8	154.00	2.16	2.06	-9.6	-0.2	60.4	60.9	0.99	161.00	1.79	1.69	-9.5	-0.3	63.1	63.2	1.00
USJ0	03/27/20	157-5	18.73	13.70	136.7	-9.8	153.00	2.21	2.12	-9.5	-0.3	61.1	61.7	0.99	162.00	1.74	1.64	-9.4	-0.4	64.5	64.5	1.00
USJ0	03/27/20	157-5	18.73	13.70	136.7	-9.8	152.00	2.27	2.17	-9.6	-0.2	62.3	62.7	0.99	163.00	1.68	1.59	-9.1	-0.7	66.2	65.8	1.01

Note: Close of 1/16/20. Swaption notional is for 1000 Board contracts. The costless swaption strike makes the DV01-neutral trade costless. The BP pick-up is the spread between the costless strike spread and invoice spread. Futures trading involves substantial risk of loss. Source: Citi Research

USD midcurve swaptions

We report levels and z-scores of at-the-money vols, implied correlations, and implied/realized ratios of correlations. We also compute the relative richness/cheapness of midcurve vols to principal components of the plain vanilla surface.

- Midcurves on 5y5y look cheap to plain vanilla swaptions if evaluated by implied correlations and direct regressions. For example, the 1y5y5y vol looks cheap to swaptions by about 3 sigmas (Figure 57).

Figure 54. Implied vols, normals (3m z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	53.5 (-1.7)	62.5 (-1.8)	64.6 (-1.9)	56.2 (-1.8)	62.7 (-1.8)	64.6 (-1.8)	60.1 (-2.1)	56.4 (-1.8)
3m	58.1 (-1.4)	65.2 (-1.6)	67.7 (-1.6)	60.1 (-1.4)	65.7 (-1.6)	66.8 (-1.6)	65.4 (-1.9)	60.1 (-1.7)
6m	59.3 (-1.5)	66.0 (-1.7)	67.3 (-1.8)	61.1 (-1.6)	66.1 (-1.7)	66.6 (-1.9)	66.5 (-1.6)	60.9 (-1.7)
1y	64.9 (-1.5)	68.2 (-1.5)	69.8 (-1.5)	65.5 (-1.5)	68.6 (-1.5)	69.5 (-1.5)	65.1 (-2.2)	60.1 (-2.1)
2y	63.8 (-1.5)	63.9 (-1.1)	63.9 (-1.2)	63.7 (-1.3)	63.4 (-1.1)	63.9 (-1.3)	66.3 (-1.6)	59.6 (-1.3)

Close of 1/16/20. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 55. Implied correlations, % (3y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	104.8 (1.8)	97.8 (0.3)	100.0 (1.0)	103.3 (1.7)	98.0 (0.8)	99.6 (0.7)	98.3 (1.6)	98.3 (0.5)
3m	101.8 (1.3)	99.2 (0.0)	100.0 (0.7)	104.8 (1.4)	99.6 (0.6)	100.2 (0.9)	98.0 (1.4)	98.6 (0.9)
6m	101.6 (1.6)	98.1 (-1.6)	100.2 (1.2)	98.4 (0.6)	98.3 (-0.1)	99.7 (0.4)	98.4 (2.0)	98.4 (0.7)
1y	97.9 (-0.2)	98.5 (-0.8)	99.8 (0.3)	93.3 (-0.3)	97.5 (-0.3)	98.1 (-0.9)	98.8 (2.8)	98.7 (1.2)
2y	100.1 (-0.5)	100.3 (1.7)	100.1 (0.6)	101.9 (0.6)	101.3 (1.1)	100.1 (-0.1)	97.5 (1.6)	97.4 (-0.4)

Close of 1/16/20. The table shows implied correlations (3y z-scores in parentheses). Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

Figure 56. Implied/realized correlation ratios (1y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	1.08 (1.4)	0.98 (0.5)	1.00 (1.3)	1.09 (1.7)	1.00 (1.2)	1.01 (0.6)	0.99 (0.1)	0.99 (-0.8)
3m	1.04 (0.9)	1.00 (0.4)	1.00 (0.7)	1.09 (1.4)	1.01 (0.8)	1.01 (0.5)	0.99 (-0.4)	0.99 (-1.1)
6m	1.03 (1.5)	0.98 (-1.1)	1.00 (1.3)	1.01 (0.7)	0.99 (0.0)	1.00 (-0.1)	0.99 (0.3)	0.99 (-0.5)
1y	0.98 (0.3)	0.99 (-0.9)	1.00 (0.0)	0.94 (-0.2)	0.98 (-0.4)	0.99 (-1.3)	0.99 (1.1)	0.99 (0.3)
2y	1.00 (-0.9)	1.01 (0.6)	1.00 (-0.4)	1.03 (0.0)	1.02 (0.2)	1.00 (-0.9)	0.98 (0.7)	0.98 (-1.2)

Close of 1/16/20. The table shows the ratios of implied to 3m realized correlations (in daily changes). The 1y z-scores of ratios are shown in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 57. Rich/cheap to plain vanilla swaptions (z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	-6.3 (-1.6)	-1.7 (-0.5)	-3.5 (-1.0)	-5.2 (-1.6)	-3.1 (-1.0)	-2.7 (-0.7)	-2.6 (-1.8)	-0.7 (-0.4)
3m	-3.7 (-1.6)	-0.9 (-0.3)	-2.4 (-0.8)	-3.4 (-1.5)	-1.8 (-0.9)	-3.1 (-1.0)	-2.2 (-1.9)	-1.2 (-0.8)
6m	-3.1 (-1.7)	2.6 (1.2)	-3.1 (-1.3)	-1.2 (-0.8)	-0.2 (-0.1)	-1.3 (-0.5)	-2.5 (-2.5)	-0.6 (-0.5)
1y	-0.4 (-0.2)	1.0 (0.5)	-0.3 (-0.1)	-0.1 (0.0)	0.5 (0.3)	2.4 (0.9)	-2.3 (-2.8)	-0.5 (-0.5)
2y	-0.2 (-0.1)	-2.0 (-1.4)	0.4 (0.2)	-0.9 (-0.8)	-2.1 (-0.9)	1.4 (0.7)	-0.6 (-1.2)	1.2 (1.1)

Close of 1/16/20. We regress midcurve vols on two corresponding plain vanilla vols using last 3y of data. For example, 1y2y1y is regressed against 1y3y and 1y2y vols. We calculate rich/cheap and z-scores in brackets. Source: Citi Research

USD curve (spread) CMS options

We report levels and z-scores of at-the-money vols, rich/cheap on PCA, implied correlations, and implied/realized vol ratios for curve options.

- The 2s10s vol has richened with implied correlations off the recent highs (Figure 60). The 2s10s vol now looks rich on the PCA of the spread surface (Figure 59). The 10s30s vol looks especially rich to the recent realized vol (Figure 61).

Figure 58. Implied vols, normals (3m z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	20.0 (-0.3)	28.9 (-0.6)	32.5 (-1.3)	12.4 (-1.8)	21.8 (-1.7)	13.5 (-2.1)
3m	20.6 (0.4)	30.7 (0.3)	34.7 (-1.0)	13.4 (-1.6)	23.6 (-1.6)	14.7 (-2.0)
6m	18.2 (-0.2)	30.9 (0.4)	34.9 (-1.4)	13.7 (-1.6)	23.8 (-1.7)	15.0 (-2.1)
1y	18.2 (0.3)	30.8 (-0.6)	35.5 (-1.8)	13.8 (-1.6)	24.5 (-1.8)	14.6 (-2.1)
2y	16.7 (-1.6)	31.3 (-0.2)	36.9 (-1.7)	14.2 (-1.5)	25.3 (-1.7)	15.0 (-2.1)
3y	17.1 (-1.0)	31.5 (0.2)	38.0 (-0.9)	14.3 (-1.4)	25.5 (-1.8)	15.2 (-1.9)
5y	17.2 (-0.3)	29.6 (-0.7)	38.1 (-0.3)	14.4 (-0.7)	25.9 (-1.6)	15.5 (-1.9)

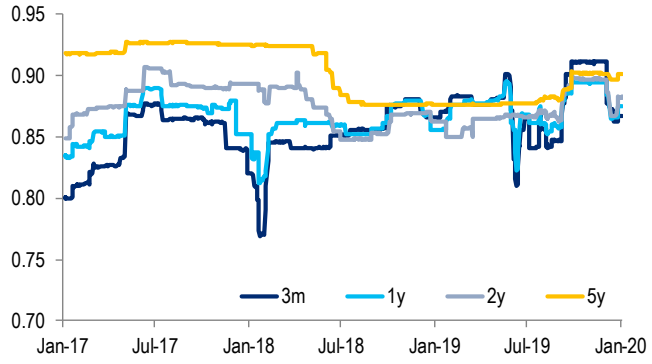
Close of 1/16/20. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 59. Rich/cheap to PCA, normals (z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	2.3 (1.4)	1.9 (0.9)	0.4 (0.1)	-2.0 (-1.1)	-2.3 (-1.2)	-0.3 (-0.2)
3m	2.9 (2.0)	3.8 (2.4)	2.0 (1.3)	-0.9 (-0.6)	-0.7 (-0.5)	0.8 (0.8)
6m	0.4 (0.3)	3.2 (2.2)	1.3 (1.0)	-0.5 (-0.4)	-0.7 (-0.7)	0.8 (0.7)
1y	-0.2 (-0.2)	1.6 (1.2)	0.3 (0.2)	-1.1 (-1.0)	-1.3 (-1.3)	0.1 (0.1)
2y	-1.7 (-2.0)	1.5 (1.3)	1.3 (0.9)	-0.7 (-1.1)	-1.1 (-1.0)	-0.9 (-1.3)
3y	-1.1 (-0.8)	1.4 (1.4)	2.6 (1.8)	-0.2 (-0.5)	-1.1 (-0.9)	-1.3 (-2.0)
5y	-0.7 (-1.4)	0.4 (0.3)	3.6 (2.6)	0.8 (1.3)	0.2 (0.2)	-1.0 (-1.6)

Close of 1/16/20. We calculate rich/cheap and z-scores in parentheses from PCA on curve vols using two principal components and 3y of data. By design, this framework only yields relative value. Source: Citi Research

Figure 60. Implied correlations of 2s10s CMS options



Note: The chart shows implied correlation of options on the 2s10s curve with different expirations. Source: Citi Research

Figure 61. Implied/realized ratios

implied/1m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.10 (-0.6)	1.03 (-0.8)	1.03 (-0.6)	1.05 (-0.9)	1.33 (0.4)	1.76 (1.4)
3m	1.23 (-0.2)	1.16 (-0.1)	1.17 (0.2)	1.19 (-0.2)	1.50 (1.2)	1.90 (2.1)
6m	1.17 (-0.4)	1.28 (0.3)	1.28 (0.7)	1.34 (0.3)	1.62 (1.8)	1.97 (2.3)
1y	1.21 (-0.5)	1.39 (0.4)	1.42 (1.0)	1.61 (0.9)	1.88 (2.2)	2.03 (2.2)

implied/3m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	0.91 (-1.4)	0.94 (-1.0)	0.95 (-0.6)	1.05 (-0.5)	1.13 (0.1)	1.22 (-0.1)
3m	1.07 (-0.9)	1.12 (0.0)	1.11 (0.4)	1.19 (0.2)	1.24 (0.8)	1.32 (0.5)
6m	1.10 (-0.6)	1.29 (0.9)	1.23 (1.3)	1.30 (0.8)	1.28 (1.2)	1.35 (0.6)
1y	1.32 (0.4)	1.51 (1.8)	1.39 (2.2)	1.46 (1.3)	1.35 (1.3)	1.34 (0.3)

Close of 1/16/20. Close-to-close realized vol is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

Eurodollar convexity adjustments

We calculate ED/FRA convexity adjustment for 1y ED packs and compare them to model values calibrated to cap/floor volatilities.

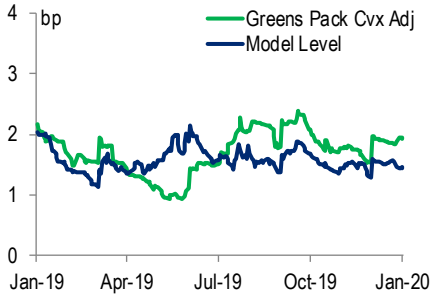
- Convexity adjustments have tightened closer to fair value in Blues and Golds over the past week (Figure 64).
- *Note: Futures trading involves substantial risk of loss.*

Figure 62. Convexity adjustments for 1y Eurodollar packs (vs CME swaps)

	convexity adjustement				vs the model				implied vs realized vol			
ED Pack	Cvx Adj (bp)	1 Week Chg (bp)	3m Z-Score	1Y Z-Score	Vs Model (bp)	3m Z-Score	1Y Z-Score	3m Roll (short cvx, bp)	Implied Vol	Realized Vol	Implied/ Realized	Cap vol Impl/Rlzd
H0-Z0	-0.02	0.04	-0.20	-0.64	-0.06	0.14	-0.49	0.01	37.2	55.9	0.7	0.7
M0-H1	0.01	0.05	-0.49	-1.46	-0.07	-0.23	-0.71	0.03	n/a	59.5	n/a	0.7
U0-M1	0.08	0.05	-0.65	-1.51	-0.07	-0.36	-0.65	0.07	n/a	64.7	n/a	0.7
Z0-U1	0.21	0.06	-0.71	-1.40	-0.05	-0.37	-0.48	0.13	48.9	69.5	0.7	0.7
H1-Z1	0.44	0.07	-0.54	-0.88	0.02	-0.08	-0.02	0.23	59.8	73.2	0.8	0.7
M1-H2	0.76	0.08	-0.15	-0.19	0.14	0.57	0.51	0.32	68.1	76.1	0.9	0.8
U1-M2	1.12	0.09	**0.34**	**0.28**	0.26	**1.33**	**0.79**	0.36	72.8	78.6	0.9	0.8
Z1-U2	1.53	0.09	**0.87**	**0.58**	0.40	**1.88**	**0.95**	0.41	75.7	79.8	0.9	0.8
H2-Z2	1.93	0.10	**1.05**	**0.63**	0.49	**2.05**	**0.94**	0.40	76.7	80.6	**1.0**	0.8
M2-H3	2.28	0.12	-0.43	0.18	0.51	-0.21	0.47	0.34	75.9	80.9	0.9	0.8
U2-M3	2.85	0.02	-0.21	0.20	**0.71**	-0.12	0.39	**0.57**	78.0	81.3	**1.0**	0.8
Z2-U3	3.23	-0.12	-0.94	-0.06	0.69	-0.90	0.09	0.38	76.8	81.4	0.9	0.8
H3-Z3	3.79	-0.18	-0.85	-0.07	**0.82**	-0.84	0.04	**0.56**	77.3	81.4	**0.9**	0.8
M3-H4	4.25	-0.26	-0.78	-0.11	**0.80**	-0.81	-0.02	0.46	76.5	81.4	0.9	0.8
U3-M4	4.54	-0.29	-0.89	-0.34	0.59	-0.91	-0.22	0.29	74.1	81.3	0.9	0.8
Z3-U4	5.08	-0.31	-0.61	-0.38	0.61	-0.67	-0.23	**0.54**	73.9	81.3	0.9	0.8
H4-Z4	5.58	-0.39	-0.55	-0.54	0.56	-0.63	-0.33	0.50	73.2	81.6	0.9	0.8

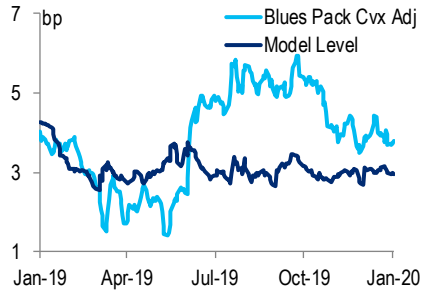
Note: Close of 1/16/20. Convexity adjustments for 1y ED packs are computed as the spread between the pack's rate (the average of 4 ED rates in the pack) and matched-maturity forward 1y CME swap rate. The model for convexity adjustment is the Ho-Lee model calibrated to cap/floor vols. Implied vol is calculated by matching the model to the observed convexity adjustment. Realized vol is 3m realized vol of the corresponding pack. For each valuation metric, we mark three best short convexity trades in **bold**.
Source: Citi Research

Figure 63. Greens pack (rolling) vs cap/floor-based model



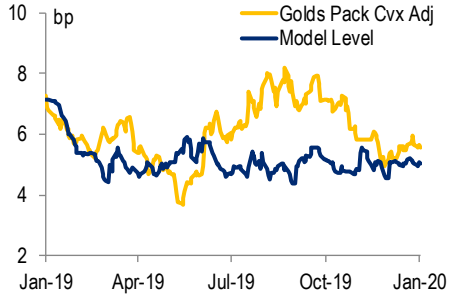
The chart shows the convexity adjustment of the Greens pack vs the (Ho-Lee) model. Source: Citi Research

Figure 64. Blues pack (rolling) vs cap/floor-based model



The chart shows the convexity adjustment of the Blues pack vs the (Ho-Lee) model. Source: Citi Research

Figure 65. Golds pack (rolling) vs cap/floor-based model



The chart shows the convexity adjustment of the Golds pack vs the (Ho-Lee) model. Source: Citi Research

USD conditional curves

We analyze conditional curve trades based on the entry point, implied/realized betas, and carry.

- We still like 3y 5s30s conditional bear steepeners as recommended in US Rates Vol Lab: CCAR and bear steepeners. The zero-cost strike spread is about 14bp (pricing as of 3pm on 1/16/2020).

Prepared for Christopher Lee

Figure 66. Conditional curves: bear trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bear steepener				bear flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.3	-8.8	-5.4	0.6	-0.7	-1.6	0.5	13	0.29	0.25	-0.04	-3.3	-4.7	-8.0	-7.6	3.3	4.7	8.0	7.6
3m 1y3y	100/33.8	-11.2	-6.0	0.5	1.1	-3.2	0.5	11	0.40	0.36	-0.04	-5.2	-7.1	-12.3	-7.6	5.2	7.1	12.3	7.6
3m 1y5y	100/20.5	-9.5	-2.8	0.4	7.5	-6.2	0.4	10	0.51	0.46	-0.06	-6.7	-10.3	-17.0	-7.6	6.7	10.3	17.0	7.6
3m 2y5y	100/40.8	-0.8	2.6	0.2	7.2	-3.8	0.3	11	0.22	0.20	-0.02	-3.4	-4.5	-7.9	-4.2	3.4	4.5	7.9	4.2
3m 2y10y	100/21.2	12.8	17.4	0.3	23.0	-3.6	0.4	15	0.26	0.25	0.00	-4.5	-5.7	-10.2	-4.2	4.5	5.7	10.2	4.2
3m 2y30y	100/8.4	31.2	35.4	0.2	40.0	-3.0	0.4	16	0.22	0.24	0.02	-4.2	-4.6	-8.8	-4.2	4.2	4.6	8.8	4.2
3m 3y5y	100/60.8	1.7	3.2	0.2	5.5	-2.3	0.2	14	0.11	0.09	-0.01	-1.6	-2.3	-3.9	-2.3	1.6	2.3	3.9	2.3
3m 3y10y	100/31.6	15.2	17.9	0.2	21.2	-2.1	0.3	16	0.15	0.14	0.00	-2.7	-3.3	-6.0	-2.3	2.7	3.3	6.0	2.3
3m 3y30y	100/12.5	33.6	36.0	0.1	38.2	-1.5	0.3	17	0.10	0.12	0.02	-2.3	-2.3	-4.6	-2.3	2.3	2.3	4.6	2.3
3m 5y10y	100/51.9	13.6	14.7	0.3	15.6	0.3	0.5	18	0.04	0.05	0.01	-1.1	-0.9	-2.0	-0.7	1.1	0.9	2.0	0.7
3m 5y30y	100/20.5	32.0	32.7	0.0	32.6	0.9	0.4	18	0.00	0.03	0.03	-0.8	0.1	-0.7	-0.6	0.8	-0.1	0.7	0.6
3m 10y30y	100/39.5	18.4	18.0	-0.4	17.0	0.6	-0.1	20	-0.04	-0.02	0.02	0.4	1.0	1.4	0.0	-0.4	-1.0	-1.4	0.0
6m 1y2y	100/50.3	-8.8	-2.7	0.4	2.6	-1.6	0.6	15	0.21	0.16	-0.06	-6.0	-5.3	-11.3	-2.2	6.0	5.3	11.3	2.2
6m 1y3y	100/33.8	-11.2	-2.1	0.3	5.3	-3.0	0.5	13	0.28	0.23	-0.05	-9.1	-7.3	-16.5	-2.7	9.1	7.3	16.5	2.7
6m 1y5y	100/20.6	-9.5	2.4	0.3	13.1	-4.7	0.5	13	0.37	0.29	-0.08	-11.9	-10.7	-22.6	-3.3	11.9	10.7	22.6	3.3
6m 2y5y	100/40.8	-0.8	5.1	0.2	9.7	-2.8	0.4	13	0.15	0.13	-0.02	-5.9	-4.6	-10.4	-1.4	5.9	4.6	10.4	1.4
6m 2y10y	100/21.2	12.8	20.6	0.2	27.0	-1.2	0.6	16	0.20	0.16	-0.04	-7.7	-6.4	-14.1	-2.0	7.7	6.4	14.1	2.0
6m 2y30y	100/8.4	31.2	38.0	0.1	42.8	-0.7	0.6	17	0.16	0.13	-0.03	-6.8	-4.8	-11.6	-1.5	6.8	4.8	11.6	1.5
6m 3y5y	100/60.8	1.7	4.4	0.2	7.1	-1.4	0.3	16	0.09	0.06	-0.02	-2.8	-2.6	-5.4	-0.8	2.8	2.6	5.4	0.8
6m 3y10y	100/31.6	15.2	19.9	0.2	24.2	0.2	0.6	18	0.13	0.09	-0.04	-4.7	-4.3	-9.0	-1.5	4.7	4.3	9.0	1.5
6m 3y30y	100/12.5	33.6	37.3	0.0	40.1	0.6	0.6	18	0.09	0.06	-0.03	-3.7	-2.8	-6.5	-1.0	3.7	2.8	6.5	1.0
6m 5y10y	100/52.0	13.6	15.5	0.2	17.0	1.5	1.0	20	0.05	0.03	-0.02	-1.9	-1.6	-3.4	-0.7	1.9	1.6	3.4	0.7
6m 5y30y	100/20.5	32.0	32.9	-0.1	32.9	1.9	0.7	19	0.00	-0.01	-0.01	-0.9	-0.1	-1.0	-0.2	0.9	0.1	1.0	0.2
6m 10y30y	100/39.6	18.4	17.4	-0.5	15.9	0.5	-0.3	18	-0.05	-0.03	0.01	1.0	1.5	2.5	0.5	-1.0	-1.5	-2.5	-0.5
1y 1y2y	100/50.2	-8.8	0.7	-0.2	5.2	-1.6	0.6	19	0.11	0.07	-0.04	-9.4	-4.5	-13.9	-0.5	9.4	4.5	13.9	0.5
1y 1y3y	100/33.8	-11.2	2.7	-0.1	8.9	-2.4	0.5	15	0.15	0.10	-0.04	-13.9	-6.2	-20.1	-0.6	13.9	6.2	20.1	0.6
1y 1y5y	100/20.6	-9.5	8.9	0.0	17.2	-2.8	0.5	14	0.19	0.13	-0.05	-18.4	-8.4	-26.7	-0.6	18.4	8.4	26.7	0.6
1y 2y5y	100/41.0	-0.8	8.2	0.2	11.6	-1.1	0.4	15	0.08	0.06	-0.01	-8.9	-3.5	-12.4	-0.2	8.9	3.5	12.4	0.2
1y 2y10y	100/21.3	12.8	24.4	0.2	28.7	0.4	0.5	17	0.09	0.06	-0.03	-11.6	-4.3	-15.9	0.0	11.6	4.3	15.9	0.0
1y 2y30y	100/8.4	31.2	40.2	0.0	41.5	0.8	0.5	17	0.03	0.02	-0.01	-9.0	-1.3	-10.3	0.8	9.0	1.3	10.3	-0.8
1y 3y5y	100/60.9	1.7	6.2	0.3	8.0	-0.2	0.5	17	0.04	0.03	-0.01	-4.5	-1.9	-6.4	-0.1	4.5	1.9	6.4	0.1
1y 3y10y	100/31.7	15.2	22.4	0.3	25.0	1.1	0.6	19	0.06	0.03	-0.03	-7.2	-2.6	-9.8	0.0	7.2	2.6	9.8	0.0
1y 3y30y	100/12.5	33.6	38.2	0.0	37.9	1.6	0.5	17	-0.01	-0.02	-0.01	-4.6	0.3	-4.3	0.8	4.6	-0.3	4.3	-0.8
1y 5y10y	100/52.0	13.6	16.2	0.2	16.9	1.4	0.7	20	0.01	0.00	-0.02	-2.7	-0.7	-3.4	0.2	2.7	0.7	3.4	-0.2
1y 5y30y	100/20.6	32.0	32.1	-0.2	29.9	1.9	0.2	16	-0.05	-0.05	0.00	-0.1	2.2	2.1	1.0	0.1	-2.2	-2.1	-1.0
1y 10y30y	100/39.6	18.4	15.8	-0.5	12.9	0.5	-0.5	13	-0.06	-0.04	0.02	2.6	2.9	5.5	0.8	-2.6	-2.9	-5.5	-0.8
2y 1y2y	100/50.6	-8.8	2.7	0.1	4.8	0.7	0.5	19	0.02	0.01	-0.01	-11.5	-2.1	-13.5	0.4	11.5	2.1	13.5	-0.4
2y 1y3y	100/34.0	-11.2	6.1	0.2	9.2	1.7	0.7	15	0.04	0.02	-0.02	-17.3	-3.1	-20.4	0.4	17.3	3.1	20.4	-0.4
2y 1y5y	100/20.7	-9.5	13.3	0.3	16.7	1.6	0.7	14	0.04	0.03	-0.02	-22.8	-3.4	-26.2	0.6	22.8	3.4	26.2	-0.6
2y 2y5y	100/41.0	-0.8	10.6	0.5	11.9	0.9	0.9	15	0.02	0.01	-0.01	-11.4	-1.3	-12.7	0.2	11.4	1.3	12.7	-0.2
2y 2y10y	100/21.3	12.8	26.6	0.4	27.4	2.2	0.8	17	0.01	-0.01	-0.02	-13.8	-0.8	-14.6	0.5	13.8	0.8	14.6	-0.5
2y 2y30y	100/8.5	31.2	38.5	0.0	34.4	2.6	0.6	14	-0.06	-0.06	0.00	-7.3	4.1	-3.2	1.3	7.3	-4.1	3.2	-1.3
2y 3y5y	100/60.9	1.7	7.2	0.4	7.5	-0.1	0.3	17	0.01	0.00	0.00	-5.5	-0.3	-5.8	0.2	5.5	0.3	5.8	-0.2
2y 3y10y	100/31.7	15.2	23.1	0.3	23.0	1.1	0.4	18	0.00	-0.02	-0.02	-7.9	0.2	-7.7	0.5	7.9	-0.2	7.7	-0.5
2y 3y30y	100/12.6	33.6	35.1	-0.1	29.9	1.6	0.0	12	-0.08	-0.07	0.00	-1.4	5.1	3.7	1.3	1.4	-5.1	-3.7	-1.3
2y 5y10y	100/52.1	13.6	15.9	0.0	15.5	1.2	0.4	18	-0.01	-0.02	-0.02	-2.4	0.5	-1.9	0.3	2.4	-0.5	1.9	-0.3
2y 5y30y	100/20.6	32.0	27.9	-0.4	22.4	1.8	-0.1	10	-0.08	-0.08	0.00	4.1	5.5	9.6	1.1	-4.1	-5.5	-9.6	-1.1
2y 10y30y	100/39.6	18.4	11.9	-0.6	6.9	0.5	-0.3	7	-0.07	-0.05	0.02	6.5	5.0	11.5	0.8	-6.5	-5.0	-11.5	-0.8
3y 1y2y	100/50.4	-8.8	3.8	0.6	4.5	1.8	1.2	18	0.01	0.01	0.00	-12.6	-0.7	-13.3	-0.1	12.6	0.7	13.3	0.1
3y 1y3y	100/33.8	-11.2	7.7	0.5	7.9	1.5	0.7	14	0.00	0.01	0.01	-18.9	-0.2	-19.1	0.0	18.9	0.2	19.1	0.0
3y 1y5y	100/20.6	-9.5	14.8	0.4	13.9	1.9	0.0	13	-0.01	0.00	0.01	-24.4	0.9	-23.4	0.3	24.4	-0.9	23.4	-0.3
3y 2y5y	100/40.9	-0.8	11.0	0.1	9.4	0.0	-0.8	13	-0.02	-0.01	0.00	-11.8	1.6	-10.1	0.4	11.8	-1.6	10.1	-0.4
3y 2y10y	100/21.4	12.8	25.7	0.2	22.9	1.1	-0.4	15	-0.03	-0.04	-0.01	-12.8	2.7	-10.1	0.7	12.8	-2.7	10.1	-0.7
3y 2y30y	100/8.5	31.2	33.5	-0.2	23.2	1.4	-0.5	8	-0.12	-0.09	0.03	-2.2	10.3	8.0	1.5	2.2	-10.3	-8.0	-1.5
3y 3y5y	100/60.9	1.7	7.1	0.0	6.0	0.4	-1.0	15	-0.01	-0.01	0.00	-5.5	1.1	-4.3	0.2	5.5	-1.1	4.3	-0.2
3y 3y10y	100/31.8	15.2	21.8	0.1	19.6	1.4	-0.4	15	-0.02	-0.04	-0.01	-6.5	2.2	-4.3	0.5	6.5	-2.2	4.3	-0.5
3y 3y30y	100/12.6	33.6	29.6	-0.3	19.8	1.7	-0.5	6	-0.12	-0.09	0.02	4.1	9.8	13.8	1.3	-4.1	-9.8	-13.8	-1.3
3y 5y10y	100/52.2	13.6	14.7	0.2	13.6	1.0	0.2	14	-0.01	-0.03	-0.01	-1.1	1.1	0.0	0.3	1.1	-1.1	0.0	-0.3
3y 5y30y	100/20.7	32.0	22.4	-0.4	13.8	1.4	-0.2	5	-0.10	-0.08	0.02	9.5	8.6	18.1	1.1	-9.5	-8.6	-18.1	-1.1
3y 10y30y	100/39.7	18.4	7.8	-0.8	0.4	0.4	-0.5	3	-0.09	-0.05	0.04	10.6	7.4	18.0	0.8	-10.6	-7.4	-18.0	-0.8

Note: Close of 1/16/20. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m carry is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best bear steepeners in red, and three best bear flatteners in green. Source: Citi Research

Prepared for Christopher Lee

Figure 67. Conditional curves: bull trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bull steepener				bull flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.3	-8.8	-5.4	0.6	-11.3	0.0	0.1	7	0.29	0.25	-0.04	-3.3	5.9	2.6	0.0	3.3	-5.9	-2.6	0.0
3m 1y3y	100/33.8	-11.2	-6.0	0.5	-14.8	-0.7	-0.3	6	0.40	0.36	-0.04	-5.2	8.8	3.6	0.0	5.2	-8.8	-3.6	0.0
3m 1y5y	100/20.5	-9.5	-2.8	0.4	-14.6	-0.2	-0.3	5	0.51	0.46	-0.06	-6.7	11.9	5.1	0.0	6.7	-11.9	-5.1	0.0
3m 2y5y	100/40.8	-0.8	2.6	0.2	-2.5	-0.5	-0.4	5	0.22	0.20	-0.02	-3.4	5.1	1.7	0.0	3.4	-5.1	-1.7	0.0
3m 2y10y	100/21.2	12.8	17.4	0.3	11.3	0.5	-0.3	7	0.26	0.25	0.00	-4.5	6.1	1.5	0.0	4.5	-6.1	-1.5	0.0
3m 2y30y	100/8.4	31.2	35.4	0.2	30.6	0.7	-0.4	12	0.22	0.24	0.02	-4.2	4.8	0.6	0.0	4.2	-4.8	-0.6	0.0
3m 3y5y	100/60.8	1.7	3.2	0.2	0.7	0.3	0.1	4	0.11	0.09	-0.01	-1.6	2.6	1.0	0.0	1.6	-2.6	-1.0	0.0
3m 3y10y	100/31.6	15.2	17.9	0.2	14.4	1.2	0.0	10	0.15	0.14	0.00	-2.7	3.5	0.8	0.0	2.7	-3.5	-0.8	0.0
3m 3y30y	100/12.5	33.6	36.0	0.1	33.6	1.4	-0.2	14	0.10	0.12	0.02	-2.3	2.4	0.0	0.0	2.3	-2.4	0.0	0.0
3m 5y10y	100/51.9	13.6	14.7	0.3	13.8	0.9	-0.2	15	0.04	0.05	0.01	-1.1	0.9	-0.2	0.0	1.1	-0.9	0.2	0.0
3m 5y30y	100/20.5	32.0	32.7	0.0	32.8	1.1	-0.4	19	0.00	0.03	0.03	-0.8	-0.1	-0.9	-0.1	0.8	0.1	0.9	0.1
3m 10y30y	100/39.5	18.4	18.0	-0.4	19.0	0.2	-0.5	25	-0.04	-0.02	0.02	0.4	-1.0	-0.7	-0.1	-0.4	1.0	0.7	0.1
6m 1y2y	100/50.3	-8.8	-2.7	0.4	-9.4	-0.4	-0.4	7	0.21	0.16	-0.06	-6.0	6.7	0.7	-0.9	6.0	-6.7	-0.7	0.9
6m 1y3y	100/33.8	-11.2	-2.1	0.3	-11.2	-1.2	-0.7	7	0.28	0.23	-0.05	-9.1	9.1	0.0	-1.5	9.1	-9.1	0.0	1.5
6m 1y5y	100/20.6	-9.5	2.4	0.3	-10.0	-1.5	-0.9	6	0.37	0.29	-0.08	-11.9	12.4	0.5	-1.9	11.9	-12.4	-0.5	1.9
6m 2y5y	100/40.8	-0.8	5.1	0.2	-0.1	-1.0	-1.0	6	0.15	0.13	-0.02	-5.9	5.2	-0.7	-1.1	5.9	-5.2	0.7	1.1
6m 2y10y	100/21.2	12.8	20.6	0.2	13.6	-1.0	-1.2	9	0.20	0.16	-0.04	-7.7	6.9	-0.8	-1.0	7.7	-6.9	0.8	1.0
6m 2y30y	100/8.4	31.2	38.0	0.1	32.9	-0.5	-1.1	13	0.16	0.13	-0.03	-6.8	5.1	-1.7	-1.0	6.8	-5.1	1.7	1.0
6m 3y5y	100/60.8	1.7	4.4	0.2	1.5	-0.2	-0.5	6	0.09	0.06	-0.02	-2.8	3.0	0.2	-0.4	2.8	-3.0	-0.2	0.4
6m 3y10y	100/31.6	15.2	19.9	0.2	15.2	-0.3	-0.9	11	0.13	0.09	-0.04	-4.7	4.7	0.0	-0.4	4.7	-4.7	0.0	0.4
6m 3y30y	100/12.5	33.6	37.3	0.0	34.4	0.2	-0.8	15	0.09	0.06	-0.03	-3.7	2.9	-0.7	-0.4	3.7	-2.9	0.7	0.4
6m 5y10y	100/52.0	13.6	15.5	0.2	13.8	0.0	-1.0	15	0.05	0.03	-0.02	-1.9	1.7	-0.2	0.0	1.9	-1.7	0.2	0.0
6m 5y30y	100/20.5	32.0	32.9	-0.1	32.8	0.4	-0.8	19	0.00	-0.01	-0.01	-0.9	0.1	-0.9	0.0	0.9	-0.1	0.9	0.0
6m 10y30y	100/39.6	18.4	17.4	-0.5	19.0	0.4	-0.5	25	-0.05	-0.03	0.01	1.0	-1.6	-0.6	0.0	-1.0	1.6	0.6	0.0
1y 1y2y	100/50.2	-8.8	0.7	-0.2	-4.8	-1.3	-1.9	10	0.11	0.07	-0.04	-9.4	5.5	-4.0	-0.8	9.4	-5.5	4.0	0.8
1y 1y3y	100/33.8	-11.2	2.7	-0.1	-4.6	-2.1	-2.2	9	0.15	0.10	-0.04	-13.9	7.3	-6.6	-1.2	13.9	-7.3	6.6	1.2
1y 1y5y	100/20.6	-9.5	8.9	0.0	-0.7	-2.5	-2.0	8	0.19	0.13	-0.05	-18.4	9.5	-8.8	-1.8	18.4	-9.5	8.8	1.8
1y 2y5y	100/41.0	-0.8	8.2	0.2	4.2	-1.1	-0.8	9	0.08	0.06	-0.01	-8.9	3.9	-5.0	-1.0	8.9	-3.9	5.0	1.0
1y 2y10y	100/21.3	12.8	24.4	0.2	19.7	-0.3	-0.9	13	0.09	0.06	-0.03	-11.6	4.7	-6.9	-1.4	11.6	-4.7	6.9	1.4
1y 2y30y	100/8.4	31.2	40.2	0.0	38.9	0.2	-0.8	16	0.03	0.02	-0.01	-9.0	1.4	-7.7	-1.4	9.0	-1.4	7.7	1.4
1y 3y5y	100/60.9	1.7	6.2	0.3	4.0	-0.3	-0.4	11	0.04	0.03	-0.01	-4.5	2.1	-2.4	-0.6	4.5	-2.1	2.4	0.6
1y 3y10y	100/31.7	15.2	22.4	0.3	19.5	0.5	-0.6	15	0.06	0.03	-0.03	-7.2	2.9	-4.3	-1.0	7.2	-2.9	4.3	1.0
1y 3y30y	100/12.5	33.6	38.2	0.0	38.6	1.0	-0.6	17	-0.01	-0.02	-0.01	-4.6	-0.3	-4.9	-1.0	4.6	0.3	4.9	1.0
1y 5y10y	100/52.0	13.6	16.2	0.2	15.5	0.7	-0.5	18	0.01	0.00	-0.02	-2.7	0.8	-1.9	-0.4	2.7	-0.8	1.9	0.4
1y 5y30y	100/20.6	32.0	32.1	-0.2	34.5	1.2	-0.5	20	-0.05	-0.05	0.00	-0.1	-2.5	-2.6	-0.4	0.1	2.5	2.6	0.4
1y 10y30y	100/39.6	18.4	15.8	-0.5	19.0	0.5	-0.4	25	-0.06	-0.04	0.02	2.6	-3.2	-0.6	0.0	-2.6	3.2	0.6	0.0
2y 1y2y	100/50.6	-8.8	2.7	0.1	0.4	-1.6	-0.8	13	0.02	0.01	-0.01	-11.5	2.3	-9.2	-0.5	11.5	-2.3	9.2	0.5
2y 1y3y	100/34.0	-11.2	6.1	0.2	2.8	-2.5	-1.0	12	0.04	0.02	-0.02	-17.3	3.3	-14.0	-0.9	17.3	-3.3	14.0	0.9
2y 1y5y	100/20.7	-9.5	13.3	0.3	9.6	-0.8	-0.5	11	0.04	0.03	-0.02	-22.8	3.7	-19.2	-1.2	22.8	-3.7	19.2	1.2
2y 2y5y	100/41.0	-0.8	10.6	0.5	9.2	0.9	-0.2	13	0.02	0.01	-0.01	-11.4	1.4	-10.0	-0.7	11.4	-1.4	10.0	0.7
2y 2y10y	100/21.3	12.8	26.6	0.4	25.7	1.9	-0.2	16	0.01	-0.01	-0.02	-13.8	0.9	-12.9	-0.9	13.8	-0.9	12.9	0.9
2y 2y30y	100/8.5	31.2	38.5	0.0	43.0	2.0	-0.4	17	-0.06	-0.06	0.00	-7.3	-4.6	-11.8	-0.6	7.3	4.6	11.8	0.6
2y 3y5y	100/60.9	1.7	7.2	0.4	6.9	1.7	0.4	16	0.01	0.00	0.00	-5.5	0.3	-5.2	-0.4	5.5	-0.3	5.2	0.4
2y 3y10y	100/31.7	15.2	23.1	0.3	23.3	2.7	0.1	18	0.00	-0.02	-0.02	-7.9	-0.2	-8.1	-0.5	7.9	0.2	8.1	0.5
2y 3y30y	100/12.6	33.6	35.1	-0.1	40.7	2.9	-0.2	18	-0.08	-0.07	0.00	-1.4	-5.6	-7.1	-0.3	1.4	5.6	7.1	0.3
2y 5y10y	100/52.1	13.6	15.9	0.0	16.5	1.0	-0.3	19	-0.01	-0.02	-0.02	-2.4	-0.5	-2.9	-0.1	2.4	0.5	2.9	0.1
2y 5y30y	100/20.6	32.0	27.9	-0.4	33.8	1.2	-0.5	19	-0.08	-0.08	0.00	4.1	-6.0	-1.9	0.1	-4.1	6.0	1.9	-0.1
2y 10y30y	100/39.6	18.4	11.9	-0.6	17.3	0.2	-0.5	20	-0.07	-0.05	0.02	6.5	-5.4	1.1	0.2	-6.5	5.4	-1.1	-0.2
3y 1y2y	100/50.4	-8.8	3.8	0.6	3.1	-1.0	-0.2	16	0.01	0.01	0.00	-12.6	0.8	-11.8	-0.4	12.6	-0.8	11.8	0.4
3y 1y3y	100/33.8	-11.2	7.7	0.5	7.5	-0.2	0.2	14	0.00	0.01	0.01	-18.9	0.2	-18.7	-0.7	18.9	-0.2	18.7	0.7
3y 1y5y	100/20.6	-9.5	14.8	0.4	15.7	1.7	0.7	14	-0.01	0.00	0.01	-24.4	-0.9	-25.3	-0.9	24.4	0.9	25.3	0.9
3y 2y5y	100/40.9	-0.8	11.0	0.1	12.7	2.7	1.1	16	-0.02	-0.01	0.00	-11.8	-1.7	-13.4	-0.5	11.8	1.7	13.4	0.5
3y 2y10y	100/21.4	12.8	25.7	0.2	28.6	3.8	0.8	17	-0.03	-0.04	-0.01	-12.8	-2.9	-15.8	-0.4	12.8	2.9	15.8	0.4
3y 2y30y	100/8.5	31.2	33.5	-0.2	44.6	3.4	0.3	17	-0.12	-0.09	0.03	-2.2	-11.1	-13.4	-0.3	2.2	11.1	13.4	0.3
3y 3y5y	100/60.9	1.7	7.1	0.0	8.3	1.9	1.0	17	-0.01	-0.01	0.00	-5.5	-1.2	-6.6	-0.2	5.5	1.2	6.6	0.2
3y 3y10y	100/31.8	15.2	21.8	0.1	24.1	3.0	0.7	18	-0.02	-0.04	-0.01	-6.5	-2.4	-8.9	-0.1	6.5	2.4	8.9	0.1
3y 3y30y	100/12.6	33.6	29.6	-0.3	40.1	2.6	0.1	18	-0.12	-0.09	0.02	4.1	-10.6	-6.5	0.0	-4.1	10.6	6.5	0.0
3y 5y10y	100/52.2	13.6	14.7	0.2	15.9	1.1	0.2	18	-0.01	-0.03	-0.01	-1.1	-1.2	-2.3	0.1	1.1	1.2	2.3	-0.1
3y 5y30y	100/20.7	32.0	22.4	-0.4	31.7	0.7	-0.5	18	-0.10	-0.08	0.02	9.5	-9.3	0.2	0.2	-9.5	9.3	-0.2	-0.2
3y 10y30y	100/39.7	18.4	7.8	-0.8	15.8	-0.4	-0.7	18	-0.09	-0.05	0.04	10.6	-8.0	2.6	0.1	-10.6	8.0	-2.6	-0.1

Note: Close of 1/16/20. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best bull steepeners in red, and three best bull flatteners in green. Source: Citi Research

Prepared for Christopher Lee

USD conditional flies

We analyze conditional fly trades based on the entry point, directionality with rates and carry.

- We still like selling 6m10y payers vs 6m2y payers and 6m30y payers initiated in Initiating 6m 2s10s30s payer fly to monetize the relative richness of the 10y vol (current levels are reported in Figure 68). The main risk is technical selling pressures on 10s.

Figure 68. Conditional payer flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long payer in the belly				short payer in the belly			
curve	notionals, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, ZC	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	148.0/100/30.4	-6.4	-4.6	0.6	84	4	3.3	1.4	2.2	-2.5	-0.4	0.8	26	-1.8	-2.2	-4.0	-4.0	1.8	2.2	4.0	4.0
3m 1y3y7y	148.0/100/22.0	-8.9	-7.4	0.4	77	0	3.3	1.2	1.9	-5.6	-0.3	0.8	45	-1.5	-1.9	-3.3	-3.4	1.5	1.9	3.3	3.4
3m 2y3y5y	74.5/100/30.4	-2.0	-1.9	-0.1	16	-37	0.4	0.5	-0.1	-2.0	0.4	0.8	75	-0.1	0.1	0.0	0.0	0.1	-0.1	0.0	0.0
3m 2y5y7y	122.5/100/36.3	-2.8	-1.5	0.1	76	-29	1.9	1.2	1.9	0.4	-1.6	0.3	10	-1.4	-1.9	-3.3	-2.1	1.4	1.9	3.3	2.1
3m 2y5y10y	122.5/100/26.0	-7.2	-6.0	0.1	74	-38	2.1	1.4	1.7	-4.3	-1.7	0.2	19	-1.1	-1.7	-2.9	-2.1	1.1	1.7	2.9	2.1
3m 2y10y30y	235.8/100/19.8	-2.8	-0.3	0.3	79	34	2.8	1.1	3.2	2.9	-1.9	0.4	11	-2.5	-3.2	-5.7	-2.1	2.5	3.2	5.7	2.1
3m 3y5y7y	82.2/100/36.3	-1.6	-1.2	0.0	56	-22	0.7	1.1	0.8	-0.3	-1.0	0.0	16	-0.5	-0.8	-1.3	-1.2	0.5	0.8	1.3	1.2
3m 3y5y10y	82.2/100/26.0	-6.0	-5.7	0.0	44	-30	0.9	0.9	0.7	-5.0	-1.1	-0.2	52	-0.2	-0.7	-0.9	-1.1	0.2	0.7	0.9	1.1
3m 3y7y10y	113.4/100/35.8	-1.0	-0.2	0.3	70	0	1.2	0.9	1.3	1.1	-1.2	0.2	11	-0.9	-1.3	-2.2	-1.2	0.9	1.3	2.2	1.2
3m 3y7y15y	113.4/100/24.9	-6.4	-5.5	0.2	72	-2	1.4	1.2	1.4	-4.1	-1.3	0.1	15	-0.9	-1.4	-2.3	-1.2	0.9	1.4	2.3	1.2
3m 3y10y30y	158.3/100/19.8	-1.6	0.0	0.3	71	58	1.5	0.9	2.1	2.1	-1.3	0.3	14	-1.5	-2.1	-3.7	-1.2	1.5	2.1	3.7	1.2
3m 5y10y15y	96.2/100/34.8	1.4	2.0	0.2	51	39	0.2	0.2	0.5	2.6	-0.1	0.4	18	-0.6	-0.5	-1.1	-0.4	0.6	0.5	1.1	0.4
3m 5y10y30y	96.2/100/19.8	-2.4	-1.7	0.5	56	87	0.7	1.1	0.9	-0.7	-0.2	0.5	20	-0.8	-0.9	-1.7	-0.4	0.8	0.9	1.7	0.4
3m 10y20y30y	91.8/100/36.3	7.3	7.3	1.4	11	9	0.1	0.2	0.0	7.3	0.4	1.3	24	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6m 1y3y5y	148.0/100/30.4	-6.4	-3.2	0.4	78	-44	2.4	1.4	2.2	-1.0	-0.7	0.6	28	-3.2	-2.2	-5.4	-1.0	3.2	2.2	5.4	1.0
6m 1y3y7y	148.0/100/22.1	-8.9	-6.3	0.2	70	-39	2.5	1.2	1.8	-4.5	-1.0	0.3	49	-2.6	-1.8	-4.4	-0.8	2.6	1.8	4.4	0.8
6m 2y3y5y	74.5/100/30.4	-2.0	-1.9	-0.1	12	-50	0.4	0.6	-0.4	-2.3	0.1	0.1	73	-0.2	0.4	0.2	0.1	0.2	-0.4	-0.2	-0.1
6m 2y5y7y	122.4/100/36.2	-2.8	-0.5	0.1	66	-43	1.3	1.0	1.8	1.3	-1.7	0.2	13	-2.4	-1.8	-4.2	-0.5	2.4	1.8	4.2	0.5
6m 2y5y10y	122.4/100/26.0	-7.2	-5.2	0.1	66	-52	1.6	1.3	1.5	-3.7	-2.0	0.1	20	-2.0	-1.5	-3.4	-0.4	2.0	1.5	3.4	0.4
6m 2y10y30y	235.6/100/19.8	-2.8	1.6	0.3	70	29	1.8	0.7	3.9	5.5	-1.0	0.6	15	-4.4	-3.9	-8.2	-1.3	4.4	3.9	8.2	1.3
6m 3y5y7y	82.2/100/36.2	-1.6	-0.8	0.1	50	-29	0.5	0.9	0.9	0.1	-1.0	0.0	19	-0.8	-0.9	-1.7	-0.2	0.8	0.9	1.7	0.2
6m 3y5y10y	82.2/100/26.0	-6.0	-5.5	0.0	43	-31	0.8	0.9	0.5	-5.0	-1.4	-0.4	53	-0.4	-0.5	-1.0	0.0	0.4	0.5	1.0	0.0
6m 3y7y10y	113.3/100/35.8	-1.0	0.5	0.2	61	-6	0.6	0.5	1.4	1.9	-0.8	0.3	13	-1.5	-1.4	-2.9	-0.4	1.5	1.4	2.9	0.4
6m 3y7y15y	113.3/100/24.9	-6.4	-4.7	0.1	66	-4	0.9	1.0	1.7	-3.1	-0.9	0.3	19	-1.6	-1.7	-3.3	-0.5	1.6	1.7	3.3	0.5
6m 3y10y30y	158.2/100/19.8	-1.6	1.2	0.3	64	58	1.0	0.6	2.9	4.1	-0.3	0.6	18	-2.8	-2.9	-5.7	-1.0	2.8	2.9	5.7	1.0
6m 5y10y15y	96.2/100/34.8	1.4	2.5	0.1	41	37	-0.1	-0.1	1.1	3.5	0.5	1.0	22	-1.1	-1.1	-2.1	-0.5	1.1	1.1	2.1	0.5
6m 5y10y30y	96.2/100/19.8	-2.4	-1.0	0.5	53	89	0.5	0.9	1.5	0.5	0.4	1.0	26	-1.4	-1.5	-3.0	-0.6	1.4	1.5	3.0	0.6
6m 10y20y30y	91.8/100/36.3	7.3	7.2	1.5	1	-1	0.1	0.1	-0.1	7.1	0.4	1.3	23	0.1	0.1	0.1	0.1	-0.1	-0.1	-0.1	-0.1
1y 1y3y5y	148.1/100/30.5	-6.4	-1.7	-0.5	69	-78	0.9	0.9	2.1	0.3	-1.0	0.4	31	-4.7	-2.1	-6.8	-0.2	4.7	2.1	6.8	0.2
1y 1y3y7y	148.1/100/22.1	-8.9	-5.0	-0.4	61	-72	1.3	1.1	1.7	-3.3	-1.3	0.3	52	-3.9	-1.7	-5.6	-0.2	3.9	1.7	5.6	0.2
1y 2y3y5y	74.4/100/30.5	-2.0	-2.1	-0.6	17	-49	0.2	0.4	-0.2	-2.2	-0.2	-0.6	73	0.0	0.2	0.2	0.0	0.0	-0.2	-0.2	0.0
1y 2y5y7y	122.0/100/36.3	-2.8	0.8	0.1	48	-25	0.5	0.4	1.4	2.2	-1.1	0.4	15	-3.6	-1.4	-5.0	-0.1	3.6	1.4	5.0	0.1
1y 2y5y10y	122.0/100/26.0	-7.2	-4.0	0.1	53	-44	0.9	0.9	1.4	-2.7	-1.3	0.3	24	-3.1	-1.4	-4.5	-0.2	3.1	1.4	4.5	0.2
1y 2y10y30y	234.7/100/19.8	-2.8	4.3	0.3	55	47	0.6	0.3	3.6	7.9	-0.4	0.6	18	-7.1	-3.6	-10.7	-0.4	7.1	3.6	10.7	0.4
1y 3y5y7y	82.1/100/36.3	-1.6	-0.2	0.3	44	-16	0.4	0.7	0.6	0.4	-0.6	0.4	22	-1.4	-0.6	-2.0	-0.1	1.4	0.6	2.0	0.1
1y 3y5y10y	82.1/100/26.0	-6.0	-5.0	0.2	45	-30	0.7	0.9	0.6	-4.5	-0.8	0.1	59	-0.9	-0.6	-1.5	-0.2	0.9	0.6	1.5	0.2
1y 3y7y10y	113.2/100/35.9	-1.0	1.5	0.2	43	8	0.0	0.0	1.3	2.8	-0.3	0.4	16	-2.5	-1.3	-3.8	-0.2	2.5	1.3	3.8	0.2
1y 3y7y15y	113.2/100/24.9	-6.4	-3.5	0.1	55	23	0.3	0.5	1.9	-1.6	-0.4	0.4	26	-2.9	-1.9	-4.8	-0.3	2.9	1.9	4.8	0.3
1y 3y10y30y	157.8/100/19.8	-1.6	3.3	0.4	51	68	0.4	0.2	2.8	6.1	0.0	0.7	23	-4.9	-2.8	-7.7	-0.4	4.9	2.8	7.7	0.4
1y 5y10y15y	96.1/100/34.8	1.4	3.1	0.0	26	48	-0.3	-0.4	1.0	4.1	0.4	0.7	24	-1.7	-1.0	-2.7	-0.1	1.7	1.0	2.7	0.1
1y 5y10y30y	96.1/100/19.8	-2.4	0.2	0.6	49	92	0.3	0.6	1.8	2.0	0.3	0.9	34	-2.6	-1.8	-4.4	-0.3	2.6	1.8	4.4	0.3
1y 10y20y30y	91.7/100/36.3	7.3	6.9	1.5	-13	-10	0.0	0.0	-0.2	6.8	0.5	1.2	21	0.3	0.2	0.5	0.1	-0.3	-0.2	-0.5	-0.1

Note: Close of 1/16/20. Conditional flies are structured by buying (selling) payers in the belly vs selling (buying) payers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best long belly payer trades in red, and three best short belly payer trades in green. Source: Citi Research

Prepared for Christopher Lee

Figure 69. Conditional receiver flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly				costless strike					long receiver in the belly				short receiver in the belly			
curve	ntl, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, zs	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	148.0/100/30.4	-6.4	-4.6	0.6	84	4	3.3	1.4	2.6	-7.2	-0.3	-0.2	19	1.8	-2.6	-0.8	0.0	-1.8	2.6	0.8	0.0
3m 1y3y7y	148.0/100/22.0	-8.9	-7.4	0.4	77	0	3.3	1.2	2.2	-9.6	-0.5	-0.2	34	1.5	-2.2	-0.8	0.0	-1.5	2.2	0.8	0.0
3m 2y3y5y	74.5/100/30.4	-2.0	-1.9	-0.1	16	-37	0.4	0.5	-0.1	-1.8	-0.4	-0.6	77	0.1	0.1	0.2	0.0	-0.1	-0.1	-0.2	0.0
3m 2y5y7y	122.5/100/36.3	-2.8	-1.5	0.1	76	-29	1.9	1.2	2.1	-3.6	-0.7	-0.4	5	1.4	-2.1	-0.7	0.0	-1.4	2.1	0.7	0.0
3m 2y5y10y	122.5/100/26.0	-7.2	-6.0	0.1	74	-38	2.1	1.4	2.0	-8.0	-0.9	-0.2	12	1.1	-2.0	-0.8	-0.2	-1.1	2.0	0.8	0.2
3m 2y10y30y	235.8/100/19.8	-2.8	-0.3	0.3	79	34	2.8	1.1	3.4	-3.8	-0.3	-0.1	4	2.5	-3.4	-1.0	0.0	-2.5	3.4	1.0	0.0
3m 3y5y7y	82.2/100/36.3	-1.6	-1.2	0.0	56	-22	0.7	1.1	0.9	-2.1	-0.2	0.1	7	0.5	-0.9	-0.5	0.0	-0.5	0.9	0.5	0.0
3m 3y5y10y	82.2/100/26.0	-6.0	-5.7	0.0	44	-30	0.9	0.9	0.8	-6.5	-0.4	0.3	40	0.2	-0.8	-0.6	-0.2	-0.2	0.8	0.6	0.2
3m 3y7y10y	113.4/100/35.8	-1.0	-0.2	0.3	70	0	1.2	0.9	1.4	-1.6	-0.2	0.3	3	0.9	-1.4	-0.6	-0.2	-0.9	1.4	0.6	0.2
3m 3y7y15y	113.4/100/24.9	-6.4	-5.5	0.2	72	-2	1.4	1.2	1.5	-7.0	-0.2	0.2	6	0.9	-1.5	-0.6	-0.2	-0.9	1.5	0.6	0.2
3m 3y10y30y	158.3/100/19.8	-1.6	0.0	0.3	71	58	1.5	0.9	2.2	-2.3	0.1	0.2	3	1.5	-2.2	-0.7	0.0	-1.5	2.2	0.7	0.0
3m 5y10y15y	96.2/100/34.8	1.4	2.0	0.2	51	39	0.2	0.2	0.6	1.5	0.2	-0.4	12	0.6	-0.6	0.0	-0.2	-0.6	0.6	0.0	0.2
3m 5y10y30y	96.2/100/19.8	-2.4	-1.7	0.5	56	87	0.7	1.1	1.0	-2.6	0.2	0.4	10	0.8	-1.0	-0.2	0.0	-0.8	1.0	0.2	0.0
3m 10y20y30y	91.8/100/36.3	7.3	7.3	1.4	11	9	0.1	0.2	0.0	7.3	0.3	1.1	24	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6m 1y3y5y	148.0/100/30.4	-6.4	-3.2	0.4	78	-44	2.4	1.4	2.7	-5.9	-0.5	-0.3	21	3.2	-2.7	0.5	0.5	-3.2	2.7	-0.5	-0.5
6m 1y3y7y	148.0/100/22.1	-8.9	-6.3	0.2	70	-39	2.5	1.2	2.2	-8.4	-0.3	-0.1	37	2.6	-2.2	0.5	0.4	-2.6	2.2	-0.5	-0.4
6m 2y3y5y	74.5/100/30.4	-2.0	-1.9	-0.1	12	-50	0.4	0.6	-0.4	-1.4	-0.2	-0.2	80	0.2	0.4	0.6	0.1	-0.2	-0.4	-0.6	-0.1
6m 2y5y7y	122.4/100/36.2	-2.8	-0.5	0.1	66	-43	1.3	1.0	2.0	-2.5	-0.5	-0.6	6	2.4	-2.0	0.3	0.5	-2.4	2.0	-0.3	-0.5
6m 2y5y10y	122.4/100/26.0	-7.2	-5.2	0.1	66	-52	1.6	1.3	1.6	-6.8	-0.4	0.0	14	2.0	-1.6	0.4	0.5	-2.0	1.6	-0.4	-0.5
6m 2y10y30y	235.6/100/19.8	-2.8	1.6	0.3	70	29	1.8	0.7	4.2	-2.6	-0.9	-1.0	6	4.4	-4.2	0.2	0.5	-4.4	4.2	-0.2	-0.5
6m 3y5y7y	82.2/100/36.2	-1.6	-0.8	0.1	50	-29	0.5	0.9	1.0	-1.8	-0.1	0.1	8	0.8	-1.0	-0.2	0.1	-0.8	1.0	0.2	-0.1
6m 3y5y10y	82.2/100/26.0	-6.0	-5.5	0.0	43	-31	0.8	0.9	0.6	-6.1	0.0	0.6	43	0.4	-0.6	-0.2	0.2	-0.4	0.6	0.2	-0.2
6m 3y7y10y	113.3/100/35.8	-1.0	0.5	0.2	61	-6	0.6	0.5	1.5	-1.0	-0.2	-0.3	4	1.5	-1.5	0.0	0.3	-1.5	1.5	0.0	-0.3
6m 3y7y15y	113.3/100/24.9	-6.4	-4.7	0.1	66	-4	0.9	1.0	1.8	-6.6	-0.4	-0.3	7	1.6	-1.8	-0.2	0.2	-1.6	1.8	0.2	-0.2
6m 3y10y30y	158.2/100/19.8	-1.6	1.2	0.3	64	58	1.0	0.6	3.1	-1.9	-0.5	-0.7	4	2.8	-3.1	-0.3	0.2	-2.8	3.1	0.3	-0.2
6m 5y10y15y	96.2/100/34.8	1.4	2.5	0.1	41	37	-0.1	-0.1	1.1	1.4	-0.3	-1.6	11	1.1	-1.1	-0.1	-0.1	-1.1	1.1	0.1	0.1
6m 5y10y30y	96.2/100/19.8	-2.4	-1.0	0.5	53	89	0.5	0.9	1.6	-2.6	-0.3	-0.7	10	1.4	-1.6	-0.2	0.0	-1.4	1.6	0.2	0.0
6m 10y20y30y	91.8/100/36.3	7.3	7.2	1.5	1	-1	0.1	0.1	-0.1	7.3	0.4	0.5	24	-0.1	0.1	0.0	0.0	0.1	-0.1	0.0	0.0
1y 1y3y5y	148.1/100/30.5	-6.4	-1.7	-0.5	69	-78	0.9	0.9	2.4	-4.2	-0.9	-1.8	23	4.7	-2.4	2.3	0.3	-4.7	2.4	-2.3	-0.3
1y 1y3y7y	148.1/100/22.1	-8.9	-5.0	-0.4	61	-72	1.3	1.1	2.0	-7.1	-1.0	-1.5	41	3.9	-2.0	1.8	0.2	-3.9	2.0	-1.8	-0.2
1y 2y3y5y	74.4/100/30.5	-2.0	-2.1	-0.6	17	-49	0.2	0.4	-0.2	-1.9	-0.2	-0.5	76	0.0	0.2	0.1	-0.1	0.0	-0.2	-0.1	0.1
1y 2y5y7y	122.0/100/36.3	-2.8	0.8	0.1	48	-25	0.5	0.4	1.6	-0.7	-0.7	-0.5	8	3.6	-1.6	2.1	0.4	-3.6	1.6	-2.1	-0.4
1y 2y5y10y	122.0/100/26.0	-7.2	-4.0	0.1	53	-44	0.9	0.9	1.5	-5.6	-0.8	-0.3	16	3.1	-1.5	1.6	0.3	-3.1	1.5	-1.6	-0.3
1y 2y10y30y	234.7/100/19.8	-2.8	4.3	0.3	55	47	0.6	0.3	3.9	0.3	-0.6	-0.8	8	7.1	-3.9	3.1	0.7	-7.1	3.9	-3.1	-0.7
1y 3y5y7y	82.1/100/36.3	-1.6	-0.2	0.3	44	-16	0.4	0.7	0.7	-0.9	-0.3	0.0	12	1.4	-0.7	0.8	0.2	-1.4	0.7	-0.8	-0.2
1y 3y5y10y	82.1/100/26.0	-6.0	-5.0	0.2	45	-30	0.7	0.9	0.7	-5.7	-0.4	0.2	46	0.9	-0.7	0.3	0.1	-0.9	0.7	-0.3	-0.1
1y 3y7y10y	113.2/100/35.9	-1.0	1.5	0.2	43	8	0.0	0.0	1.4	0.1	-0.1	-0.4	8	2.5	-1.4	1.1	0.3	-2.5	1.4	-1.1	-0.3
1y 3y7y15y	113.2/100/24.9	-6.4	-3.5	0.1	55	23	0.3	0.5	2.1	-5.5	-0.3	-0.6	10	2.9	-2.1	0.8	0.2	-2.9	2.1	-0.8	-0.2
1y 3y10y30y	157.8/100/19.8	-1.6	3.3	0.4	51	68	0.4	0.2	3.0	0.2	-0.2	-0.5	9	4.9	-3.0	1.8	0.5	-4.9	3.0	-1.8	-0.5
1y 5y10y15y	96.1/100/34.8	1.4	3.1	0.0	26	48	-0.3	-0.4	1.0	2.1	0.1	-0.9	16	1.7	-1.0	0.7	0.2	-1.7	1.0	-0.7	-0.2
1y 5y10y30y	96.1/100/19.8	-2.4	0.2	0.6	49	92	0.3	0.6	2.0	-1.8	0.0	-0.3	14	2.6	-2.0	0.6	0.2	-2.6	2.0	-0.6	-0.2
1y 10y20y30y	91.7/100/36.3	7.3	6.9	1.5	-13	-10	0.0	0.0	-0.2	7.1	0.5	0.3	23	-0.3	0.2	-0.1	0.0	0.3	-0.2	0.1	0.0

Note: Close of 1/16/20. Conditional flies are structured by buying (selling) receivers in the belly vs selling (buying) receivers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **long belly receiver trades in red**, and three **best short belly receiver trades in green**. Source: Citi Research

Prepared for Christopher Lee

EUR at-the-money swaption vols

EUR relative value on the swaption surface

We report levels and z-scores of at-the-money vols, as well as the relative richness/cheapness of vols to rates (based on principal components of the curve).

- Similar to USD vols, implieds have cheapened led by short expires. The 30y vols have especially underperformed (Figure 71).
- The volatilities in 5y and 7y tails continue to look relatively rich on the macro-vol model (Figure 72) and the PCA of the vol surface (Figure 73). For example, the 1y5y is about 2.8 sigmas rich on the PCA.
- Similar to USD vols, upper-left EUR vols look relatively cheap on the surface, if evaluated by the macro-vol model and PCA of the surface (Figure 72 and Figure 73).
- The 30y tail also looks cheap on the PCA of the vol surface (Figure 73).

Figure 70. Implied vols, normals (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	8.0 (-1.2)	12.1 (-1.2)	17.8 (-1.0)	29.4 (-0.6)	34.9 (-0.7)	38.3 (-1.1)	43.7 (-1.3)
3m	10.8 (-1.4)	14.0 (-1.7)	19.7 (-1.4)	30.4 (-0.6)	35.0 (-1.3)	39.5 (-1.7)	45.5 (-1.8)
6m	12.3 (-1.7)	16.2 (-1.8)	21.5 (-1.6)	31.5 (-0.2)	36.0 (-1.3)	40.6 (-1.9)	46.4 (-1.9)
1y	15.9 (-1.6)	20.4 (-1.3)	25.4 (-0.6)	33.5 (0.4)	37.8 (-0.9)	42.0 (-2.3)	47.6 (-2.2)
2y	24.5 (0.1)	28.4 (0.7)	32.6 (1.0)	37.8 (0.3)	41.4 (-0.5)	44.3 (-2.3)	48.7 (-1.9)
3y	32.7 (0.8)	35.3 (0.7)	37.8 (0.5)	41.7 (-0.1)	44.1 (-1.7)	46.4 (-2.6)	49.0 (-1.8)
5y	42.6 (-0.3)	43.4 (-0.8)	44.6 (-1.1)	47.1 (-1.5)	48.2 (-2.3)	49.5 (-2.8)	49.1 (-2.3)
10y	52.1 (-2.4)	51.9 (-2.4)	52.3 (-2.5)	52.4 (-2.9)	52.6 (-2.9)	53.0 (-2.6)	49.0 (-1.8)

Close of 1/16/20. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 71. 2w change in implied vols, normals

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-1.5	-1.9	-2.5	-3.6	-4.8	-8.0	-9.2
3m	-1.4	-2.2	-2.1	-2.7	-4.1	-6.1	-6.9
6m	-1.2	-1.4	-1.4	-2.4	-3.4	-4.8	-5.7
1y	-0.9	-0.9	-1.1	-1.8	-2.6	-3.7	-4.8
2y	-0.5	-0.5	-0.6	-1.6	-2.1	-3.3	-3.6
3y	-0.9	-0.9	-1.2	-1.5	-2.1	-2.9	-3.1
5y	-1.0	-1.1	-1.2	-1.3	-1.9	-2.9	-2.4
10y	-2.0	-2.0	-1.9	-2.3	-2.4	-2.5	-2.2

Note: The table shows changes in ATM implied vols from 1/6/20 to 1/16/20. Source: Citi Research

Figure 72. Macro-vol (rich/cheap to rates), normals (z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-8.4 (-1.5)	-7.4 (-1.4)	-6.9 (-1.3)	-4.1 (-0.6)	-5.3 (-0.7)	-10.0 (-1.1)	-18.9 (-1.4)
3m	-6.7 (-1.5)	-6.5 (-1.5)	-5.3 (-1.2)	-2.6 (-0.5)	-4.8 (-0.7)	-7.4 (-0.9)	-14.3 (-1.2)
6m	-5.5 (-1.5)	-5.3 (-1.4)	-4.2 (-1.1)	-1.9 (-0.4)	-3.6 (-0.6)	-6.0 (-0.8)	-10.9 (-1.0)
1y	-4.8 (-1.4)	-3.9 (-1.2)	-3.2 (-0.9)	-1.8 (-0.4)	-3.3 (-0.6)	-5.4 (-0.8)	-7.9 (-0.8)
2y	-3.8 (-1.1)	-3.3 (-1.0)	-2.9 (-0.9)	-2.8 (-0.7)	-3.7 (-0.7)	-5.7 (-0.8)	-6.0 (-0.7)
3y	-3.7 (-1.0)	-3.5 (-0.9)	-3.8 (-1.0)	-3.9 (-0.9)	-5.1 (-1.0)	-6.2 (-0.9)	-5.2 (-0.7)
5y	-6.1 (-1.4)	-5.8 (-1.4)	-5.8 (-1.4)	-5.7 (-1.2)	-6.1 (-1.2)	-6.6 (-1.1)	-4.3 (-0.7)
10y	-6.4 (-1.8)	-6.5 (-1.7)	-6.3 (-1.6)	-6.2 (-1.5)	-6.1 (-1.4)	-5.5 (-1.3)	-2.8 (-0.8)

Close of 1/16/20. We regress each ATM point on 3 principal components of the swap curve. We report rich/cheap for each vol as a residual from this regression together with z-scores in parentheses. We use the 2016+ period, i.e. the negative rates regime. Source: Citi Research

Figure 73. Rich/cheap to PCA of the vol surface, normals (z-score)

exp\tls	3m	1y	2y	3y	5y	7y	10y	30y
1m	-1.6 (-1.0)	0.6 (0.4)	-0.8 (-0.6)	-0.5 (-0.3)	2.6 (1.1)	2.1 (0.9)	-1.0 (-0.4)	-5.8 (-1.7)
3m	-1.7 (-1.0)	0.1 (0.1)	-1.2 (-1.4)	-0.1 (-0.1)	2.7 (2.0)	1.3 (1.0)	-0.7 (-0.6)	-3.8 (-2.3)
6m	-1.8 (-1.0)	-0.2 (-0.2)	-0.3 (-0.4)	0.5 (0.9)	2.5 (2.3)	1.4 (1.5)	-0.2 (-0.2)	-2.1 (-2.1)
1y	0.1 (0.1)	-0.6 (-0.8)	0.3 (0.4)	0.9 (1.7)	2.3 (2.8)	1.5 (2.4)	0.0 (0.1)	-0.5 (-0.6)
2y	-0.2 (-0.1)	0.2 (0.1)	0.8 (0.9)	1.3 (1.5)	1.6 (2.2)	1.2 (2.3)	0.0 (0.1)	0.6 (0.8)
3y	0.9 (0.6)	0.6 (0.5)	0.9 (1.1)	0.8 (1.2)	0.9 (1.8)	0.4 (1.0)	-0.2 (-0.5)	0.8 (1.1)
5y	-0.2 (-0.2)	-0.4 (-0.6)	-0.2 (-0.4)	-0.3 (-0.6)	-0.3 (-0.6)	-0.6 (-1.4)	-0.9 (-1.7)	0.4 (0.6)
10y	-3.1 (-2.8)	-2.2 (-2.4)	-2.2 (-2.7)	-1.9 (-2.6)	-1.9 (-2.9)	-1.8 (-2.7)	-1.5 (-2.1)	-0.5 (-0.7)

Close of 1/16/20. We calculate rich/cheap and z-scores in parentheses from PCA on the vol surface based on 3 principal components using 2y of data. By design, this framework only yields relative value. Source: Citi Research

EUR implied vs realized volatilities

We calculate the relative richness of short-dated implied vols to 1m and 3m realized vols. We calculate realized vols using both intraday (hourly) and close-to-close sampling.

- Intraday realized volatilities have underperformed close-to-close volatilities over the last month, indicating a trending intraday market (Figure 74 and Figure 75).
- Consistent with the macro-vol model and PCA, volatilities in the top right of the surface look relatively cheap if compared to recent realized vols (Figure 77).

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Figure 74. 1m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.00 (-0.5)	0.82 (-1.0)	0.80 (-1.0)	0.89 (-0.8)	0.92 (-0.5)	0.95 (-0.1)	0.92 (0.1)
3m	0.98 (-0.4)	0.79 (-1.1)	0.79 (-1.0)	0.89 (-0.8)	0.92 (-0.6)	0.95 (-0.1)	0.92 (0.1)
6m	0.93 (-0.5)	0.79 (-1.0)	0.80 (-1.0)	0.90 (-0.8)	0.92 (-0.5)	0.95 (-0.1)	0.92 (0.1)
1y	0.82 (-1.0)	0.80 (-1.0)	0.83 (-0.9)	0.91 (-0.8)	0.93 (-0.5)	0.95 (-0.0)	0.91 (0.1)

Note: Close of 1/16/20. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 75. 3m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.24 (0.9)	1.01 (0.4)	0.93 (0.1)	0.99 (-0.2)	0.99 (0.1)	0.99 (0.5)	0.90 (0.0)
3m	1.14 (0.7)	0.98 (0.3)	0.92 (0.0)	0.99 (-0.2)	0.99 (0.1)	0.99 (0.6)	0.90 (0.1)
6m	1.06 (0.5)	0.96 (0.1)	0.92 (-0.1)	0.99 (-0.2)	0.99 (0.1)	0.99 (0.6)	0.90 (0.0)
1y	0.99 (0.2)	0.93 (-0.0)	0.94 (-0.2)	1.00 (-0.2)	0.99 (0.2)	0.99 (0.7)	0.90 (0.0)

Note: Close of 1/16/20. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 76. Intraday realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.55 (0.8)	1.35 (0.8)	1.24 (0.5)	1.03 (0.1)	0.99 (-0.1)	0.97 (-0.4)	1.03 (-0.3)
3m	1.79 (1.7)	1.35 (1.2)	1.23 (0.9)	1.00 (0.4)	0.96 (0.0)	0.98 (-0.0)	1.07 (0.4)
6m	1.52 (1.6)	1.24 (1.2)	1.12 (0.8)	0.95 (0.3)	0.95 (0.1)	0.98 (0.1)	1.08 (0.6)
1y	1.18 (1.2)	1.07 (0.9)	0.96 (0.4)	0.88 (0.1)	0.91 (0.0)	0.95 (0.1)	1.08 (0.7)

Note: Close of 1/16/20. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 77. Intraday realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.92 (-0.8)	0.97 (-0.4)	1.03 (-0.4)	0.96 (-0.3)	0.93 (-0.6)	0.89 (-0.9)	0.92 (-0.9)
3m	1.07 (-0.2)	1.00 (-0.1)	1.04 (-0.0)	0.94 (-0.1)	0.90 (-0.5)	0.90 (-0.7)	0.95 (-0.6)
6m	1.03 (0.3)	0.98 (0.4)	0.98 (0.1)	0.90 (-0.0)	0.89 (-0.4)	0.90 (-0.5)	0.97 (-0.2)
1y	0.94 (0.5)	0.93 (0.4)	0.89 (-0.0)	0.84 (-0.2)	0.85 (-0.4)	0.87 (-0.4)	0.97 (0.0)

Note: Close of 1/16/20. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 78. Close-to-close realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.55 (0.2)	1.10 (-0.1)	0.99 (-0.3)	0.91 (-0.6)	0.91 (-0.5)	0.92 (-0.4)	0.94 (-0.1)
3m	1.76 (1.0)	1.06 (0.1)	0.98 (-0.1)	0.89 (-0.4)	0.89 (-0.4)	0.93 (-0.1)	0.97 (0.3)
6m	1.42 (0.8)	0.98 (0.2)	0.90 (-0.2)	0.85 (-0.4)	0.87 (-0.3)	0.93 (0.1)	0.99 (0.5)
1y	0.96 (0.2)	0.85 (-0.0)	0.80 (-0.4)	0.80 (-0.4)	0.84 (-0.2)	0.91 (0.1)	0.99 (0.6)

Note: Close of 1/16/20. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 79. Close-to-close realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.13 (-0.4)	0.98 (-0.2)	0.96 (-0.3)	0.95 (-0.4)	0.92 (-0.5)	0.88 (-0.7)	0.83 (-0.8)
3m	1.22 (0.2)	0.98 (0.1)	0.96 (0.1)	0.93 (-0.2)	0.89 (-0.4)	0.89 (-0.4)	0.86 (-0.5)
6m	1.10 (0.4)	0.94 (0.4)	0.91 (0.1)	0.89 (-0.1)	0.88 (-0.3)	0.89 (-0.3)	0.87 (-0.2)
1y	0.93 (0.7)	0.86 (0.4)	0.83 (-0.1)	0.83 (-0.2)	0.84 (-0.3)	0.87 (-0.2)	0.87 (-0.0)

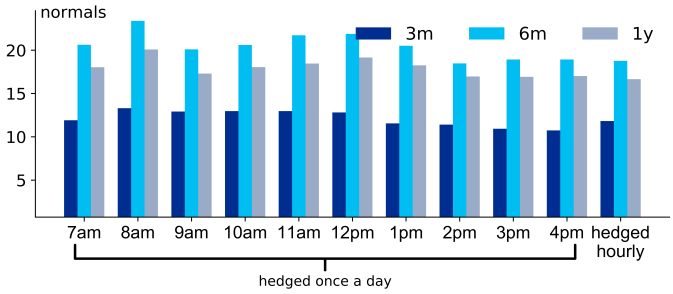
Note: Close of 1/16/20. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

EUR day-to-day realized volatilities for a given time snapshot

We calculate day-to-day realized volatilities measured at a given hour within a day for the last 3m, 6m and 1y (London time)

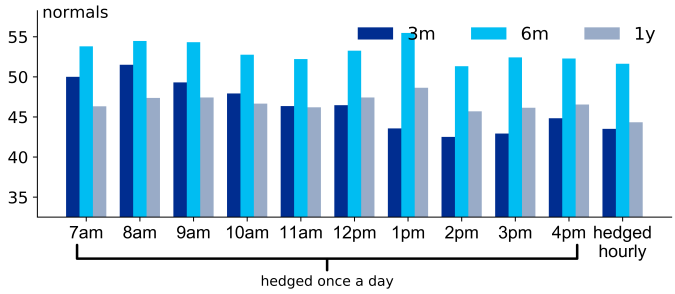
- Hedging around 2pm has delivered relatively low realized volatility of the 10y rate over the past 3 and 6 months (consistent with the USD vol low at 9am NY time; Figure 81). In contrast, hedging around 8am has delivered the highest realized vol over the recent past.

Figure 80. Realized vol of the 2y rate



Note: Close of 1/16/20. Time is London. Source: Citi Research

Figure 81. Realized vol of the 10y rate



Note: Close of 1/16/20. Time is London. Source: Citi Research

EUR expiry and tail switches

We analyze ATM vol switches across expiries and tails.

- Consistent with our macro-vol model, short tails in short expiries are trading cheap to longer expiries. For example, the 6m2y/1y2y vol ratio is 1.8 sigmas low (Figure 82).

- The 5y vols have richened to longer tails. For example, the 3m5y/3m10y vol ratio is 2.2 sigmas high and the 3m5y/3m30y vol ratio is 2.1 sigmas high (Figure 83).

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Figure 82. Short/long expiry ratio (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m/3m	0.74 (-1.0)	0.86 (-0.4)	0.90 (-0.6)	0.97 (-0.4)	1.00 (-0.2)	0.97 (-0.5)	0.96 (-0.6)
3m/6m	0.88 (-0.9)	0.87 (-1.4)	0.92 (-1.1)	0.97 (-0.8)	0.97 (-1.1)	0.97 (-1.3)	0.98 (-1.6)
6m/1y	0.77 (-1.7)	0.79 (-1.8)	0.85 (-1.8)	0.94 (-1.7)	0.95 (-1.4)	0.97 (-1.1)	0.97 (-1.4)
1y/3y	0.49 (-1.6)	0.58 (-1.4)	0.67 (-1.0)	0.80 (0.8)	0.86 (0.2)	0.91 (-1.5)	0.97 (-2.0)
3y/5y	0.77 (1.2)	0.81 (1.4)	0.85 (1.3)	0.89 (1.3)	0.92 (0.9)	0.94 (0.3)	1.00 (-0.7)
5y/10y	0.82 (1.9)	0.84 (1.9)	0.85 (1.9)	0.90 (2.5)	0.92 (2.1)	0.93 (0.2)	1.00 (-0.1)

Note: Close of 1/16/20. The table reports shorter expiry/longer expiry ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

Figure 83. Short/long tail ratio (3m z-score)

exp\tails	1y/2y	1y/3y	2y/5y	2y/10y	5y/10y	5y/30y	10y/30y
1m	0.67 (-1.4)	0.45 (-1.5)	0.41 (-1.4)	0.31 (-1.0)	0.77 (2.5)	0.67 (2.2)	0.88 (1.7)
3m	0.77 (-0.5)	0.55 (-1.4)	0.46 (-1.7)	0.36 (-1.4)	0.77 (2.2)	0.67 (2.1)	0.87 (1.6)
1y	0.78 (-1.7)	0.63 (-1.7)	0.61 (-1.5)	0.49 (0.7)	0.80 (2.2)	0.70 (2.1)	0.88 (1.6)
2y	0.86 (-1.4)	0.75 (-1.4)	0.75 (1.5)	0.64 (2.1)	0.85 (1.9)	0.78 (1.6)	0.91 (0.8)
5y	0.98 (1.2)	0.96 (1.3)	0.92 (1.1)	0.88 (2.2)	0.95 (2.3)	0.96 (0.9)	1.01 (-1.4)
10y	1.00 (2.6)	1.00 (-1.1)	0.99 (-0.5)	0.98 (-0.5)	0.99 (-0.5)	1.07 (-1.1)	1.08 (-1.7)

Note: Close of 1/16/20. The table reports shorter tail/longer tail ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

EUR forward volatilities

We calculate forward 1y2y, 1y5y, 1y10y and 1y30y volatilities implied from the spot swaption surface. Forward volatilities reflect market expectations about future short-dated vols and may highlight dislocations on the spot vol surface.

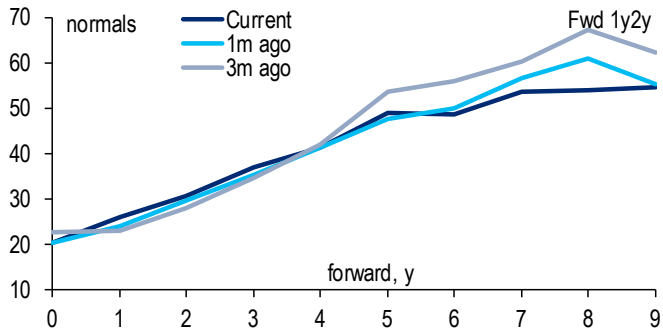
- Forwards volatilities have generally continued to cheapen over the past week, led by longer tails (Figure 84).

Figure 84. Forward volatilities

	1y2y							1y5y							1y10y							1y30y						
Yrs		1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y	
Fwd	Vol	chg	chg	avg	z-scr	spt	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt	z-scr
spt	20.4	(0.5)	0.0	(2.6)	(1.3)	(33.6)	(1.4)	33.5	(0.8)	1.1	0.1	0.1	(31.9)	(1.6)	42.0	(2.4)	(1.2)	(0.4)	(0.1)	(32.2)	(2.3)	47.6	(2.7)	(2.8)	0.5	0.1	(30.0)	(1.9)
1	25.8	0.5	1.7	(3.0)	(0.7)	(28.1)	(1.2)	35.3	(0.0)	1.3	(1.9)	(0.5)	(30.2)	(1.5)	42.9	(0.9)	(1.0)	(0.8)	(0.4)	(31.3)	(2.2)	48.5	(1.3)	(1.2)	2.8	0.7	(29.2)	(1.9)
2	30.7	0.0	1.1	(4.7)	(0.8)	(23.3)	(1.0)	39.0	0.6	0.6	(3.5)	(0.7)	(26.4)	(1.4)	44.9	(0.5)	(0.2)	(2.3)	(0.7)	(29.3)	(2.1)	47.5	(0.9)	0.6	1.4	0.5	(30.2)	(1.9)
3	36.9	0.4	1.5	(5.5)	(0.7)	(17.0)	(0.7)	43.6	0.9	1.8	(3.3)	(0.6)	(21.8)	(1.1)	47.4	(0.1)	(1.0)	(3.3)	(0.8)	(26.9)	(1.9)	47.4	(0.2)	(0.3)	0.4	0.1	(30.3)	(1.9)
4	41.3	1.2	0.0	(5.7)	(0.8)	(12.7)	(0.5)	45.8	1.4	(0.6)	(4.5)	(0.8)	(19.6)	(1.0)	49.4	(0.1)	(2.9)	(4.2)	(0.9)	(24.9)	(1.8)	46.6	(0.2)	(1.2)	(1.3)	(0.5)	(31.1)	(2.0)
5	49.0	(0.6)	1.5	(6.9)	(1.2)	(5.0)	(0.2)	48.9	0.3	(1.5)	(5.2)	(0.9)	(16.5)	(0.8)	51.8	(0.4)	(3.2)	(4.7)	(1.0)	(22.5)	(1.6)	45.5	(0.5)	(2.0)	(2.6)	(1.0)	(32.2)	(2.1)
6	48.5	3.3	(1.5)	(7.8)	(1.6)	(5.4)	(0.2)	49.8	(0.6)	1.3	(5.0)	(1.1)	(15.6)	(0.8)	53.9	(1.1)	(0.7)	(4.0)	(0.9)	(20.3)	(1.4)	48.4	(1.2)	(1.0)	(1.6)	(0.7)	(29.2)	(1.9)
7	53.5	(1.6)	(3.0)	(5.5)	(1.5)	(0.5)	(0.0)	53.2	(0.3)	(0.4)	(4.7)	(1.2)	(12.2)	(0.6)	55.0	(0.4)	(1.8)	(4.1)	(1.0)	(19.2)	(1.4)	49.4	(0.6)	(2.2)	(1.3)	(0.6)	(28.3)	(1.8)
8	54.0	(0.4)	(6.9)	(8.1)	(2.5)	0.0	0.0	51.7	(0.7)	(3.3)	(5.8)	(1.5)	(13.7)	(0.7)	55.8	(0.6)	(3.2)	(5.1)	(1.4)	(18.4)	(1.3)	47.4	(0.8)	(3.3)	(3.1)	(1.5)	(30.3)	(1.9)
9	54.6	0.1	(0.7)	(4.2)	(1.2)	0.6	0.0	51.4	(0.9)	(3.1)	(5.9)	(1.6)	(14.0)	(0.7)	57.0	(0.5)	(2.9)	(4.3)	(1.2)	(17.2)	(1.2)	50.8	(0.7)	(2.8)	(0.1)	(0.0)	(26.8)	(1.7)

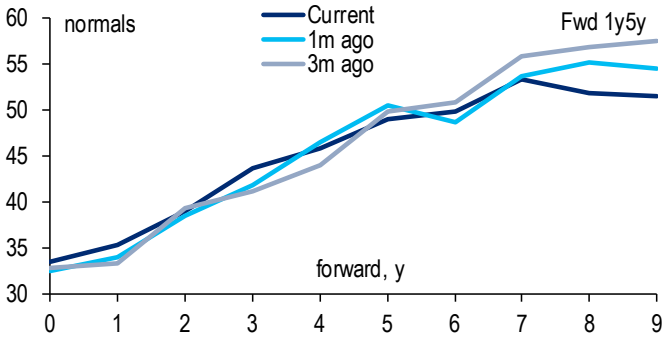
Note: Close of 1/16/20. Forward vols are computed using the standard triangular approximation from spot vols and using 6m realized correlations as a proxy for implied correlations. In addition to levels of forward vols, we calculate 1w and 1m changes, the difference from 1y average (with corresponding z-scores), and the difference from the 10y average of the corresponding 1y expiry spot vol (together with z-scores). Source: Citi Research

Figure 85. Forward vol of 1y2y



Note: Close of 1/16/20. Source: Citi Research

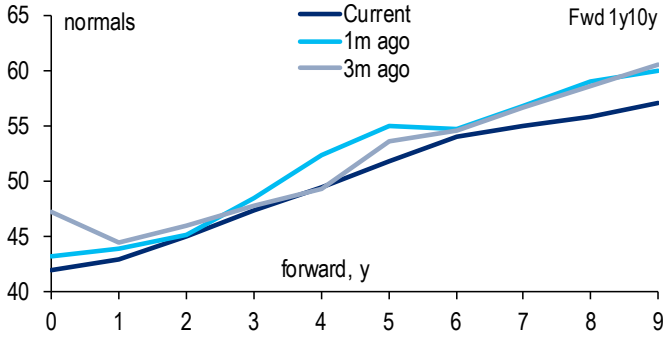
Figure 86. Forward vol of 1y5y



Note: Close of 1/16/20. Source: Citi Research

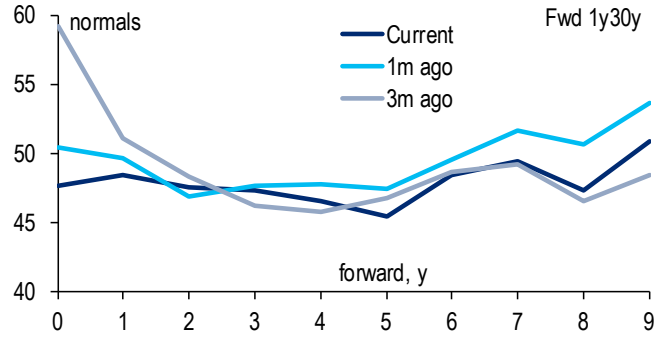
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Figure 87. Forward vol of 1y10y



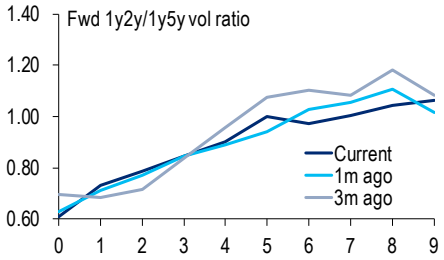
Note: Close of 1/16/20. Source: Citi Research

Figure 88. Forward vol of 1y30y



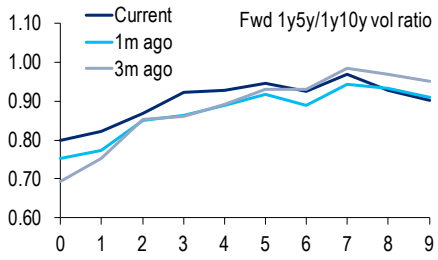
Note: Close of 1/16/20. Source: Citi Research

Figure 89. Fwd 1y2y/1y5y vol ratio



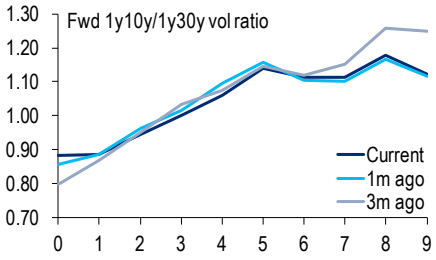
Note: Close of 1/16/20. Source: Citi Research

Figure 90. Fwd 1y5y/1y10y vol ratio



Note: Close of 1/16/20. Source: Citi Research

Figure 91. Fwd 1y10y/1y30y vol ratio



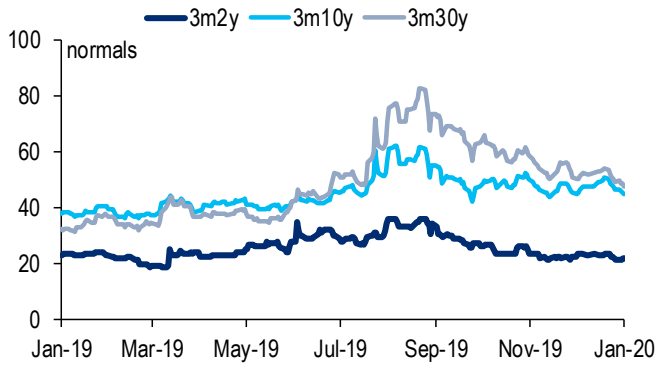
Note: Close of 1/16/20. Source: Citi Research

EUR swaption skew

We calculate z-scores of payer and receiver swaption skews across the surface.

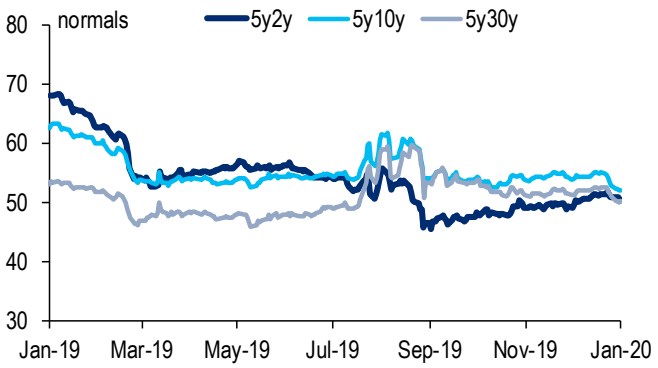
- Short-dated risk reversals have richened over the past two weeks and trade in the positive territory (except for the 30y vol; Figure 96).
- Longer-dated payer and receiver skews have been little changed (Figure 93 and Figure 95).

Figure 92. 3m expiry payer skew



Note: 25bp-out payer skews for 3m expiry payers. Source: Citi Research

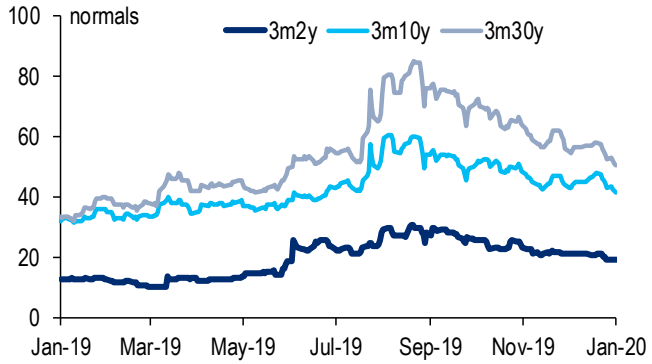
Figure 93. 5y expiry payer skew



Note: 100bp-out payer skews for 5y expiry payers. Source: Citi Research

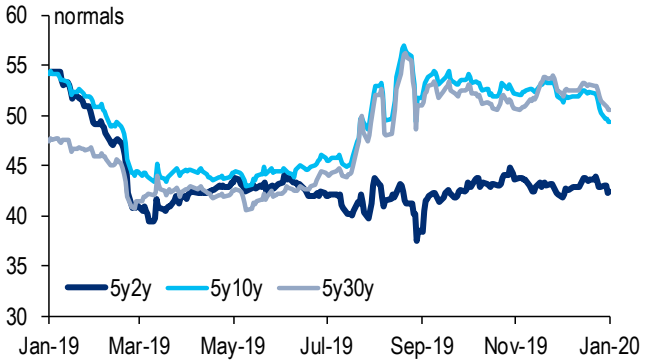
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Figure 94. 3m expiry receiver skew



Note: 25bp-out receiver skews for 3m expiry receivers. Source: Citi Research

Figure 95. 5y expiry receiver skew



Note: 100bp-out receiver skews for 5y expiry receivers. Source: Citi Research

Figure 96. Skews (1y z-scores) of risk reversals, OTM payer and OTM receivers

	Risk Reversals						Payer skew						Receiver skew					
expr\tail	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y
3m	2.3 (-0.8)	2.5 (-0.9)	4.6 (-0.3)	4.3 (0.0)	3.5 (0.6)	-2.7 (1.3)	19.1 (-0.7)	21.8 (-1.0)	37.1 (0.2)	41.1 (0.0)	44.9 (-0.1)	47.7 (-0.1)	16.8 (0.1)	19.2 (-0.1)	32.5 (0.3)	36.8 (0.0)	41.4 (-0.2)	50.4 (-0.3)
6m	3.4 (-0.7)	3.9 (-0.8)	5.9 (-0.2)	5.5 (0.0)	4.5 (0.3)	-3.7 (0.6)	26.0 (-0.7)	29.3 (-1.0)	43.4 (1.1)	47.0 (0.8)	50.3 (0.3)	52.0 (0.1)	22.6 (0.3)	25.4 (0.2)	37.4 (0.9)	41.5 (0.5)	45.9 (0.1)	55.7 (0.0)
1y	4.0 (-0.5)	4.5 (-0.5)	5.2 (-0.3)	4.3 (-0.3)	2.6 (-0.2)	-3.7 (-0.5)	26.1 (-1.1)	29.6 (-0.7)	41.3 (0.6)	44.4 (0.5)	47.2 (0.1)	49.5 (0.2)	22.1 (0.0)	25.2 (0.2)	36.0 (0.9)	40.1 (0.5)	44.6 (0.2)	53.1 (0.3)
2y	5.2 (-0.2)	5.3 (-0.3)	4.6 (-0.2)	3.5 (-0.4)	1.9 (-0.5)	-2.2 (-1.5)	30.8 (-0.4)	34.1 (-0.2)	42.1 (0.1)	44.8 (0.1)	46.6 (-0.3)	48.9 (0.2)	25.5 (-0.4)	28.8 (0.0)	37.5 (0.4)	41.3 (0.4)	44.7 (0.1)	51.1 (0.5)
3y	10.1 (-0.1)	9.3 (-0.3)	7.4 (-0.4)	5.6 (-0.7)	3.0 (-0.8)	-2.1 (-1.5)	45.2 (0.2)	45.9 (0.1)	49.9 (0.2)	50.7 (-0.1)	50.7 (-0.6)	50.7 (0.2)	35.2 (0.8)	36.6 (0.8)	42.6 (0.9)	45.1 (0.7)	47.8 (0.4)	52.8 (0.9)
5y	10.6 (0.0)	8.0 (-0.7)	6.0 (-0.8)	4.5 (-0.9)	2.7 (-1.0)	-0.4 (-1.5)	51.4 (-0.4)	50.4 (-0.7)	52.2 (-0.6)	52.2 (-0.9)	52.1 (-1.2)	50.2 (-0.3)	40.8 (-0.6)	42.4 (-0.3)	46.2 (0.0)	47.7 (0.0)	49.4 (0.1)	50.6 (0.7)
10y	7.6 (-0.1)	5.3 (-1.5)	4.1 (-1.5)	3.2 (-1.6)	1.9 (-1.6)	0.3 (-1.6)	56.7 (-1.4)	55.7 (-1.6)	55.4 (-1.5)	55.2 (-1.4)	55.0 (-1.4)	50.5 (-1.0)	49.1 (-1.3)	50.5 (-0.9)	51.3 (-0.6)	52.0 (-0.3)	53.1 (0.0)	50.2 (0.8)

Note: Close of 1/16/20. The table reports skews and 1y z-scores in parentheses for risk reversals, payer and receiver skews, defined in normal vol terms. The following OTM conventions are used: 25bp out for 3m expiries, 50bp out for 6m-2y expiries, 100bp out for 3y+ expiries. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

EUR midcurve swaptions

We report levels and z-scores of at-the-money vols, implied correlations, and implied/realized ratios of correlations. We also compute the relative richness/cheapness of midcurve vols to principal components of the plain vanilla surface.

- Consistent with the repricing of the plain vanilla surface, midcurves on 5y5y and 10y10y have cheapened notably (Figure 97). Midcurves on 1y2y may look rich judging by implied/realized ratio of correlations (Figure 99).

Figure 97. Implied vols, normals (3m z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	18.6 (-1.1)	32.2 (-0.5)	47.2 (-0.1)	24.3 (-0.9)	38.8 (-0.3)	49.6 (0.1)	50.8 (-1.4)	53.8 (-1.4)
3m	19.7 (-1.9)	33.8 (0.0)	49.2 (1.0)	25.8 (-1.4)	40.6 (0.6)	49.4 (0.6)	52.7 (-1.9)	55.5 (-1.9)
6m	22.5 (-1.9)	34.8 (0.5)	48.4 (1.0)	27.8 (-1.2)	40.6 (0.9)	49.3 (0.6)	54.2 (-2.1)	56.3 (-2.1)
1y	27.1 (0.0)	37.9 (0.6)	47.7 (1.2)	31.5 (0.4)	41.8 (1.0)	48.6 (0.8)	55.0 (-2.7)	57.2 (-2.2)
2y	34.0 (1.1)	43.4 (1.2)	47.8 (0.9)	37.7 (1.2)	44.6 (1.2)	48.6 (-0.3)	55.5 (-2.4)	56.2 (-1.7)

Close of 1/16/20. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 98. Implied correlations, % (3y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	77.8 (-0.6)	92.7 (-0.7)	96.9 (-1.2)	63.2 (-0.4)	84.8 (-1.0)	92.7 (-1.2)	92.2 (-0.2)	93.5 (-1.4)
3m	84.7 (-0.3)	94.4 (-1.0)	97.3 (-1.3)	73.2 (-0.2)	87.9 (-1.1)	93.5 (-1.3)	91.4 (-1.0)	93.1 (-1.7)
6m	86.4 (-0.7)	95.7 (-1.5)	97.7 (-2.2)	76.5 (-0.8)	90.9 (-1.5)	94.5 (-2.0)	90.9 (-1.2)	93.2 (-1.7)
1y	91.1 (-1.8)	97.0 (-2.2)	98.3 (-2.3)	85.3 (-1.4)	93.7 (-1.9)	95.7 (-2.1)	91.7 (-1.2)	93.6 (-1.7)
2y	95.9 (-1.5)	98.1 (-1.6)	98.9 (-1.7)	93.1 (-1.0)	96.0 (-1.5)	97.1 (-1.6)	92.9 (-1.4)	95.0 (-1.8)

Close of 1/16/20. The table shows implied correlations (3y z-scores in parentheses). Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

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Figure 99. Implied/realized correlation ratios (1y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	0.87 (-0.6)	0.95 (-0.9)	0.99 (-0.7)	0.78 (-1.0)	0.93 (-0.8)	0.98 (-0.9)	0.94 (-1.9)	0.95 (-1.9)
3m	0.93 (0.1)	0.97 (-1.0)	0.99 (-0.7)	0.85 (-0.8)	0.95 (-0.7)	0.97 (-0.8)	0.93 (-1.9)	0.94 (-1.9)
6m	0.92 (0.1)	0.98 (-0.6)	0.99 (-0.7)	0.85 (-0.9)	0.97 (-0.4)	0.97 (-1.0)	0.93 (-1.9)	0.94 (-1.6)
1y	0.93 (-1.7)	0.99 (-0.3)	0.99 (-1.2)	0.90 (-1.3)	0.98 (-0.4)	0.97 (-1.6)	0.93 (-1.6)	0.94 (-1.4)
2y	0.98 (-1.1)	0.99 (-1.5)	0.99 (-2.4)	0.97 (-0.9)	0.97 (-1.8)	0.98 (-2.4)	0.94 (-1.2)	0.96 (-1.2)

Close of 1/16/20. The table shows the ratios of implied to 3m realized correlations (in daily changes). The 1y z-scores of ratios are shown in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 100. Rich/cheap to plain vanilla swaptions (z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	0.1 (0.1)	-0.1 (0.0)	-1.0 (-1.0)	0.0 (0.0)	-0.2 (-0.3)	-0.8 (-1.1)	-0.5 (-0.5)	0.0 (0.0)
3m	0.4 (0.4)	0.2 (0.3)	-0.4 (-0.5)	0.2 (0.4)	0.0 (-0.1)	-0.4 (-0.7)	-0.1 (-0.2)	0.2 (0.3)
6m	0.2 (0.3)	0.3 (0.5)	-0.1 (-0.1)	0.2 (0.5)	0.1 (0.2)	-0.2 (-0.6)	0.0 (0.1)	0.3 (0.6)
1y	0.2 (0.5)	0.1 (0.2)	-0.1 (-0.2)	0.2 (0.6)	0.0 (-0.1)	0.0 (-0.2)	0.0 (0.0)	0.2 (0.4)
2y	0.2 (0.5)	-0.2 (-0.5)	-0.1 (-0.3)	0.2 (0.5)	-0.2 (-0.6)	-0.1 (-0.2)	0.0 (0.1)	0.1 (0.3)

Close of 1/16/20. We regress midcurve vols on two corresponding plain vanilla vols using last 3y of data. For example, 1y2y1y is regressed against 1y3y and 1y2y vols. We calculate rich/cheap and z-scores in brackets. Source: Citi Research

EUR curve (spread) CMS options

We report levels and z-scores of at-the-money vols, rich/cheap on PCA, implied correlations, and implied/realized vol ratios for curve options.

- EUR curve vols have cheapened notably, directionally with swaptions. Short-dated implied correlation have increased as well (Figure 102). The 10s30s vol looks very rich to recent realized vols (Figure 104).

Figure 101. Implied vols, normals (3m z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	21.7 (0.3)	30.8 (-1.3)	40.1 (-1.3)	14.0 (-2.0)	25.4 (-1.4)	14.9 (-1.3)
3m	20.5 (1.2)	30.6 (-2.0)	40.0 (-1.9)	14.0 (-2.4)	25.7 (-1.9)	15.3 (-1.7)
6m	19.6 (1.0)	29.7 (-2.5)	39.0 (-2.2)	14.4 (-2.5)	25.6 (-2.2)	15.5 (-1.9)
1y	17.6 (1.2)	28.3 (-3.3)	37.0 (-2.6)	14.6 (-2.8)	25.2 (-2.7)	15.6 (-2.1)
2y	16.6 (-0.1)	25.9 (-3.6)	34.1 (-2.9)	14.6 (-2.7)	25.4 (-2.8)	15.9 (-3.0)
3y	16.4 (-0.9)	24.8 (-2.9)	31.3 (-2.9)	14.7 (-3.1)	24.9 (-2.1)	16.0 (-3.2)
5y	16.3 (-1.9)	22.6 (-3.0)	29.3 (-4.0)	14.4 (-3.5)	25.0 (-3.0)	15.9 (-4.0)

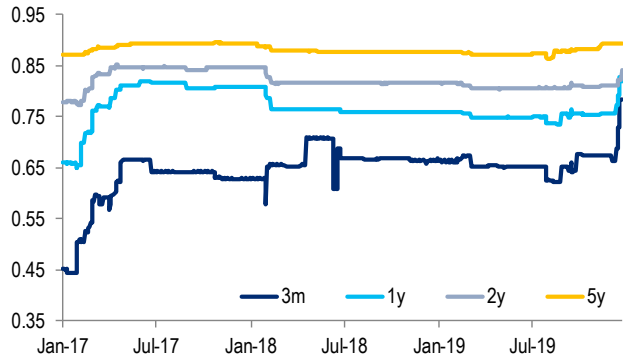
Close of 1/16/20. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 102. Rich/cheap to PCA, normals (z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	2.0 (0.7)	1.0 (0.4)	2.6 (1.1)	0.2 (0.1)	-2.9 (-1.4)	-3.6 (-1.6)
3m	3.2 (1.9)	3.2 (2.1)	5.2 (2.2)	0.8 (0.5)	-1.2 (-0.9)	-2.3 (-1.8)
6m	3.1 (2.7)	3.0 (2.0)	5.2 (2.2)	0.9 (0.7)	-0.4 (-0.4)	-1.7 (-1.8)
1y	2.5 (3.3)	2.1 (2.0)	4.3 (2.1)	0.8 (1.0)	-0.4 (-0.4)	-1.5 (-2.0)
2y	1.0 (1.7)	0.3 (0.5)	2.0 (1.4)	0.4 (0.9)	-0.9 (-1.2)	-1.3 (-1.7)
3y	0.2 (0.4)	-0.6 (-0.7)	0.0 (0.0)	0.1 (0.2)	-1.3 (-1.5)	-1.1 (-1.8)
5y	-0.2 (-0.3)	-3.0 (-2.7)	-2.7 (-3.3)	-0.3 (-1.1)	-2.7 (-3.1)	-1.3 (-1.4)

Close of 1/16/20. We calculate rich/cheap and z-scores in parentheses from PCA on curve vols using two principal components and 3y of data. By design, this framework only yields relative value. Source: Citi Research

Figure 103. Implied correlations of 2s10s CMS options



Note: The chart shows implied correlation of options on the 2s10s curve with different expirations. Source: Citi Research

Figure 104. Implied/realized ratios

implied/1m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	0.98 (-0.7)	0.95 (-0.6)	1.08 (0.0)	1.24 (1.1)	1.47 (1.2)	1.81 (1.3)
3m	0.95 (-0.4)	0.99 (-0.1)	1.13 (0.5)	1.33 (1.6)	1.59 (1.4)	1.93 (1.6)
6m	0.92 (-0.4)	1.01 (0.1)	1.19 (0.9)	1.53 (2.0)	1.77 (1.8)	2.02 (1.5)
1y	0.92 (-0.2)	1.14 (0.5)	1.36 (1.2)	1.94 (2.4)	2.12 (2.1)	2.22 (1.7)

implied/3m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.00 (-0.8)	0.85 (-1.1)	0.88 (-1.0)	0.87 (-0.8)	0.94 (-0.7)	1.18 (-0.1)
3m	0.94 (-0.6)	0.87 (-1.0)	0.90 (-0.8)	0.92 (-0.5)	0.99 (-0.4)	1.24 (0.3)
6m	0.90 (-0.6)	0.87 (-0.8)	0.92 (-0.5)	1.01 (-0.1)	1.06 (0.0)	1.33 (0.6)
1y	0.85 (-0.5)	0.91 (-0.5)	0.97 (-0.1)	1.18 (0.6)	1.20 (0.9)	1.51 (1.2)

Close of 1/16/20. Close-to-close realized vol is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

EUR conditional curves

We analyze conditional trades based on the entry point, implied/realized betas, and carry.

Prepared for Christopher Lee

Figure 105. Conditional curves: bear trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bear steepener				bear flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/49.9	9.2	9.9	0.9	11.4	0.5	0.7	30	0.27	0.45	0.18	-0.8	-1.5	-2.3	0.0	0.8	1.5	2.3	0.0
3m 1y3y	100/33.2	13.3	14.5	0.8	19.1	1.8	1.1	32	0.62	0.83	0.21	-1.2	-4.7	-5.9	0.0	1.2	4.7	5.9	0.0
3m 1y5y	100/19.8	23.5	25.2	0.6	37.6	4.1	1.1	31	1.05	1.25	0.20	-1.7	-12.4	-14.1	0.0	1.7	12.4	14.1	0.0
3m 2y5y	100/39.8	14.4	15.3	0.4	24.6	3.2	1.1	30	0.77	0.86	0.09	-0.9	-9.4	-10.3	0.0	0.9	9.4	10.3	0.0
3m 2y10y	100/19.9	44.5	45.8	0.2	61.5	4.1	0.4	27	1.02	1.15	0.13	-1.3	-15.7	-17.0	0.0	1.3	15.7	17.0	0.0
3m 2y30y	100/7.1	88.1	87.4	0.6	105.9	5.2	0.2	28	1.15	1.29	0.14	0.7	-18.5	-17.8	0.0	-0.7	18.5	17.8	0.0
3m 3y5y	100/59.8	10.2	10.7	0.3	16.1	1.5	0.9	28	0.44	0.49	0.05	-0.5	-5.4	-5.9	0.0	0.5	5.4	5.9	0.0
3m 3y10y	100/29.9	40.4	41.3	0.0	52.0	2.2	-0.2	25	0.70	0.81	0.11	-0.9	-10.7	-11.6	0.0	0.9	10.7	11.6	0.0
3m 3y30y	100/10.6	84.0	82.9	0.6	96.4	3.5	-0.4	28	0.84	0.97	0.13	1.1	-13.5	-12.4	0.0	-1.1	13.5	12.4	0.0
3m 5y10y	100/50.1	30.1	30.5	-0.2	34.7	0.7	-1.3	24	0.26	0.33	0.07	-0.4	-4.1	-4.5	0.0	0.4	4.1	4.5	0.0
3m 5y30y	100/17.7	73.7	72.2	0.7	78.9	2.2	-0.8	34	0.41	0.51	0.10	1.6	-6.7	-5.1	0.0	-1.6	6.7	5.1	0.0
3m 10y30y	100/35.4	43.6	41.6	1.7	44.0	1.7	-0.1	50	0.14	0.17	0.03	2.0	-2.4	-0.4	0.0	-2.0	2.4	0.4	0.0
6m 1y2y	100/49.9	9.2	10.8	0.5	13.4	0.5	0.8	37	0.28	0.42	0.14	-1.6	-2.6	-4.2	-0.5	1.6	2.6	4.2	0.5
6m 1y3y	100/33.2	13.3	15.7	0.5	22.6	1.8	1.0	37	0.57	0.75	0.18	-2.4	-6.9	-9.3	-0.6	2.4	6.9	9.3	0.6
6m 1y5y	100/19.9	23.5	26.8	0.3	43.7	4.9	1.0	36	0.95	1.13	0.18	-3.3	-16.9	-20.2	-0.6	3.3	16.9	20.2	0.6
6m 2y5y	100/39.8	14.4	16.0	0.2	28.0	3.6	1.0	34	0.67	0.76	0.09	-1.6	-12.0	-13.7	-0.3	1.6	12.0	13.7	0.3
6m 2y10y	100/19.9	44.5	46.9	0.0	67.5	4.4	0.2	29	0.91	1.01	0.10	-2.4	-20.6	-23.0	-0.4	2.4	20.6	23.0	0.4
6m 2y30y	100/7.1	88.1	86.4	0.5	110.7	5.2	-0.1	29	1.05	1.15	0.10	1.7	-24.3	-22.6	-0.3	-1.7	24.3	22.6	0.3
6m 3y5y	100/59.8	10.2	11.1	0.1	18.1	1.9	0.8	30	0.38	0.42	0.04	-0.9	-6.9	-7.8	-0.2	0.9	6.9	7.8	0.2
6m 3y10y	100/30.0	40.4	42.0	-0.1	56.3	2.4	-0.4	27	0.64	0.70	0.06	-1.6	-14.4	-16.0	-0.5	1.6	14.4	16.0	0.5
6m 3y30y	100/10.6	84.0	81.6	0.5	99.5	3.5	-0.7	29	0.77	0.85	0.07	2.4	-18.0	-15.6	-0.3	-2.4	18.0	15.6	0.3
6m 5y10y	100/50.1	30.1	30.9	-0.3	36.7	0.5	-1.4	25	0.26	0.28	0.03	-0.7	-5.9	-6.6	-0.5	0.7	5.9	6.6	0.5
6m 5y30y	100/17.8	73.7	70.4	0.7	79.8	1.8	-0.9	34	0.39	0.44	0.04	3.3	-9.4	-6.1	-0.1	-3.3	9.4	6.1	0.1
6m 10y30y	100/35.5	43.6	39.6	1.7	42.9	1.5	-0.2	48	0.14	0.15	0.01	4.0	-3.3	0.7	0.5	-4.0	3.3	-0.7	-0.5
1y 1y2y	100/49.9	9.2	12.0	-0.1	16.2	0.6	0.8	45	0.25	0.35	0.10	-2.9	-4.2	-7.1	-0.3	2.9	4.2	7.1	0.3
1y 1y3y	100/33.2	13.3	17.4	-0.1	27.3	1.4	0.9	43	0.47	0.62	0.15	-4.2	-9.8	-14.0	-0.1	4.2	9.8	14.0	0.1
1y 1y5y	100/19.9	23.5	29.1	-0.1	49.6	3.6	0.9	41	0.75	0.90	0.15	-5.6	-20.5	-26.0	0.1	5.6	20.5	26.0	-0.1
1y 2y5y	100/39.9	14.4	17.1	-0.1	30.8	2.0	0.8	37	0.50	0.57	0.06	-2.7	-13.7	-16.4	0.3	2.7	13.7	16.4	-0.3
1y 2y10y	100/20.0	44.5	48.3	-0.3	71.8	0.4	-0.7	31	0.73	0.77	0.04	-3.8	-23.5	-27.3	0.3	3.8	23.5	27.3	-0.3
1y 2y30y	100/7.1	88.1	83.6	0.4	112.0	1.4	-0.7	29	0.85	0.88	0.03	4.5	-28.4	-23.9	0.7	-4.5	28.4	23.9	-0.7
1y 3y5y	100/60.0	10.2	11.6	-0.2	19.4	1.3	0.7	32	0.28	0.30	0.02	-1.4	-7.7	-9.1	0.1	1.4	7.7	9.1	-0.1
1y 3y10y	100/30.1	40.4	42.8	-0.3	59.4	0.1	-1.4	28	0.50	0.50	0.00	-2.5	-16.5	-19.0	0.1	2.5	16.5	19.0	-0.1
1y 3y30y	100/10.7	84.0	78.2	0.5	99.6	1.1	-1.0	29	0.63	0.62	-0.01	5.8	-21.4	-15.6	0.6	-5.8	21.4	15.6	-0.6
1y 5y10y	100/50.2	30.1	31.2	-0.4	38.8	-0.8	-1.9	26	0.23	0.21	-0.02	-1.1	-7.6	-8.6	-0.1	1.1	7.6	8.6	0.1
1y 5y30y	100/17.8	73.7	66.6	0.7	78.8	0.2	-1.1	34	0.36	0.32	-0.03	7.2	-12.3	-5.1	0.5	-7.2	12.3	5.1	-0.5
1y 10y30y	100/35.5	43.6	35.4	1.9	39.9	1.1	0.0	45	0.13	0.11	-0.01	8.2	-4.5	3.7	0.8	-8.2	4.5	-3.7	-0.8
2y 1y2y	100/49.8	9.2	13.3	-0.7	18.2	0.1	0.3	50	0.15	0.22	0.07	-4.1	-4.9	-9.0	0.1	4.1	4.9	9.0	-0.1
2y 1y3y	100/33.3	13.3	19.2	-0.4	30.3	1.0	0.6	47	0.29	0.37	0.09	-5.9	-11.1	-17.0	0.1	5.9	11.1	17.0	-0.1
2y 1y5y	100/19.9	23.5	31.5	-0.6	50.5	1.2	0.1	42	0.44	0.49	0.05	-8.0	-19.0	-27.0	0.4	8.0	19.0	27.0	-0.4
2y 2y5y	100/40.1	14.4	18.2	-0.4	31.1	1.0	-0.1	37	0.29	0.27	-0.02	-3.9	-12.8	-16.7	0.3	3.9	12.8	16.7	-0.3
2y 2y10y	100/20.2	44.5	49.2	-0.4	70.9	-1.5	-1.9	31	0.45	0.38	-0.07	-4.7	-21.8	-26.4	0.6	4.7	21.8	26.4	-0.6
2y 2y30y	100/7.2	88.1	75.8	0.5	102.8	-0.3	-1.1	27	0.55	0.43	-0.12	12.2	-26.9	-14.7	1.4	-12.2	26.9	14.7	-1.4
2y 3y5y	100/60.0	10.2	12.3	-0.6	19.0	0.2	-1.1	31	0.15	0.12	-0.03	-2.1	-6.6	-8.7	0.2	2.1	6.6	8.7	-0.2
2y 3y10y	100/30.2	40.4	43.2	-0.5	58.3	-1.9	-2.1	28	0.31	0.22	-0.09	-2.9	-15.1	-18.0	0.6	2.9	15.1	18.0	-0.6
2y 3y30y	100/10.8	84.0	69.9	0.6	90.2	-0.7	-1.2	26	0.41	0.27	-0.13	14.0	-20.2	-6.2	1.4	-14.0	20.2	6.2	-1.4
2y 5y10y	100/50.3	30.1	30.9	-0.3	38.8	-1.6	-1.7	26	0.16	0.10	-0.06	-0.8	-7.9	-8.7	0.4	0.8	7.9	8.7	-0.4
2y 5y30y	100/17.9	73.7	57.6	1.0	70.6	-0.4	-0.8	28	0.26	0.15	-0.11	16.1	-13.0	3.1	1.3	-16.1	13.0	-3.1	-1.3
2y 10y30y	100/35.6	43.6	26.7	2.2	31.7	1.3	0.6	37	0.09	0.05	-0.05	16.9	-5.0	11.9	1.1	-16.9	5.0	-11.9	-1.1
3y 1y2y	100/50.2	9.2	14.0	0.0	18.6	1.1	0.6	51	0.08	0.11	0.03	-4.8	-4.7	-9.5	0.1	4.8	4.7	9.5	-0.1
3y 1y3y	100/33.5	13.3	20.1	-0.2	28.8	0.9	0.0	45	0.15	0.15	0.00	-6.9	-8.6	-15.5	0.4	6.9	8.6	15.5	-0.4
3y 1y5y	100/20.1	23.5	33.2	-0.5	48.3	0.8	-0.8	40	0.25	0.19	-0.06	-9.6	-15.1	-24.7	0.6	9.6	15.1	24.7	-0.6
3y 2y5y	100/40.0	14.4	19.2	-0.7	29.1	-0.3	-1.2	35	0.17	0.08	-0.09	-4.8	-9.9	-14.7	0.4	4.8	9.9	14.7	-0.4
3y 2y10y	100/20.2	44.5	48.6	-0.4	65.6	-2.4	-1.7	28	0.27	0.12	-0.15	-4.1	-16.9	-21.1	0.9	4.1	16.9	21.1	-0.9
3y 2y30y	100/7.2	88.1	66.5	0.7	87.0	-0.4	-0.9	24	0.33	0.14	-0.20	21.6	-20.5	1.1	1.9	-21.6	20.5	-1.1	-1.9
3y 3y5y	100/60.1	10.2	13.0	-0.7	18.9	0.1	-1.1	31	0.10	0.04	-0.06	-2.8	-5.9	-8.7	0.2	2.8	5.9	8.7	-0.2
3y 3y10y	100/30.3	40.4	42.4	-0.3	55.3	-1.9	-1.7	26	0.20	0.09	-0.12	-2.1	-12.9	-14.9	0.6	2.1	12.9	14.9	-0.6
3y 3y30y	100/10.8	84.0	60.4	0.9	76.8	0.1	-0.8	22	0.26	0.10	-0.16	23.6	-16.4	7.2	1.7	-23.6	16.4	-7.2	-1.7
3y 5y10y	100/50.5	30.1	29.4	0.0	36.2	-1.6	-1.7	25	0.11	0.05	-0.06	0.7	-6.8	-6.0	0.5	-0.7	6.8	6.0	-0.5
3y 5y30y	100/18.0	73.7	47.3	1.5	57.7	0.3	-0.4	21	0.16	0.06	-0.10	26.4	-10.4	16.0	1.6	-26.4	10.4	-16.0	-1.6
3y 10y30y	100/35.7	43.6	17.9	2.5	21.6	2.0	1.4	22	0.06	0.01	-0.04	25.7	-3.7	22.0	1.3	-25.7	3.7	-22.0	-1.3

Note: Close of 1/16/20. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m carry is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best bear steepeners in red, and three best bear flatteners in green. Source: Citi Research

Prepared for Christopher Lee

Figure 106. Conditional curves: bull trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bull steepener				bull flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/49.9	9.2	9.9	0.9	8.6	1.0	1.3	25	0.27	0.45	0.18	-0.8	1.3	0.6	0.5	0.8	-1.3	-0.6	-0.5
3m 1y3y	100/33.2	13.3	14.5	0.8	10.3	0.5	0.6	21	0.62	0.83	0.21	-1.2	4.1	3.0	0.6	1.2	-4.1	-3.0	-0.6
3m 1y5y	100/19.8	23.5	25.2	0.6	14.7	-0.3	-0.1	19	1.05	1.25	0.20	-1.7	10.5	8.9	0.6	1.7	-10.5	-8.9	-0.6
3m 2y5y	100/39.8	14.4	15.3	0.4	7.2	-1.0	-0.5	17	0.77	0.86	0.09	-0.9	8.1	7.2	1.4	0.9	-8.1	-7.2	-1.4
3m 2y10y	100/19.9	44.5	45.8	0.2	31.8	1.5	0.4	17	1.02	1.15	0.13	-1.3	14.0	12.7	1.4	1.3	-14.0	-12.7	-1.4
3m 2y30y	100/7.1	88.1	87.4	0.6	67.1	6.4	1.3	18	1.15	1.29	0.14	0.7	20.3	21.0	1.4	-0.7	-20.3	-21.0	-1.4
3m 3y5y	100/59.8	10.2	10.7	0.3	6.0	-0.2	-0.4	17	0.44	0.49	0.05	-0.5	4.7	4.2	1.8	0.5	-4.7	-4.2	-1.8
3m 3y10y	100/29.9	40.4	41.3	0.0	31.6	2.5	0.7	17	0.70	0.81	0.11	-0.9	9.7	8.8	1.8	0.9	-9.7	-8.8	-1.8
3m 3y30y	100/10.6	84.0	82.9	0.6	68.3	7.5	1.5	20	0.84	0.97	0.13	1.1	14.6	15.7	1.8	-1.1	-14.6	-15.7	-1.8
3m 5y10y	100/50.1	30.1	30.5	-0.2	26.7	2.6	1.2	19	0.26	0.33	0.07	-0.4	3.8	3.4	2.3	0.4	-3.8	-3.4	-2.3
3m 5y30y	100/17.7	73.7	72.2	0.7	65.0	7.2	1.8	24	0.41	0.51	0.10	1.6	7.2	8.8	2.3	-1.6	-7.2	-8.8	-2.3
3m 10y30y	100/35.4	43.6	41.6	1.7	39.0	4.0	2.1	43	0.14	0.17	0.03	2.0	2.6	4.6	2.7	-2.0	-2.6	-4.6	-2.7
6m 1y2y	100/49.9	9.2	10.8	0.5	8.5	0.6	0.5	25	0.28	0.42	0.14	-1.6	2.3	0.6	0.1	1.6	-2.3	-0.6	-0.1
6m 1y3y	100/33.2	13.3	15.7	0.5	9.7	-0.2	-0.1	21	0.57	0.75	0.18	-2.4	6.0	3.6	0.4	2.4	-6.0	-3.6	-0.4
6m 1y5y	100/19.9	23.5	26.8	0.3	12.5	-1.5	-0.7	18	0.95	1.13	0.18	-3.3	14.3	11.0	0.8	3.3	-14.3	-11.0	-0.8
6m 2y5y	100/39.8	14.4	16.0	0.2	5.6	-1.6	-0.9	17	0.67	0.76	0.09	-1.6	10.3	8.7	0.9	1.6	-10.3	-8.7	-0.9
6m 2y10y	100/19.9	44.5	46.9	0.0	28.4	1.1	0.4	17	0.91	1.01	0.10	-2.4	18.5	16.1	1.4	2.4	-18.5	-16.1	-1.4
6m 2y30y	100/7.1	88.1	86.4	0.5	59.4	6.4	1.6	18	1.05	1.15	0.10	1.7	27.1	28.7	1.8	-1.7	-27.1	-28.7	-1.8
6m 3y5y	100/59.8	10.2	11.1	0.1	5.0	-0.4	-0.6	17	0.38	0.42	0.04	-0.9	6.1	5.2	0.7	0.9	-6.1	-5.2	-0.7
6m 3y10y	100/30.0	40.4	42.0	-0.1	28.9	2.4	0.8	17	0.64	0.70	0.06	-1.6	13.0	11.4	1.3	1.6	-13.0	-11.4	-1.3
6m 3y30y	100/10.6	84.0	81.6	0.5	61.7	7.7	1.7	18	0.77	0.85	0.07	2.4	19.9	22.3	2.1	-2.4	-19.9	-22.3	-2.1
6m 5y10y	100/50.1	30.1	30.9	-0.3	25.4	2.8	1.3	18	0.26	0.28	0.03	-0.7	5.4	4.7	0.9	0.7	-5.4	-4.7	-0.9
6m 5y30y	100/17.8	73.7	70.4	0.7	60.3	7.6	1.9	22	0.39	0.44	0.04	3.3	10.2	13.5	2.1	-3.3	-10.2	-13.5	-2.1
6m 10y30y	100/35.5	43.6	39.6	1.7	36.0	4.1	2.2	41	0.14	0.15	0.01	4.0	3.6	7.6	1.6	-4.0	-3.6	-7.6	-1.6
1y 1y2y	100/49.9	9.2	12.0	-0.1	8.3	-1.1	-1.1	25	0.25	0.35	0.10	-2.9	3.7	0.8	0.3	2.9	-3.7	-0.8	-0.3
1y 1y3y	100/33.2	13.3	17.4	-0.1	9.0	-1.7	-1.1	20	0.47	0.62	0.15	-4.2	8.4	4.3	0.3	4.2	-8.4	-4.3	-0.3
1y 1y5y	100/19.9	23.5	29.1	-0.1	11.8	-2.0	-1.3	18	0.75	0.90	0.15	-5.6	17.3	11.8	0.5	5.6	-17.3	-11.8	-0.5
1y 2y5y	100/39.9	14.4	17.1	-0.1	5.2	-0.4	-0.8	17	0.50	0.57	0.06	-2.7	11.9	9.2	0.4	2.7	-11.9	-9.2	-0.4
1y 2y10y	100/20.0	44.5	48.3	-0.3	26.4	4.1	0.8	16	0.73	0.77	0.04	-3.8	21.9	18.1	0.7	3.8	-21.9	-18.1	-0.7
1y 2y30y	100/7.1	88.1	83.6	0.4	51.7	9.8	1.7	17	0.85	0.88	0.03	4.5	31.9	36.4	1.3	-4.5	-31.9	-36.4	-1.3
1y 3y5y	100/60.0	10.2	11.6	-0.2	4.8	0.1	-0.8	17	0.28	0.30	0.02	-1.4	6.8	5.4	0.3	1.4	-6.8	-5.4	-0.3
1y 3y10y	100/30.1	40.4	42.8	-0.3	27.3	4.3	1.1	16	0.50	0.50	0.00	-2.5	15.5	13.1	0.8	2.5	-15.5	-13.1	-0.8
1y 3y30y	100/10.7	84.0	78.2	0.5	54.4	9.8	2.0	17	0.63	0.62	-0.01	5.8	23.8	29.6	1.6	-5.8	-23.8	-29.6	-1.6
1y 5y10y	100/50.2	30.1	31.2	-0.4	24.0	3.9	1.7	18	0.23	0.21	-0.02	-1.1	7.2	6.1	0.6	1.1	-7.2	-6.1	-0.6
1y 5y30y	100/17.8	73.7	66.6	0.7	53.1	8.9	2.3	20	0.36	0.32	-0.03	7.2	13.5	20.6	1.7	-7.2	-13.5	-20.6	-1.7
1y 10y30y	100/35.5	43.6	35.4	1.9	30.5	4.2	2.4	35	0.13	0.11	-0.01	8.2	4.9	13.1	1.4	-8.2	-4.9	-13.1	-1.4
2y 1y2y	100/49.8	9.2	13.3	-0.7	9.0	-0.6	-1.1	26	0.15	0.22	0.07	-4.1	4.2	0.1	0.0	4.1	-4.2	-0.1	0.0
2y 1y3y	100/33.3	13.3	19.2	-0.4	9.8	-0.6	-0.9	21	0.29	0.37	0.09	-5.9	9.4	3.5	0.2	5.9	-9.4	-3.5	-0.2
2y 1y5y	100/19.9	23.5	31.5	-0.6	15.1	0.9	-0.4	19	0.44	0.49	0.05	-8.0	16.4	8.4	0.1	8.0	-16.4	-8.4	-0.1
2y 2y5y	100/40.1	14.4	18.2	-0.4	7.1	1.6	0.1	17	0.29	0.27	-0.02	-3.9	11.1	7.2	0.2	3.9	-11.1	-7.2	-0.2
2y 2y10y	100/20.2	44.5	49.2	-0.4	28.7	6.8	1.3	17	0.45	0.38	-0.07	-4.7	20.5	15.8	0.2	4.7	-20.5	-15.8	-0.2
2y 2y30y	100/7.2	88.1	75.8	0.5	47.0	10.8	1.6	17	0.55	0.43	-0.12	12.2	28.8	41.1	0.9	-12.2	-28.8	-41.1	-0.9
2y 3y5y	100/60.0	10.2	12.3	-0.6	6.5	1.4	0.5	17	0.15	0.12	-0.03	-2.1	5.9	3.8	0.0	2.1	-5.9	-3.8	0.0
2y 3y10y	100/30.2	40.4	43.2	-0.5	28.9	6.3	1.7	17	0.31	0.22	-0.09	-2.9	14.3	11.5	0.1	2.9	-14.3	-11.5	-0.1
2y 3y30y	100/10.8	84.0	69.9	0.6	48.3	10.4	2.0	17	0.41	0.27	-0.13	14.0	21.7	35.7	0.9	-14.0	-21.7	-35.7	-0.9
2y 5y10y	100/50.3	30.1	30.9	-0.3	23.4	4.5	1.8	17	0.16	0.10	-0.06	-0.8	7.5	6.8	0.2	0.8	-7.5	-6.8	-0.2
2y 5y30y	100/17.9	73.7	57.6	1.0	43.8	8.3	2.1	17	0.26	0.15	-0.11	16.1	13.8	29.9	1.1	-16.1	-13.8	-29.9	-1.1
2y 10y30y	100/35.6	43.6	26.7	2.2	21.4	3.3	2.1	22	0.09	0.05	-0.05	16.9	5.2	22.2	1.1	-16.9	-5.2	-22.2	-1.1
3y 1y2y	100/50.2	9.2	14.0	0.0	10.2	0.3	0.0	27	0.08	0.11	0.03	-4.8	3.8	-1.0	-0.1	4.8	-3.8	1.0	0.1
3y 1y3y	100/33.5	13.3	20.1	-0.2	12.9	1.4	0.3	24	0.15	0.15	0.00	-6.9	7.2	0.3	-0.3	6.9	-7.2	-0.3	0.3
3y 1y5y	100/20.1	23.5	33.2	-0.5	20.1	3.1	0.6	21	0.25	0.19	-0.06	-9.6	13.0	3.4	-0.3	9.6	-13.0	-3.4	0.3
3y 2y5y	100/40.0	14.4	19.2	-0.7	10.5	2.8	0.9	19	0.17	0.08	-0.09	-4.8	8.7	3.9	-0.2	4.8	-8.7	-3.9	0.2
3y 2y10y	100/20.2	44.5	48.6	-0.4	32.4	7.2	1.8	17	0.27	0.12	-0.15	-4.1	16.2	12.1	-0.1	4.1	-16.2	-12.1	0.1
3y 2y30y	100/7.2	88.1	66.5	0.7	45.1	10.1	1.9	17	0.33	0.14	-0.20	21.6	21.4	43.0	0.6	-21.6	-21.4	-43.0	-0.6
3y 3y5y	100/60.1	10.2	13.0	-0.7	7.8	1.6	0.9	18	0.10	0.04	-0.06	-2.8	5.2	2.4	-0.1	2.8	-5.2	-2.4	0.1
3y 3y10y	100/30.3	40.4	42.4	-0.3	30.1	5.8	1.9	17	0.20	0.09	-0.12	-2.1	12.3	10.2	0.1	2.1	-12.3	-10.2	-0.1
3y 3y30y	100/10.8	84.0	60.4	0.9	43.2	8.5	2.1	16	0.26	0.10	-0.16	23.6	17.2	40.8	0.9	-23.6	-17.2	-40.8	-0.9
3y 5y10y	100/50.5	30.1	29.4	0.0	23.0	3.9	2.1	17	0.11	0.05	-0.06	0.7	6.5	7.2	0.2	-0.7	-6.5	-7.2	-0.2
3y 5y30y	100/18.0	73.7	47.3	1.5	36.6	6.5	2.2	16	0.16	0.06	-0.10	26.4	10.8	37.2	1.0	-26.4	-10.8	-37.2	-1.0
3y 10y30y	100/35.7	43.6	17.9	2.5	14.2	2.3	2.0	15	0.06	0.01	-0.04	25.7	3.7	29.4	0.9	-25.7	-3.7	-29.4	-0.9

Note: Close of 1/16/20. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best bull steepeners in red, and three best bull flatteners in green. Source: Citi Research

Prepared for Christopher Lee

EUR conditional flies

We analyze conditional fly trades based on the entry point, directionality with rates and carry.

Figure 107. Conditional payer flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long payer in the belly				short payer in the belly			
curve	notionals, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, ZC	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	150.6/100/29.9	1.5	1.9	1.3	16	63	1.0	1.5	-0.4	1.5	0.5	1.0	53	-0.3	0.4	0.0	0.0	0.3	-0.4	0.0	0.0
3m 1y3y7y	150.6/100/21.4	-4.3	-4.0	1.1	-34	-63	3.1	1.8	-1.3	-5.3	0.6	1.2	66	-0.2	1.3	1.0	0.0	0.2	-1.3	-1.0	0.0
3m 2y3y5y	75.1/100/29.9	-3.1	-3.1	0.5	-55	-81	1.2	1.7	-1.0	-4.1	0.3	0.1	73	0.0	1.0	1.1	0.0	0.0	-1.0	-1.1	0.0
3m 2y5y7y	125.7/100/35.7	1.4	1.7	0.6	93	96	0.4	0.8	2.7	4.4	0.8	1.3	39	-0.3	-2.7	-3.0	0.0	0.3	2.7	3.0	0.0
3m 2y5y10y	125.7/100/25.0	-7.9	-7.6	0.6	71	-27	2.8	1.9	1.6	-6.0	0.6	1.4	43	-0.2	-1.6	-1.9	0.0	0.2	1.6	1.9	0.0
3m 2y10y30y	251.0/100/17.7	0.5	2.1	-0.5	94	95	-1.1	-0.7	4.5	6.6	0.1	0.3	23	-1.6	-4.5	-6.1	0.0	1.6	4.5	6.1	0.0
3m 3y5y7y	83.6/100/35.7	-0.7	-0.5	0.5	81	77	0.6	1.6	1.4	0.8	0.3	1.3	44	-0.1	-1.4	-1.5	0.0	0.1	1.4	1.5	0.0
3m 3y5y10y	83.6/100/25.0	-9.9	-9.9	0.6	22	-76	2.9	1.9	0.4	-9.5	0.2	1.4	59	0.0	-0.4	-0.4	0.0	0.0	0.4	0.4	0.0
3m 3y7y10y	117.1/100/35.0	1.6	1.9	0.1	93	96	-0.3	-0.5	2.4	4.3	0.0	0.8	26	-0.3	-2.4	-2.7	0.0	0.3	2.4	2.7	0.0
3m 3y7y15y	117.1/100/23.6	-11.2	-10.6	0.0	82	29	1.2	1.0	1.5	-9.1	0.2	1.0	24	-0.6	-1.5	-2.1	0.0	0.6	1.5	2.1	0.0
3m 3y10y30y	167.0/100/17.7	-1.6	-0.2	-1.0	90	93	-1.3	-0.9	3.1	2.9	-0.4	-0.1	20	-1.4	-3.1	-4.5	0.0	1.4	3.1	4.5	0.0
3m 5y10y15y	99.9/100/33.7	2.3	2.8	-0.6	87	94	-1.6	-1.8	1.1	3.9	0.2	-1.0	18	-0.5	-1.1	-1.6	0.0	0.5	1.1	1.6	0.0
3m 5y10y30y	99.9/100/17.7	-6.7	-5.5	-2.3	61	68	-0.8	-0.6	0.7	-4.9	-0.6	-1.5	10	-1.2	-0.7	-1.8	0.0	1.2	0.7	1.8	0.0
3m 10y20y30y	97.2/100/34.4	17.4	17.2	0.2	70	97	-0.8	-1.2	1.0	18.1	-0.7	-1.5	53	0.3	-1.0	-0.7	0.0	-0.3	1.0	0.7	0.0
6m 1y3y5y	150.6/100/29.9	1.5	2.3	0.8	33	62	0.7	1.2	-0.2	2.1	0.2	0.8	56	-0.8	0.2	-0.5	-0.3	0.8	-0.2	0.5	0.3
6m 1y3y7y	150.6/100/21.4	-4.3	-3.8	0.8	-25	-65	2.9	1.7	-1.5	-5.3	0.3	1.2	66	-0.5	1.5	1.0	-0.1	0.5	-1.5	-1.0	0.1
6m 2y3y5y	75.1/100/29.9	-3.1	-3.1	0.4	-59	-82	1.2	1.7	-1.3	-4.4	0.0	0.0	71	0.1	1.3	1.4	0.0	-0.1	-1.3	-1.4	0.0
6m 2y5y7y	125.6/100/35.7	1.4	2.0	0.4	92	96	0.1	0.1	3.5	5.4	0.8	1.2	42	-0.5	-3.5	-4.0	-0.2	0.5	3.5	4.0	0.2
6m 2y5y10y	125.6/100/25.0	-7.9	-7.4	0.5	73	-26	2.4	1.8	1.9	-5.5	0.7	1.3	45	-0.5	-1.9	-2.4	-0.1	0.5	1.9	2.4	0.1
6m 2y10y30y	250.7/100/17.7	0.5	3.6	-0.7	94	96	-1.7	-1.1	6.0	9.6	0.2	0.2	26	-3.2	-6.0	-9.2	-0.9	3.2	6.0	9.2	0.9
6m 3y5y7y	83.6/100/35.7	-0.7	-0.5	0.4	82	81	0.4	1.3	1.7	1.2	0.4	1.2	47	-0.2	-1.7	-1.9	0.0	0.2	1.7	1.9	0.0
6m 3y5y10y	83.6/100/25.0	-9.9	-9.9	0.5	28	-76	2.7	1.9	0.3	-9.6	0.3	1.4	59	-0.1	-0.3	-0.3	0.1	0.1	0.3	0.3	-0.1
6m 3y7y10y	117.0/100/35.1	1.6	2.2	0.0	93	96	-0.6	-0.9	3.1	5.3	0.2	0.7	28	-0.6	-3.1	-3.7	-0.2	0.6	3.1	3.7	0.2
6m 3y7y15y	117.0/100/23.7	-11.2	-10.0	-0.2	82	40	0.8	0.8	1.9	-8.1	0.5	1.0	26	-1.2	-1.9	-3.1	-0.3	1.2	1.9	3.1	0.3
6m 3y10y30y	166.8/100/17.7	-1.6	1.2	-1.2	90	94	-1.7	-1.2	4.1	5.3	-0.3	-0.2	24	-2.8	-4.1	-6.9	-0.8	2.8	4.1	6.9	0.8
6m 5y10y15y	99.8/100/33.7	2.3	3.2	-0.7	85	93	-1.8	-1.9	1.6	4.8	0.2	-1.2	20	-0.9	-1.6	-2.6	-0.3	0.9	1.6	2.6	0.3
6m 5y10y30y	99.8/100/17.7	-6.7	-4.4	-2.4	60	73	-1.0	-0.8	1.0	-3.4	-0.6	-1.8	12	-2.4	-1.0	-3.3	-0.6	2.4	1.0	3.3	0.6
6m 10y20y30y	97.2/100/34.5	17.4	16.9	0.2	66	96	-0.7	-1.1	1.3	18.1	-0.8	-1.7	53	0.5	-1.3	-0.7	0.0	-0.5	1.3	0.7	0.0
1y 1y3y5y	150.8/100/30.0	1.5	2.9	0.0	76	64	0.5	0.9	0.6	3.5	0.0	0.7	61	-1.4	-0.6	-2.0	-0.2	1.4	0.6	2.0	0.2
1y 1y3y7y	150.8/100/21.4	-4.3	-3.4	0.3	10	-65	2.6	1.7	-1.2	-4.6	0.2	1.4	68	-0.9	1.2	0.3	-0.2	0.9	-1.2	-0.3	0.2
1y 2y3y5y	75.2/100/30.0	-3.1	-3.1	0.4	-55	-83	1.2	1.8	-1.3	-4.4	-0.1	-0.4	71	0.0	1.3	1.3	0.0	0.0	-1.3	-1.3	0.0
1y 2y5y7y	125.4/100/35.7	1.4	2.3	0.0	89	96	-0.4	-0.7	4.0	6.3	0.3	1.1	45	-0.8	-4.0	-4.9	0.1	0.8	4.0	4.9	-0.1
1y 2y5y10y	125.4/100/25.1	-7.9	-7.1	0.1	76	-20	1.7	1.5	2.0	-5.0	0.6	1.4	47	-0.8	-2.0	-2.9	0.1	0.8	2.0	2.9	-0.1
1y 2y10y30y	249.9/100/17.8	0.5	6.4	-1.1	92	96	-2.6	-1.6	7.1	13.5	-1.0	-0.6	32	-6.0	-7.1	-13.1	-0.5	6.0	7.1	13.1	0.5
1y 3y5y7y	83.3/100/35.7	-0.7	-0.5	0.2	83	87	0.2	0.7	1.7	1.2	0.3	1.4	47	-0.2	-1.7	-1.9	0.0	0.2	1.7	1.9	0.0
1y 3y5y10y	83.3/100/25.1	-9.9	-9.8	0.3	35	-77	2.3	1.8	-0.1	-9.9	0.6	1.6	57	-0.2	0.1	0.0	0.1	0.2	-0.1	0.0	-0.1
1y 3y7y10y	116.6/100/35.1	1.6	2.8	-0.3	93	95	-1.1	-1.4	3.5	6.3	0.1	0.7	30	-1.2	-3.5	-4.7	0.0	1.2	3.5	4.7	0.0
1y 3y7y15y	116.6/100/23.7	-11.2	-8.9	-0.7	83	59	0.1	0.1	2.2	-6.7	0.4	1.0	30	-2.3	-2.2	-4.5	-0.2	2.3	2.2	4.5	0.2
1y 3y10y30y	166.1/100/17.8	-1.6	3.7	-1.6	88	94	-2.3	-1.6	4.7	8.5	-1.0	-1.3	30	-5.3	-4.7	-10.1	-0.5	5.3	4.7	10.1	0.5
1y 5y10y15y	99.6/100/33.8	2.3	4.0	-0.9	79	93	-2.0	-2.0	2.3	6.3	-0.3	-2.0	23	-1.7	-2.3	-4.0	-0.3	1.7	2.3	4.0	0.3
1y 5y10y30y	99.6/100/17.8	-6.7	-2.1	-2.6	55	79	-1.2	-1.0	1.2	-0.9	-1.0	-3.3	26	-4.6	-1.2	-5.8	-0.5	4.6	1.2	5.8	0.5
1y 10y20y30y	97.1/100/34.5	17.4	16.3	0.4	58	95	-0.5	-0.8	1.7	17.9	-0.5	-0.7	52	1.1	-1.7	-0.5	0.1	-1.1	1.7	0.5	-0.1

Note: Close of 1/16/20. Conditional flies are structured by buying (selling) payers in the belly vs selling (buying) payers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **long belly payer trades in red**, and three **best short belly payer trades in green**. Source: Citi Research

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Figure 108. Conditional receiver flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly				costless strike					long receiver in the belly				short receiver in the belly			
curve	ntl, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, zs	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	150.6/100/29.9	1.5	1.9	1.3	16	63	1.0	1.5	-0.4	2.2	0.9	1.5	57	0.3	0.4	0.7	0.7	-0.3	-0.4	-0.7	-0.7
3m 1y3y7y	150.6/100/21.4	-4.3	-4.0	1.1	-34	-63	3.1	1.8	-1.2	-2.9	0.7	0.7	72	0.2	1.2	1.4	1.4	-0.2	-1.2	-1.4	-1.4
3m 2y3y5y	75.1/100/29.9	-3.1	-3.1	0.5	-55	-81	1.2	1.7	-0.9	-2.2	0.3	0.5	83	0.0	0.9	0.9	0.9	0.0	-0.9	-0.9	-0.9
3m 2y5y7y	125.7/100/35.7	1.4	1.7	0.6	93	96	0.4	0.8	2.4	-0.7	-1.1	-0.8	20	0.3	-2.4	-2.1	-2.0	-0.3	2.4	2.1	2.0
3m 2y5y10y	125.7/100/25.0	-7.9	-7.6	0.6	71	-27	2.8	1.9	1.5	-9.1	-1.5	-1.7	32	0.2	-1.5	-1.2	-1.2	-0.2	1.5	1.2	1.2
3m 2y10y30y	251.0/100/17.7	0.5	2.1	-0.5	94	95	-1.1	-0.7	4.1	-2.0	-1.9	-1.1	13	1.6	-4.1	-2.5	-1.1	-1.6	4.1	2.5	1.1
3m 3y5y7y	83.6/100/35.7	-0.7	-0.5	0.5	81	77	0.6	1.6	1.2	-1.8	-0.7	-1.4	18	0.1	-1.2	-1.1	-1.1	-0.1	1.2	1.1	1.1
3m 3y5y10y	83.6/100/25.0	-9.9	-9.9	0.6	22	-76	2.9	1.9	0.3	-10.2	-1.2	-2.0	55	0.0	-0.3	-0.3	-0.3	0.0	0.3	0.3	0.3
3m 3y7y10y	117.1/100/35.0	1.6	1.9	0.1	93	96	-0.3	-0.5	2.2	-0.3	-0.9	-0.4	16	0.3	-2.2	-1.9	-1.9	-0.3	2.2	1.9	1.9
3m 3y7y15y	117.1/100/23.6	-11.2	-10.6	0.0	82	29	1.2	1.0	1.4	-12.0	-1.3	-1.7	16	0.6	-1.4	-0.8	-0.8	-0.6	1.4	0.8	0.8
3m 3y10y30y	167.0/100/17.7	-1.6	-0.2	-1.0	90	93	-1.3	-0.9	2.9	-3.0	-1.6	-1.5	10	1.4	-2.9	-1.4	-1.3	-1.4	2.9	1.4	1.3
3m 5y10y15y	99.9/100/33.7	2.3	2.8	-0.6	87	94	-1.6	-1.8	1.1	1.7	0.3	0.5	15	0.5	-1.1	-0.6	-0.6	-0.5	1.1	0.6	0.6
3m 5y10y30y	99.9/100/17.7	-6.7	-5.5	-2.3	61	68	-0.8	-0.6	0.6	-6.1	-1.0	-2.2	8	1.2	-0.6	0.6	0.6	-1.2	0.6	-0.6	-0.6
3m 10y20y30y	97.2/100/34.4	17.4	17.2	0.2	70	97	-0.8	-1.2	1.0	16.2	0.4	1.1	43	-0.3	-1.0	-1.2	-1.2	0.3	1.0	1.2	1.2
6m 1y3y5y	150.6/100/29.9	1.5	2.3	0.8	33	62	0.7	1.2	-0.2	2.5	0.6	0.8	58	0.8	0.2	1.0	0.2	-0.8	-0.2	-1.0	-0.2
6m 1y3y7y	150.6/100/21.4	-4.3	-3.8	0.8	-25	-65	2.9	1.7	-1.4	-2.4	0.4	0.1	73	0.5	1.4	1.9	0.4	-0.5	-1.4	-1.9	-0.4
6m 2y3y5y	75.1/100/29.9	-3.1	-3.1	0.4	-59	-82	1.2	1.7	-1.2	-1.9	0.2	0.4	84	-0.1	1.2	1.1	0.3	0.1	-1.2	-1.1	-0.3
6m 2y5y7y	125.6/100/35.7	1.4	2.0	0.4	92	96	0.1	0.1	3.1	-1.1	-1.5	-1.3	19	0.5	-3.1	-2.5	-0.4	-0.5	3.1	2.5	0.4
6m 2y5y10y	125.6/100/25.0	-7.9	-7.4	0.5	73	-26	2.4	1.8	1.7	-9.2	-1.9	-2.0	32	0.5	-1.7	-1.3	-0.1	-0.5	1.7	1.3	0.1
6m 2y10y30y	250.7/100/17.7	0.5	3.6	-0.7	94	96	-1.7	-1.1	5.6	-1.9	-2.3	-1.4	13	3.2	-5.6	-2.4	-0.1	-3.2	5.6	2.4	0.1
6m 3y5y7y	83.6/100/35.7	-0.7	-0.5	0.4	82	81	0.4	1.3	1.5	-2.0	-0.9	-1.6	16	0.2	-1.5	-1.3	-0.2	-0.2	1.5	1.3	0.2
6m 3y5y10y	83.6/100/25.0	-9.9	-9.9	0.5	28	-76	2.7	1.9	0.2	-10.1	-1.4	-1.9	56	0.1	-0.2	-0.2	0.1	-0.1	0.2	0.2	-0.1
6m 3y7y10y	117.0/100/35.1	1.6	2.2	0.0	93	96	-0.6	-0.9	2.8	-0.6	-1.1	-0.9	16	0.6	-2.8	-2.2	-0.3	-0.6	2.8	2.2	0.3
6m 3y7y15y	117.0/100/23.7	-11.2	-10.0	-0.2	82	40	0.8	0.8	1.8	-11.8	-1.4	-2.1	17	1.2	-1.8	-0.6	0.0	-1.2	1.8	0.6	0.0
6m 3y10y30y	166.8/100/17.7	-1.6	1.2	-1.2	90	94	-1.7	-1.2	3.8	-2.7	-1.7	-1.6	11	2.8	-3.8	-1.0	0.1	-2.8	3.8	1.0	-0.1
6m 5y10y15y	99.8/100/33.7	2.3	3.2	-0.7	85	93	-1.8	-1.9	1.5	1.6	0.3	0.5	15	0.9	-1.5	-0.6	-0.1	-0.9	1.5	0.6	0.1
6m 5y10y30y	99.8/100/17.7	-6.7	-4.4	-2.4	60	73	-1.0	-0.8	0.9	-5.3	-0.9	-2.2	10	2.4	-0.9	1.5	0.4	-2.4	0.9	-1.5	-0.4
6m 10y20y30y	97.2/100/34.5	17.4	16.9	0.2	66	96	-0.7	-1.1	1.3	15.6	0.5	1.4	40	-0.5	-1.3	-1.8	-0.4	0.5	1.3	1.8	0.4
1y 1y3y5y	150.8/100/30.0	1.5	2.9	0.0	76	64	0.5	0.9	0.5	2.4	-0.6	-0.5	58	1.4	-0.5	0.9	0.0	-1.4	0.5	-0.9	0.0
1y 1y3y7y	150.8/100/21.4	-4.3	-3.4	0.3	10	-65	2.6	1.7	-1.1	-2.3	-1.0	-1.2	73	0.9	1.1	2.0	0.1	-0.9	-1.1	-2.0	-0.1
1y 2y3y5y	75.2/100/30.0	-3.1	-3.1	0.4	-55	-83	1.2	1.8	-1.2	-1.9	0.0	0.6	84	0.0	1.2	1.1	0.1	0.0	-1.2	-1.1	-0.1
1y 2y5y7y	125.4/100/35.7	1.4	2.3	0.0	89	96	-0.4	-0.7	3.6	-1.3	-1.5	-1.7	18	0.8	-3.6	-2.7	-0.1	-0.8	3.6	2.7	0.1
1y 2y5y10y	125.4/100/25.1	-7.9	-7.1	0.1	76	-20	1.7	1.5	1.8	-8.9	-2.2	-2.6	33	0.8	-1.8	-1.0	0.1	-0.8	1.8	1.0	-0.1
1y 2y10y30y	249.9/100/17.8	0.5	6.4	-1.1	92	96	-2.6	-1.6	6.7	-0.3	-1.5	-1.1	16	6.0	-6.7	-0.7	0.3	-6.0	6.7	0.7	-0.3
1y 3y5y7y	83.3/100/35.7	-0.7	-0.5	0.2	83	87	0.2	0.7	1.5	-2.0	-0.9	-2.1	16	0.2	-1.5	-1.3	-0.1	-0.2	1.5	1.3	0.1
1y 3y5y10y	83.3/100/25.1	-9.9	-9.8	0.3	35	-77	2.3	1.8	-0.1	-9.7	-1.7	-2.4	58	0.2	0.1	0.3	0.1	-0.2	-0.1	-0.3	-0.1
1y 3y7y10y	116.6/100/35.1	1.6	2.8	-0.3	93	95	-1.1	-1.4	3.2	-0.5	-0.8	-1.3	16	1.2	-3.2	-2.1	0.0	-1.2	3.2	2.1	0.0
1y 3y7y15y	116.6/100/23.7	-11.2	-8.9	-0.7	83	59	0.1	0.1	2.0	-10.9	-1.3	-2.3	19	2.3	-2.0	0.3	0.2	-2.3	2.0	-0.3	-0.2
1y 3y10y30y	166.1/100/17.8	-1.6	3.7	-1.6	88	94	-2.3	-1.6	4.5	-0.8	-1.0	-1.2	15	5.3	-4.5	0.8	0.3	-5.3	4.5	-0.8	-0.3
1y 5y10y15y	99.6/100/33.8	2.3	4.0	-0.9	79	93	-2.0	-2.0	2.2	1.8	0.8	1.1	15	1.7	-2.2	-0.5	-0.1	-1.7	2.2	0.5	0.1
1y 5y10y30y	99.6/100/17.8	-6.7	-2.1	-2.6	55	79	-1.2	-1.0	1.1	-3.2	-0.5	-1.3	13	4.6	-1.1	3.5	0.5	-4.6	1.1	-3.5	-0.5
1y 10y20y30y	97.1/100/34.5	17.4	16.3	0.4	58	95	-0.5	-0.8	1.8	14.5	0.3	1.2	37	-1.1	-1.8	-2.9	-0.3	1.1	1.8	2.9	0.3

Note: Close of 1/16/20. Conditional flies are structured by buying (selling) receivers in the belly vs selling (buying) receivers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **long belly receiver trades in red**, and three **best short belly receiver trades in green**. Source: Citi Research

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Appendix A-1

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